Gravitational entanglement: an explicit proper gauge invariant field theoretic approach

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We calculate the gravitationally induced entanglement of two distinguishable massive particles in the effective quantum field theory of general relativity. Each particle is prepared in a superposition of two freely moving Gaussian wavepackets whose branches reconverge in free flight, while the two particles remain well separated. The gravitational initial data are fixed by solving the Einstein constraints with operator-valued matter sources on the initial Cauchy slice. The constrained part of the field is then a function of the matter operators, the transverse-traceless (TT) modes are the only independent gravitational degrees of freedom, and they are prepared in their vacuum. The measured subsystems are physical mode observables, defined by returning every configuration to a reference gauge section.

We show that the predictions do not depend on the proper gauge used to compute them. The canonical measure and action reduce to the same probability functional on every admitted section. An explicit finite-time Ward identity shows how the dressing of the initial state and of the detector cancels the boundary variation of the interaction, for changes of the lapse as well as of the shift. For the declared state, detector probabilities and the leading negativity computed in different sections agree up to $O(G^2)$.

The finite-time channel contains instantaneous constraint exchange, virtual TT exchange, and TT gain and loss. In a benchmark, the selected two-particle state has smallest partial-transpose eigenvalue $\simeq -5.16 \times 10^{-6}$ at leading order in G. At this order the TT part and the completed constraint part of the entangling amplitude are separately gauge invariant. For tuned preparations, the completed constraint part vanishes while the TT sector still produces a negative partial transpose; a model with quantum constraint interactions and a classical TT mean field would give no leading-order entanglement there. Finally, an entanglement witness built from local mode observables gives the calculated entanglement an operational meaning in principle.

I. INTRODUCTION

Two massive particles, each prepared in a superposition of spatially distinct alternatives, can acquire gravitational phases that depend jointly on both alternatives. That joint phase entangles their motional states. This mechanism underlies the proposals to test the quantum nature of gravity through gravitationally induced entanglement [1, 2], reviewed in Refs. [3, 4].

Because the leading entangling interaction in such proposals is Newtonian, its interpretation has been debated. In a canonical description the Newtonian potential is fixed by the constraints of general relativity rather than by an independent gravitational degree of freedom. Several analyses therefore argue that entanglement generated in this way does not by itself require quantized gravitational degrees of freedom [5, 6]. Others argue that consistency with causality and complementarity ties the Newtonian entanglement to the quantization of the radiative field [7, 8], or describe the entanglement as locally mediated by the linearized field [9]. A related concern is that independently gauge-fixed quantizations of weak-field gravity coupled to matter may give gauge-dependent predictions [5].

Such questions can only be settled for specified observables. A prediction for gravitationally induced entanglement must state which physical preparation and which detector observables it refers to. It must also show that the result does not depend on the gauge used to compute it. This paper does both for a concrete problem, in the effective field theory of general relativity [10].

The ingredients exist in the literature. Open-system descriptions in which matter is coupled to linearized gravity and the gravitons are traced out have been developed for gravitational decoherence [11–14]. The gravitational analogue of Dirac's dressed charges [15], and the nonlocal algebra of diffeomorphism-invariant observables it implies, are developed in Refs. [16, 17]. An operational field-theoretic model of gravitationally induced entanglement is given in Ref. [18]. Our contribution is their concrete integration for one freely propagating two-particle problem, in which the preparation, the dynamics, the observables, and the gauge argument are all specified explicitly.

We consider two distinguishable particles, A and B. Each is prepared in an equal superposition of two Gaussian wavepackets ("branches") aimed so that they reconverge in free flight, while A and B remain separated. No trap, applied force, or recombination pulse enters the model. Finite packets overlap in Hilbert space, and gravity can move amplitude between the packet modes or emit a graviton. The prediction therefore refers to specified outgoing modes: we compute the two-particle state conditioned on finding each particle in them. The contributions are the following.

1. **A physical Cauchy problem.** We solve the Hamiltonian and momentum constraints on the initial slice with operator-valued matter sources. The constrained gravitational field is then an operator function of the matter. The independent gravitational degrees of freedom are the TT modes, prepared in their vacuum. A bare product of a gravitational state and a finite-width matter packet fails the sourced constraint (Sec. II C).

2. **Proper-gauge independence.** The experiment is specified relative to prescribed asymptotic data, the

laboratory frame and the asymptotic clock, which define its space of admissible solutions. Diffeomorphism covariance alone does not decide which transformations are gauge redundancies in such a problem: *proper* gauge transformations act trivially on its physical states, whereas asymptotic symmetries, such as translations, boosts, and the time evolution, preserve the problem but act on its states (Sec. II B). The measured mode observables are defined by returning each configuration to a reference section. Following the Faddeev measure [19] and the Fradkin–Vilkovisky gauge-independence theorem [20–23], we show that the canonical measure and action reduce to the same probability functional on every admitted section, with the finite-time boundary kernels included. An explicit endpoint Ward identity shows how the dressing of the preparation and the detector cancels the boundary variation of the interaction, for lapse and shift. For the declared state, the detector probabilities and the leading negativity agree between sections up to $O(G^2)$ (Sec. II E).

3. **The finite-time channel and its entanglement.** The reduced channel contains instantaneous constraint exchange, virtual TT exchange, and TT gain and loss, followed by conditioning on the selected outgoing modes. The leading negativity is fixed by five numbers, and the selected state is entangled in a benchmark (Secs. III and IV).

4. **Entanglement from the TT sector.** At this order the TT and completed constraint parts of the entangling amplitude are separately gauge invariant. For tuned preparations the constraint part vanishes and the TT sector still entangles the selected modes at leading order (Sec. IV C).

5. **Operational meaning.** An entanglement witness built from local mode observables certifies the calculated state, including conditioning on selection (Sec. V).

The readout establishes operational meaning in principle; it is not an experimental proposal.

The initial state in covariant calculations. A covariant bulk action does not by itself specify a physical finite-time transition amplitude: its quantum boundary data must also satisfy the constraints. In Ref. [9], the boundary states are products of a field state and a sharp position, with the field given by the sourced static Newtonian field of the masses. Before such boundary data can describe finite-width quantum matter, they need an explicit constraint-compatible interpretation: a product of a gravitational state with a finite-width packet fails the sourced constraint (Sec. II C). We specify the reduced Cauchy state and its dressed representatives from the outset. Reference [5] stresses that independently gauge-fixed quantizations need not be unitarily equivalent. Our statement is narrower: within one perturbative quantization, the probabilities of the same physical observables, for the same physical preparation, do not depend on the proper-gauge section used to compute them.

Outline. Section II defines the physical problem and proves gauge independence. Section III derives the finite-time channel and its Newtonian limit. Section IV evaluates the entanglement of the selected state, including the TT-sector result. Section V constructs the witness, and Section VI summarizes the results and their scope. Appendix A reviews the canonical formulation; Appendix B gives the second-order constraint and dressing construction and the explicit order-G argument. Appendix C compares with the covariant closed-time-path calculation. Appendices D–G contain point-history limits, the selected matrix elements and their numerical evaluation, the perturbative partial transpose, and the constraint zeros. Appendix H explains why position statistics at exact overlap cannot certify the state. Readers who want the physics first can read Sec. II A, Sec. III D, and Sec. IV, and return to the gauge argument afterwards.

A. Notation

Symbol	Meaning	Where defined
$\hbar = c = 1$, $\eta_{\mu\nu} = \mathrm{diag}(-1,1,1,1)$	natural units, mostly-plus signature	Eq. (2.1)
G, $\kappa_g = \sqrt{32\pi G}$	Newton's constant; graviton coupling ($g_{\mu\nu} = \eta_{\mu\nu} + \kappa_g h_{\mu\nu}$)	Eq. (2.1)
$g = Gm_A m_B$	strength of the Newtonian interaction (potential $-g/r$), dimensionless for $\hbar = c = 1$	Eq. (2.1)
$\int_{\mathbf{p}}$, $\int_{\mathbf{k}}$	$\int d^3p/(2\pi)^3$, $\int d^3k/(2\pi)^3$	Eq. (2.1)
$\alpha \in \{A, B\}$	particle (species) label	Sec. II A
$s, r \in \{+,-\}$ (also ± 1)	branch label of one particle	Eq. (2.2)
$\mathbf{a}_{\alpha s}$, $\mathbf{v}_{\alpha s}$, $\mathbf{p}_{\alpha s}$	initial center, velocity, and central momentum of branch s	Eqs. (2.2), (2.5)
L, b, d_α	initial half-distance along x; impact separation; initial branch separation ("arm length")	Eq. (2.2)
T, t_f	designed crossing time; actual detection time	Sec. II A
ℓ_α, $\sigma_{x,\alpha}(t)$	initial rms position width; nonrelativistic rms width at time t	Eqs. (2.5), (2.16)
$f_{\alpha s}$, $w_{\alpha s}$	raw Gaussian packet; orthonormalized packet mode	Eqs. (2.5), (2.8)
S_α	overlap of the two raw branch packets	Eq. (2.7)

Symbol	Meaning	Where defined
$c^\dagger_{\alpha s}$, $\lvert s\rangle_\alpha$	creation operator of mode $w_{\alpha s}$; one-particle state in that mode	Eqs. (2.8), (2.11)
X, Y, Z; $\lvert\pm_x\rangle$	Pauli operators on $\{\lvert+\rangle, \lvert-\rangle\}$; eigenstates $(\lvert+\rangle \pm \lvert-\rangle)/\sqrt{2}$ of X	Eq. (2.10)
$P_{\rm sel}$, $p_{\rm sel}$, $\rho_{\rm det}$	projector onto the selected modes; selection probability; conditional state	Eq. (2.14)
$N, N_i, \gamma_{ij}, K_{ij}$	lapse, shift, spatial metric, extrinsic curvature	Eqs. (2.17), (2.22)
ξ^μ; $(\xi^\perp, \xi^i)$	descriptor (generating vector field) of an infinitesimal transformation; its normal and tangential parts on a slice	Sec. II B
Γ; $\bar h_{ij}$, $\bar\pi^{ij}$	space of admissible initial data and solutions; solution-dependent leading asymptotic coefficients of γ_{ij} and π^{ij}	Sec. II B, Eq. (2.19)
$\mathscr{G}_\Sigma[\xi]$, $B_{\partial\Sigma}[\xi]$; $E_{\rm ADM}$	canonical generator of a transformation and its boundary contribution (charge); ADM energy	Eq. (2.20), Sec. II B
$\mathbf{\Omega}_\Sigma$	presymplectic form; its null directions are the proper gauge directions	Eq. (2.21)
ψ, $h^{\rm TT}_{ij}$	scalar (Newtonian) metric potential; TT graviton field	Eq. (2.22)
ε, j_i, T_{ij}	energy density, momentum density, spatial stress	Eqs. (2.18), (3.4)
π, F_f	return map to the reference section; physical extension of an observable f	Eq. (2.29)
Q_a, P_a, C_a, J_a	gauge coordinates, their momenta, constraints, and sources ($a = 0, \dots, 3$)	Eq. (2.30), Appendix A C
D, E_t	first-order dressing generator; dressing map at time t	Eqs. (2.31), (2.33)
A_a, $\bar C_a$	gauge momenta solved from the constraints; Abelianized constraints	Eqs. (2.36), (2.37)
S_1, b_ξ	linear matter–gravity action; its endpoint generator	Eq. (2.43)
$\tau^{\mu\nu}_\alpha(p', p)$	one-particle matrix element of the stress tensor ("stress vertex")	Eqs. (3.1)–(3.3)
$\mathcal{C}_{AB}$, H_C	kernel of the constraint interaction; constraint (instantaneous) interaction	Eqs. (3.5), (3.6)
$\beta_{\lambda\mathbf{k}}$, e^λ_{ij}	graviton annihilation operator; polarization tensor	Eq. (3.10)
$\mathsf{v}_{\lambda\mathbf{k}}$, $L_{\lambda\mathbf{k}}$	graviton emission vertex; its time integral (one-graviton Kraus operator)	Eq. (3.12)
$\mathcal{U}_0$	no-graviton amplitude, an operator on the matter	Sec. III B
K (sans-serif)	time-integrated Hamiltonian, $\mathsf{K} = \int_0^{t_f} H\, dt$	Eq. (3.13)
$\Gamma_{\rm TT}$	loss operator from graviton emission	Eq. (3.13)
$M_{\lambda\mathbf{k}}$, $B_{\lambda\mathbf{k}}$	emission amplitude inside the selected modes; probability operator for leaving them	Eq. (3.15)
Θ	entangling (two-body) phase	Eq. (3.21)
$\mathcal{N}$	negativity	Eq. (3.23)
$\mathcal{E}$, $\mathcal{E}_C$, $\mathcal{E}_{\rm TT}$, ν_α, $\gamma_{\rm cross}$	double-flip amplitude and its constraint and TT parts; local gain; cross loss	Eq. (4.1)
D	leading partial-transpose block	Eq. (4.2)
W	entanglement witness	Eq. (5.1)

Symbols with more than one meaning. Several letters are unavoidably reused; the context always decides.

- T is the designed crossing time; $T^{\mu\nu}$ (always with indices) is the stress tensor; T (sans-serif) is the spatial-stress matrix.
- K_{ij}, K (italic) are the extrinsic curvature and its trace. K (sans-serif) is an integrated generator. K_R, K_N, K_Q are time kernels (Appendices C and E). $K_a{}^b$ is the matrix relating two constraint bases (Sec. II E).
- L is a length. $L_{\lambda\mathbf{k}}$ (always with subscripts) is an emission operator. $L_{g,\rm SV}$ is a Lagrangian (Appendix A).
- b is the impact separation. In Appendix C, a and b denote the recoil frequencies ω_A and ω_B, and b_ξ (Sec. II E) is an endpoint generator. In Appendix F, A, B, D are coherence factors, not particle labels. Besides the particle label B, the letter appears as $B_{\lambda\mathbf{k}}$ (the leakage operator of Eq. (3.15)), as B_f, B_i (endpoint functions of the action, Sec. II E), as

B^1, B^2, B^n (projected endpoint integrals, Appendix C), and as $B_{\partial\Sigma}[\xi]$ (the surface term of a canonical generator, Eq. (2.20)).

- D is the first-order dressing generator, D_2 its second-order completion, D_B a retarded distance (Appendix D), and D the leading partial-transpose block.
- Θ is the entangling phase; $\Theta(t)$ with a time argument (Sec. III C) is the Heaviside step function.
- γ_{ij} is the spatial metric, $\gamma_{\rm cross}$ a cross-loss amplitude, and γ_A, γ_B are Lorentz factors (Appendices C–D).
- Γ is the space of admissible initial data (Sec. II B); $\Gamma_{\rm TT}$, $\Gamma_{\rm cross}$, and $\Gamma_{\alpha,\rm self}$ are loss operators (Eqs. (3.13), (4.1), (E.19)).
- G is Newton's constant; $G_{\rm proper}$, $G_{\rm allowed}$, and $G_{\rm asym}$ are groups of transformations (Sec. II B), G_R and G_H are Green functions (Appendix C A), and G_α is the Gram matrix of Eq. (2.8).
- $E_{\alpha,\mathbf{p}}$ is a single-particle energy, E_t a dressing map, $E_\alpha(\mathbf{x})$ the position POVM of Appendix H, and $E_{\rm ADM}$ the ADM energy (Sec. II B). The bold $\mathbf{E}$ is the electric field of Appendix A B.
- π is the return map to the reference section; π^{ij} the ADM momentum; $\pi^{\rm TT}_{ij}$ the TT momentum.
- ψ is the scalar metric potential. $|\psi\rangle$, $|\psi_\Theta\rangle$ are matter states, and Ψ is a wavefunctional.
- $u = 1 - t/T$ is the remaining fraction of the flight. In Eq. (E.18), u is an energy-mismatch variable.
- N is the lapse, $\mathcal{N}$ the negativity (and, in Eq. (A.9) only, a normalization constant).
- n is the lapse perturbation and n^μ the unit normal to a slice. The bold $\mathbf{n}$ is always a unit vector: radial, $\mathbf{n} = \mathbf{x}/r$, in Eq. (2.19) and the BMS discussion of Sec. II B, and otherwise along a wavevector or a separation.
- ξ^μ is the descriptor of a transformation (Secs. II B and II E); the bold $\boldsymbol{\xi}_A, \boldsymbol{\xi}_B$ of Appendix D A are branch-separation vectors.
- χ denotes a gauge section in Sec. II E, the observation window $\chi(t)$ in Sec. III C and Appendix C, and the doubly flipped state $|\chi\rangle$ in Sec. IV C.
- F_f is the physical extension of an observable f, Eq. (2.29); $F(\delta)$ is the branch-phase integral of Appendix D A; F_i is a metric combination in Appendix A C; and F_{rs}, $F_{\rm TT}$ are a transition-density transform and an angular factor in Appendices E and D.
- M_{ij} is the vector operator of Appendix A C, $\mathcal{M}$ the constraint matrix of Eq. (A.4), and $M_{\lambda\mathbf{k}}$ the selected emission amplitude, Eq. (3.15).
- X, Y are Pauli operators, except in Eq. (E.15), where they are scalar coefficients, and in Eq. (4.5) and Appendix B E, where they are generic local operators. X also denotes a point of phase space in Eq. (2.29) and Sec. II E.

II. PHYSICAL SETUP AND GAUGE-INDEPENDENT FORMULATION

This section defines the physical problem and shows that its predictions do not depend on the gauge used to compute them. Subsection A specifies the matter preparation and the measured modes. Subsection B explains why the setup fixes asymptotic boundary conditions and which transformations are then gauge redundancies. Subsection C solves the constraints for the Cauchy data and defines the physical observables. Subsection D relates the reduced description to one that keeps the gauge variables. Subsection E proves gauge independence and states its scope.

A. Geometry, matter sector, and selected modes

We use natural units and the conventions

$$\begin{aligned} &\hbar = c = 1, \\ &\quad \eta_{\mu\nu} = \mathrm{diag}(-1,1,1,1), \\ &\quad \kappa_g^2 = 32\pi G, \\ &\quad g = G m_A m_B, \\ &\quad \int_{\mathbf{p}} = \int \frac{d^3p}{(2\pi)^3}. \end{aligned} \tag{2.1}$$

Here G is Newton's constant. κ_g is the conventional graviton coupling. If the metric is written $g_{\mu\nu} = \eta_{\mu\nu} + \kappa_g h_{\mu\nu}$, the choice $\kappa_g^2 = 32\pi G$ makes the Einstein–Hilbert action for $h_{\mu\nu}$ canonically normalized (the standard convention of gravitational effective field theory [10]), so that each graviton polarization is an ordinary field with kinetic term $\frac{1}{2}\dot{h}^2$. The product $g = G m_A m_B$ is the strength of the Newtonian interaction between the two particles (potential energy $-g/r$). With $\hbar = c = 1$ it is dimensionless, and it is the expansion parameter of the calculation.

Matter model. Each particle is modeled as a single quantum of its own free, real, massive scalar field ϕ_α, one for each species $\alpha \in \{A, B\}$. A scalar field is the simplest relativistic matter with a well-defined, conserved stress tensor, and spin plays no role at the order considered. Because A and B are different fields, they are distinguishable and no exchange symmetrization is needed. The masses m_α are the physical (renormalized) masses, and

the stress tensor is normal ordered (vacuum subtracted), so the vacuum carries no energy.

We keep only the *number-preserving* part of every matter operator. These are the terms of the form $a^\dagger a$, which move a particle from one momentum to another. Terms of the form aa and $a^\dagger a^\dagger$, which would create or annihilate particle–antiparticle pairs, are dropped. For slow particles with momenta far below m_α they are strongly suppressed, since their energy denominators are at least of order $2m_\alpha$. Within this "one-particle sector" we compute the gravitational effect to first order in G. The interaction calculation uses the exact relativistic energies $E = \sqrt{m^2 + \mathbf{p}^2}$. A nonrelativistic approximation is made only later, when modeling the position detector.

Geometry. Everything happens in the xy plane (Fig. 1). The mean approach is along x, the impact separation is along y, and both particles split along y. Each particle is split into two branches labeled by $s = \pm 1$. The center of branch s of particle α starts at $\mathbf{a}_{\alpha,s}$ and moves with constant velocity $\mathbf{v}_{\alpha,s}$:

$$\begin{aligned} \mathbf{a}_{A,s} &= (-L, b/2 + sd_A/2, 0), \\ \mathbf{v}_{A,s} &= (L/T, -sd_A/(2T), 0), \\ \mathbf{a}_{B,s} &= (L, -b/2 + sd_B/2, 0), \\ \mathbf{v}_{B,s} &= (-L/T, -sd_B/(2T), 0), \\ \mathbf{x}_{\alpha,s}(t) &= \mathbf{a}_{\alpha,s} + \mathbf{v}_{\alpha,s} t, \\ L, b, T &> 0. \end{aligned} \tag{2.2}$$

Reading off Eq. (2.2): particle A starts at $x = -L$ and moves toward $+x$, particle B starts at $x = +L$ and moves toward $-x$, both with longitudinal speed L/T. The two branches of A start at $y = b/2 \pm d_A/2$, separated by the *arm length* d_A. They have opposite transverse velocities $\mp d_A/(2T)$ chosen so that they converge. The same holds for B around $y = -b/2$. The centers are:

So at the designed crossing time T, the two A centers coincide at $(0, b/2, 0)$ and the two B centers coincide at $(0, -b/2, 0)$. The two particles never meet: their closest approach is set by b (Eq. (2.15)). We denote the actual detection time by t_f and initially take $t_f = T$. Equation (2.2) specifies the nonrelativistic zeroth-order geometry about which the interaction is calculated. Gravity is allowed to perturb this motion, but the resulting displacement must remain small on the scales specified in Eq. (2.16).

The trajectories in Eq. (2.2) are the centers of wavepackets, not classical paths or position eigenstates. Gravity couples to the stress tensor of a field, so we build the corresponding quantum states directly in the Fock space of the scalar fields. For each species, the free field has the standard mode expansion

$$\begin{aligned} \phi_\alpha(t, \mathbf{x}) &= \int_{\mathbf{p}} \frac{1}{\sqrt{2E_{\alpha,\mathbf{p}}}} \left[a_{\alpha\mathbf{p}} e^{-iE_{\alpha,\mathbf{p}} t + i\mathbf{p}\cdot\mathbf{x}} + a^\dagger_{\alpha\mathbf{p}} e^{iE_{\alpha,\mathbf{p}} t - i\mathbf{p}\cdot\mathbf{x}} \right], \\ [a_{\alpha\mathbf{p}}, a^\dagger_{\beta\mathbf{q}}] &= \delta_{\alpha\beta}(2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}), \qquad |\mathbf{p}\rangle_\alpha = a^\dagger_{\alpha\mathbf{p}} |0_\alpha\rangle, \\ \langle \mathbf{q}|\mathbf{p}\rangle_\alpha &= (2\pi)^3 \delta^3(\mathbf{q} - \mathbf{p}), \qquad E_{\alpha,\mathbf{p}} = \sqrt{m_\alpha^2 + \mathbf{p}^2}. \end{aligned} \tag{2.3}$$

Note the nonrelativistic normalization $\langle \mathbf{q}|\mathbf{p}\rangle = (2\pi)^3 \delta^3(\mathbf{q} - \mathbf{p})$, without a factor $2E$. It is why $1/\sqrt{2E_{\alpha,\mathbf{p}}}$ appears in the field expansion. With this normalization $\int_{\mathbf{p}} |\mathbf{p}\rangle\langle\mathbf{p}|_\alpha$ is the identity on the one-particle sector.

One-body operators as field bilinears. On the one-particle sector of species α, the operator $|\mathbf{p}'\rangle_\alpha\langle\mathbf{p}|$ coincides with the number-preserving bilinear $a^\dagger_{\alpha\mathbf{p}'} a_{\alpha\mathbf{p}}$. Consequently, any operator O built from the field ϕ_α, restricted to its number-preserving part and to one-particle states, is completely determined by its momentum-space matrix elements:

$$O\big|_{1\text{ particle}} = \int_{\mathbf{p},\mathbf{p}'} \langle \mathbf{p}'|O|\mathbf{p}\rangle_\alpha \, a^\dagger_{\alpha\mathbf{p}'} a_{\alpha\mathbf{p}} = \int_{\mathbf{p},\mathbf{p}'} \langle \mathbf{p}'|O|\mathbf{p}\rangle_\alpha \, |\mathbf{p}'\rangle_\alpha \langle\mathbf{p}|. \tag{2.4}$$

We use this repeatedly. The energy density, momentum density, and stress of each particle enter the gravitational interaction only through such matrix elements (Sec. III A,

Branch	Start ($t = 0$)	At $t = T$
$A, +$	$(-L,\ b/2 + d_A/2,\ 0)$	$(0,\ b/2,\ 0)$
$A, -$	$(-L,\ b/2 - d_A/2,\ 0)$	$(0,\ b/2,\ 0)$
$B, +$	$(+L,\ -b/2 + d_B/2,\ 0)$	$(0,\ -b/2,\ 0)$
$B, -$	$(+L,\ -b/2 - d_B/2,\ 0)$	$(0,\ -b/2,\ 0)$

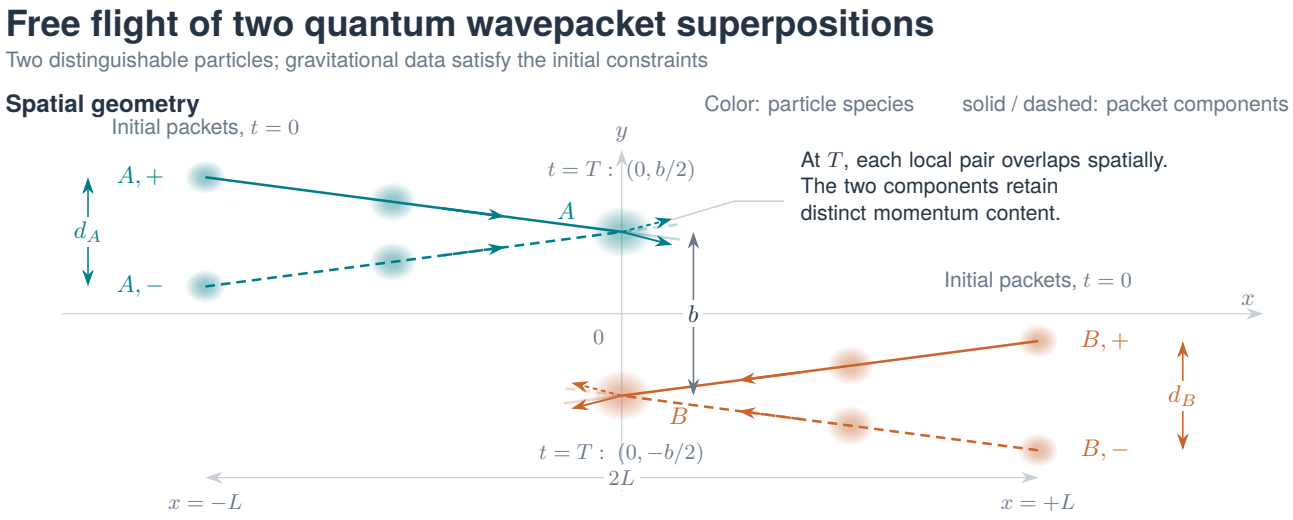


FIG. 1. Planar geometry of Eq. (2.2) (schematic, not to scale). Particle A (teal) starts at $x = -L$ and particle B (orange) at $x = +L$, each in a superposition of two packets (solid and dashed) separated by the arm length d_α. The two branches of each particle reconverge at the designed crossing time T, at $(0, b/2)$ for A and at $(0, -b/2)$ for B. There they overlap in space but keep distinct momenta, so the two packet modes of each particle remain nearly orthogonal (Eq. (2.7)).

Eq. (3.2)). No separate first-quantized position operator is ever introduced.

Branch packets. A branch packet is obtained by smearing the field creation operator with a momentum-space wavefunction. Define

$$
\begin{aligned}
a_\alpha^\dagger[f] &= \int_{\mathbf{p}} f(\mathbf{p}) a_{\alpha\mathbf{p}}^\dagger, \qquad |f_{\alpha s}\rangle = a_\alpha^\dagger[f_{\alpha s}]|0_\alpha\rangle, \\
f_{\alpha s}(\mathbf{p}) &= (8\pi\ell_\alpha^2)^{3/4} \exp[-\ell_\alpha^2(\mathbf{p} - \mathbf{p}_{\alpha s})^2 - i\mathbf{p}\cdot\mathbf{a}_{\alpha s}], \\
\mathbf{p}_{\alpha s} &= m_\alpha \mathbf{v}_{\alpha s}, \qquad \int_{\mathbf{p}} |f_{\alpha s}(\mathbf{p})|^2 = 1.
\end{aligned}
\tag{2.5}
$$

Here $f_{\alpha s}(\mathbf{p})$ is the momentum-space wavefunction of branch s. It is a Gaussian centered at $\mathbf{p}_{\alpha s} = m_\alpha \mathbf{v}_{\alpha s}$, whose width is set by ℓ_α, and the phase $e^{-i\mathbf{p}\cdot\mathbf{a}_{\alpha s}}$ translates the packet so that it is centered at position $\mathbf{a}_{\alpha s}$ at $t = 0$. The prefactor normalizes it with the measure $\int_{\mathbf{p}}$. Thus $|f_{\alpha s}\rangle$ is a normalized one-quantum field state localized near $\mathbf{a}_{\alpha s}$ with mean momentum $\mathbf{p}_{\alpha s}$. The commutator in Eq. (2.3) and the packet norm in Eq. (2.5) fix the normalization convention used throughout the calculation.

The coordinate rms width in the associated nonrelativistic one-particle envelope is ℓ_α, and each momentum component has standard deviation $1/(2\ell_\alpha)$. This field state has a nonrelativistic particle interpretation when its momentum support satisfies $|\mathbf{p}|/m_\alpha \ll 1$, equivalently when both the central speed and $1/(m_\alpha \ell_\alpha)$ are small. Under exact free evolution, its mean velocity is the packet average of $\mathbf{p}/E_{\alpha,\mathbf{p}}$. The momentum centers in Eq. (2.5) therefore reproduce Eq. (2.2) up to corrections of order v^3 and $v(\Delta p/m)^2$. We retain this declared preparation rather than retuning the momenta after replacing the nonrelativistic dispersion by the exact one.

Overlap of the two branches. The two raw branch states of the same particle need not be exactly orthogonal.

Their overlap is a Gaussian integral, which equals the field commutator of the corresponding smeared modes:

$$\begin{aligned}\mathcal{S}_{\alpha,rs} &= \langle f_{\alpha r}|f_{\alpha s}\rangle = [a_\alpha[f_{\alpha r}], a_\alpha^\dagger[f_{\alpha s}]] \\ &= \exp\left[-\frac{|\mathbf{a}_{\alpha r}-\mathbf{a}_{\alpha s}|^2}{8\ell_\alpha^2} - \frac{\ell_\alpha^2|\mathbf{p}_{\alpha r}-\mathbf{p}_{\alpha s}|^2}{2} + i\frac{\mathbf{p}_{\alpha r}+\mathbf{p}_{\alpha s}}{2}\cdot(\mathbf{a}_{\alpha r}-\mathbf{a}_{\alpha s})\right].\end{aligned} \tag{2.6}$$

The first term in the exponent comes from the position separation of the packets, the second from their momentum separation, and the third is a phase. In the geometry of Eq. (2.2), the average branch momentum is perpendicular to the branch displacement, so the phase vanishes and $S_\alpha = \mathcal{S}_{\alpha,+-}$ is real. Superposing the two branches with equal weight gives the normalized local "cat" state of particle α (a superposition of two well-separated packets):

$$\begin{aligned}S_\alpha &= \exp\left[-\frac{d_\alpha^2}{8\ell_\alpha^2} - \frac{\ell_\alpha^2 m_\alpha^2 d_\alpha^2}{2T^2}\right], \\ f_\alpha(\mathbf{p}) &= \frac{f_{\alpha+}(\mathbf{p}) + f_{\alpha-}(\mathbf{p})}{\sqrt{2(1+S_\alpha)}}, \qquad |\mathrm{cat}_\alpha\rangle = a_\alpha^\dagger[f_\alpha]|0_\alpha\rangle.\end{aligned} \tag{2.7}$$

Orthonormal branch modes. Because $S_\alpha \neq 0$, the two raw packets are not orthogonal, so "which branch" is not yet a two-level system with an orthonormal basis. Equivalently, the two raw creation operators do not define independent modes, since their commutator matrix is the Gram (overlap) matrix G_α rather than the identity. We fix this by *symmetric (Löwdin) orthonormalization* [24], which mixes the two raw packets as little as possible:

$$\begin{aligned}\mathsf{G}_\alpha &= \begin{pmatrix} 1 & S_\alpha \\ S_\alpha & 1 \end{pmatrix}, \\ R_\alpha &= \mathsf{G}_\alpha^{-1/2}, \\ w_{\alpha s}(\mathbf{p}) &= \sum_{r=\pm} f_{\alpha r}(\mathbf{p})(R_\alpha)_{rs},\end{aligned} \tag{2.8}$$

Explicitly, since G_α has eigenvalues $1 \pm S_\alpha$ with eigenvectors $(1, \pm 1)/\sqrt{2}$,

$$\begin{aligned}c_{\alpha s}^\dagger &= a_\alpha^\dagger[w_{\alpha s}], \\ [c_{\alpha r}, c_{\beta s}^\dagger] &= \delta_{\alpha\beta}\delta_{rs}, \\ R_\alpha &= \frac{1}{2}\left[\frac{1}{\sqrt{1+S_\alpha}} + \frac{1}{\sqrt{1-S_\alpha}}\right] I + \frac{1}{2}\left[\frac{1}{\sqrt{1+S_\alpha}} - \frac{1}{\sqrt{1-S_\alpha}}\right]\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.\end{aligned} \tag{2.9}$$

In the benchmark of Sec. IV B, $S_\alpha = e^{-16} \approx 10^{-7}$, so $w_{\alpha s}$ is practically $f_{\alpha s}$. Keeping the exact transformation nevertheless makes all later formulas exact. The construction requires $S_\alpha < 1$: if the raw modes coincided, their span would contain only one field mode and could not encode a local two-level system.

The two qubits. Let $|s\rangle_\alpha = c_{\alpha s}^\dagger|0_\alpha\rangle$ denote the state with particle α in orthonormal mode s. From here on, each particle is described by the two-dimensional space spanned by $|+\rangle_\alpha$ and $|-\rangle_\alpha$, i.e. by *which orthonormal packet mode the particle occupies*. On this space we use the Pauli operators

$$\begin{aligned}Z &= |+\rangle\langle+| - |-\rangle\langle-|, \\ X &= |+\rangle\langle-| + |-\rangle\langle+|, \\ Y &= -i|+\rangle\langle-| + i|-\rangle\langle+|,\end{aligned} \tag{2.10}$$

so that $Z|s\rangle = s|s\rangle$ and the eigenstates of X are $|\pm_x\rangle = (|+\rangle \pm |-\rangle)/\sqrt{2}$. A subscript A or B (e.g. Z_A) indicates which particle an operator acts on. Do not

confuse the branch states $|+\rangle, |-\rangle$ with their symmetric and antisymmetric superpositions $|+_x\rangle, |-_x\rangle$. Since the symmetric vector $(1,1)$ is an eigenvector of G_α, the cat state of Eq. (2.7) is exactly the equal superposition $|+_x\rangle_\alpha$ of the orthonormal modes. The initial two-particle state is therefore

$$\begin{aligned} |\mathrm{in}\rangle &= a_A^\dagger[f_A]a_B^\dagger[f_B]|0_\phi\rangle = |+_x\rangle_A \otimes |+_x\rangle_B, \\ |+_x\rangle_\alpha &= \frac{|+\rangle_\alpha + |-\rangle_\alpha}{\sqrt{2}}, \\ |s_A, s_B\rangle &= c^\dagger_{As_A} c^\dagger_{Bs_B}|0_\phi\rangle, \\ s_A, s_B &\in \{+,-\}, \end{aligned} \tag{2.11}$$

$$\rho_0 = |\mathrm{in}\rangle\langle\mathrm{in}| = \frac{1}{4}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}_{(++,+-,-+,--)} .$$

where the matrix is written in the ordered basis $(|++\rangle, |+-\rangle, |-+\rangle, |--\rangle)$ with the first label for A. Here $|0_\phi\rangle = |0_A\rangle \otimes |0_B\rangle$ is the scalar-field vacuum. Acting once with each species' creation operator fixes the sector to one A quantum and one B quantum. The two-level label records which orthonormal packet mode contains that species' excitation; it is not an occupation-number qubit in an unrestricted Fock space. Except in Sec. IV C, ρ_0 denotes the real-cat input of Eq. (2.11). Section IV C replaces it by a product of locally phase-shifted one-particle mode states, without changing the Fock-space mode algebra.

Selected outgoing modes. We must also specify what is measured at the end. We idealize the detector as asking, for each particle, whether it is found in one of the two *freely propagated* packet modes, and if so which one. Free evolution preserves particle number and the inner products of the packet modes. With

$$\begin{aligned} H_{0,\phi} &= \sum_{\alpha=A,B}\int_{\mathbf{p}} E_{\alpha,\mathbf{p}}\, a^\dagger_{\alpha\mathbf{p}} a_{\alpha\mathbf{p}}, \\ w^{\mathrm{out}}_{\alpha s}(\mathbf{p}, t_f) &= e^{-iE_{\alpha,\mathbf{p}}t_f} w_{\alpha s}(\mathbf{p}), \\ c^{\dagger,\mathrm{out}}_{\alpha s}(t_f) &= a^\dagger_\alpha[w^{\mathrm{out}}_{\alpha s}(t_f)], \end{aligned} \tag{2.12}$$

the state $c^{\dagger,\mathrm{out}}_{\alpha s}(t_f)|0_\alpha\rangle$ is the freely propagated image of incoming mode s. The two outgoing operators retain canonical commutators. They define a mode-selective number measurement at the final time, rather than a projection onto a point in space.

Let $|w^{\mathrm{out}}_{\alpha s}; 1\rangle = c^{\dagger,\mathrm{out}}_{\alpha s}|0_\alpha\rangle$. In the retained one-particle sector, the local projector onto the selected modes and the number-preserving mode operators are

$$\begin{aligned} P_\alpha(t_f) &= \sum_{s=\pm} |w^{\mathrm{out}}_{\alpha s}; 1\rangle\langle w^{\mathrm{out}}_{\alpha s}; 1|, \\ P_{\mathrm{sel}} &= P_A \otimes P_B, \\ \mathcal{O}^\alpha_{rs}(t_f) &= c^{\dagger,\mathrm{out}}_{\alpha r} c^{\mathrm{out}}_{\alpha s}. \end{aligned} \tag{2.13}$$

P_α projects particle α's one-particle space onto the two-dimensional span of its outgoing modes, and P_{sel} means "both particles were found in their selected modes." On the selected one-particle subspace, $\mathcal{O}^\alpha_{rs}$ acts as the matrix unit $|r\rangle_\alpha\langle s|$. Gravity can scatter a particle out of this span, for example through the recoil from an emitted graviton. Such events fall outside P_{sel}; they are discarded and reduce the selection probability.

Let $\rho_{\mathrm{matter}}(t_f) = \mathrm{Tr}_{\mathrm{TT}}\,\rho_{\mathrm{tot}}(t_f)$ be the matter state after the TT (graviton) field has been traced out. The unnormalized 4×4 detector matrix has entries $(\widetilde{\rho}_{\mathrm{sel}})_{\mathbf{r},\mathbf{s}} = \langle\mathbf{r}|\rho_{\mathrm{matter}}|\mathbf{s}\rangle = \mathrm{Tr}[\rho_{\mathrm{matter}}\,|\mathbf{s}\rangle\langle\mathbf{r}|]$. Written with field-mode operators, for joint indices $\mathbf{r} = (r_A, r_B)$ and $\mathbf{s} = (s_A, s_B)$,

$$\begin{aligned} (\widetilde{\rho}_{\mathrm{sel}})_{\mathbf{r},\mathbf{s}} &= \mathrm{Tr}\big[\rho_{\mathrm{matter}}(t_f)\mathcal{O}^A_{s_A r_A}(t_f)\mathcal{O}^B_{s_B r_B}(t_f)\big]\,, \\ \widetilde{\rho}_{\mathrm{sel}} &= P_{\mathrm{sel}}\rho_{\mathrm{matter}}(t_f)P_{\mathrm{sel}}, \\ p_{\mathrm{sel}} &= \mathrm{Tr}\,\widetilde{\rho}_{\mathrm{sel}}, \\ \rho_{\mathrm{det}} &= \frac{\widetilde{\rho}_{\mathrm{sel}}}{p_{\mathrm{sel}}}. \end{aligned} \tag{2.14}$$

p_{sel} is the probability that both particles are found in their selected modes, and ρ_{det} is the normalized two-qubit state conditioned on that event. ρ_{det} **is the object**

whose entanglement we compute. The second line identifies the same 4×4 block as an operator on the fixed $(N_A, N_B) = (1, 1)$ sector, while the first line states what its entries mean in the scalar field theory. In the interaction picture the free evolution is removed from the states, so the outgoing modes are "pulled back" to the incoming $w_{\alpha s}$. Hence $P_{\rm sel}$ and the matrix units are time independent in that picture. A generic position click does not by itself implement $P_{\rm sel}$; this operational issue is addressed in Section V.

Distance between the particles. The relevant interspecies distance is the minimum over all branch pairs and all times in the observation interval. Let $u = 1 - t/T$ be the remaining fraction of the flight ($u = 1$ at $t = 0$, $u = 0$ at $t = T$) and $\delta = (s_A d_A - s_B d_B)/2$. From Eq. (2.2), the center of branch s_A of A and the center of branch s_B of B are separated by $2Lu$ along x and by $b + \delta u$ along y, so their distance r obeys

$$\begin{aligned} r^2(u) &= (2L)^2 u^2 + (b + \delta u)^2, \\ u_* &= -\frac{b\delta}{(2L)^2 + \delta^2}, \\ r(u_*) &= \frac{2Lb}{\sqrt{(2L)^2 + \delta^2}}. \end{aligned} \tag{2.15}$$

The stationary point u_* is used only if it lies in the interval; otherwise the minimum is at an endpoint. For $t_f > T$, the minimum must be recomputed on the extended histories. Positive L and b prevent a center collision, while suppression of the Gaussian tails requires the combined width to remain small relative to $r_{\rm min}$.

Regime of validity. A sufficient regime for the one-particle interpretation, narrow-packet geometry, and undistorted outgoing modes is

$$\begin{gathered} \ell_\alpha \ll d_\alpha, \quad S_\alpha \ll 1, \quad \frac{L}{T}, \frac{d_\alpha}{T}, \frac{\hbar}{m_\alpha \ell_\alpha} \ll c, \qquad \max_{0 \le t \le t_f} \sqrt{\sigma_{x,A}(t)^2 + \sigma_{x,B}(t)^2} \ll r_{\rm min}, \\ \frac{G(m_A + m_B)}{c^2 r_{\rm min}} \ll 1, \qquad \Delta x_{\rm grav} \lesssim \frac{G(m_A + m_B) t_f^2}{r_{\rm min}^2} \ll \min(\ell_A, \ell_B, r_{\rm min}), \\ \Delta p_{{\rm grav},\alpha} \lesssim \frac{G m_A m_B t_f}{r_{\rm min}^2} \ll \frac{\hbar}{\ell_\alpha}. \end{gathered} \tag{2.16}$$

Here $\sigma_{x,\alpha}(t)^2 = \ell_\alpha^2 + t^2/(4m_\alpha^2 \ell_\alpha^2)$ is the squared rms width of the nonrelativistic position envelope of a packet at time t (Appendix H). In words: (i) $\ell_\alpha \ll d_\alpha$ and $S_\alpha \ll 1$ mean the two branches start out distinct. (ii) The velocities and the velocity spread $\hbar/(m_\alpha \ell_\alpha)$ are nonrelativistic. (iii) The combined width of the A and B packets stays small compared with their separation, so the two particles' wavefunctions never overlap. (iv) The gravitational field is weak. (v) The gravitational displacement, estimated as acceleration $G(m_A + m_B)/r_{\rm min}^2$ times t_f^2, is small compared with the packet widths and separation. (vi) The gravitational momentum kick is small compared with the momentum spread $\hbar/\ell_\alpha$, so the outgoing modes are not distorted. Conditions (v) and (vi) justify treating gravity perturbatively about free motion.

Small raw overlap is useful for interpreting separated arms; Gram orthonormalization and the exact channel retain its finite value. A near-zone calculation additionally requires $r_{\rm relevant}/(cT) \ll 1$. A strict first-order state expansion requires the relevant interaction matrix elements to remain small after controlled common and local phases are removed.

The scalar and gravitational effective theories also require energies below their ultraviolet cutoff and $\kappa_g E_{\rm hard} \ll 1$. Small central velocities alone do not control the high-frequency content of a sharply prepared state. The number-preserving sector must therefore be understood as a low-energy field preparation, or as a matched one-particle effective theory with pair channels suppressed. When the spatial readout model of Appendix H replaces the exact dispersion by its nonrelativistic form, $t_f p_{\rm support}^4/(8m^3) \ll 1$ is a sufficient condition on the retained momentum region. The exact relativistic Gaussian channel makes no such replacement.

B. The gravitational boundary problem and its transformations

ADM (3+1) variables. We describe spacetime by a foliation into spatial slices Σ_t of constant coordinate time t. (A *Cauchy slice* is one on which initial data determine the entire future evolution.) In the Arnowitt–Deser–Misner (ADM) form [25], the line element is

$$ds^2 = -N^2 dt^2 + \gamma_{ij}\,(dx^i + N^i dt)(dx^j + N^j dt). \tag{2.17}$$

The quantities appearing here, and the related ones we need, are:

- $\gamma_{ij}(t, \mathbf{x})$, the **spatial metric**: the metric induced on the slice Σ_t, measuring distances within the slice.

- N, the **lapse**: the proper time elapsed between neighboring slices is $d\tau = N\,dt$ for an observer moving normal to the slice.
- N^i, the **shift**: how the spatial coordinates are displaced from one slice to the next, relative to the normal direction. Indices are lowered with γ_{ij}, so $N_i = \gamma_{ij}N^j$.
- n^μ, the future-pointing **unit normal** to Σ_t.
- K_{ij}, the **extrinsic curvature**: how the slice is bent inside spacetime. Equivalently, it is the rate of change of γ_{ij} along the normal, $K_{ij} = -\frac{1}{2}\mathcal{L}_n\gamma_{ij} = -\frac{1}{2N}\left(\partial_t\gamma_{ij} - D_iN_j - D_jN_i\right)$, where $\mathcal{L}_n$ is the Lie derivative along n^μ and D_i the covariant derivative of γ_{ij}. Its trace $K = \gamma^{ij}K_{ij}$ is minus the local expansion rate of the normal observers' volume element.

The canonical momentum conjugate to γ_{ij} is proportional to $K^{ij} - \gamma^{ij}K$ [25, 26]. Thus (γ_{ij}, K_{ij}) play the role of the gravitational "position and velocity" on a slice. Einstein's equations split into evolution equations and four *constraints*, which involve no time derivatives of K_{ij}:

$$\begin{aligned} \text{Hamiltonian constraint:}\quad & {}^{(3)}R + K^2 - K_{ij}K^{ij} = 16\pi G\,\varepsilon, \\ \text{momentum constraint:}\quad & D_j\left(K^{ij} - \gamma^{ij}K\right) = 8\pi G\,j^i. \end{aligned} \tag{2.18}$$

Here ${}^{(3)}R$ is the scalar curvature of γ_{ij}, and $\varepsilon = T_{\mu\nu}n^\mu n^\nu$ and $j_i = -\gamma_i{}^\mu n^\nu T_{\mu\nu}$ are the energy density and momentum density measured by the normal observer. The constraints must hold on every slice, including the initial one. They are the gravitational analogue of Gauss's law. In Dirac's terminology they are first-class constraints, which generate changes of the slicing and of the spatial coordinates [23, 26, 27].

Pedagogical treatments of this 3+1 decomposition, with the same sign conventions as here ($K_{ij} = -\frac{1}{2}\mathcal{L}_n\gamma_{ij}$, $\varepsilon = T_{\mu\nu}n^\mu n^\nu$, $j_i = -\gamma_i{}^\mu n^\nu T_{\mu\nu}$), are given in Refs. [28, 29]. See also Chap. 21 of Ref. [30], Appendix E of Ref. [31], Chaps. 3–4 of Ref. [32], and the review [33].

Boundary conditions and the asymptotic reference. A boundary condition is a condition on a space of fields: it specifies which configurations belong to the problem. Our problem acquires one from the way the experiment is specified. The initial centers and velocities of Eq. (2.2), the arm lengths, the crossing and detection times, and the outgoing modes of Eq. (2.12) are all stated in coordinates, and in general relativity coordinates acquire a meaning only relative to prescribed reference data. For an isolated system the reference is supplied by the asymptotic region, where the gravitational field of the particles dies away. The laboratory frame is the asymptotic rest frame, and the clock is the asymptotic time t. In the laboratory coordinates the metric approaches the Minkowski metric far away, with $N \to 1$, $N^i \to 0$, and $\gamma_{ij} \to \delta_{ij}$; the gauge conditions of Sec. II C fix the coordinates in the interior. These prescribed data are held fixed when the fields vary. They are not an additional dynamical field of general relativity. They are comparison data: they identify the asymptotic regions of all configurations of the problem with one reference, so that the asymptotic data, energies, and detector modes of different configurations can be compared. Specifying the experiment therefore already imposes an asymptotic boundary condition.

The precise conditions follow Regge and Teitelboim [34]. With $\mathbf{n} = \mathbf{x}/r$, the data on each slice behave as

$$\begin{aligned} \gamma_{ij} &= \delta_{ij} + \frac{\bar{h}_{ij}(\mathbf{n})}{r} + O(r^{-2}), & \pi^{ij} &= \frac{\bar{\pi}^{ij}(\mathbf{n})}{r^2} + O(r^{-3}), \\ \bar{h}_{ij}(-\mathbf{n}) &= \bar{h}_{ij}(\mathbf{n}), & \bar{\pi}^{ij}(-\mathbf{n}) &= -\bar{\pi}^{ij}(\mathbf{n}), \end{aligned} \tag{2.19}$$

with the corresponding falloff of derivatives, rapidly decaying matter fields, and $N \to 1$, $N^i \to 0$. The leading term δ_{ij}, the asymptotic values of the lapse and shift, and the parity conditions are prescribed. The coefficients $\bar{h}_{ij}$ and $\bar{\pi}^{ij}$ are solution-dependent data. They vary among configurations and carry the total energy and momentum; the scalar potential of Sec. II C, $\psi \simeq (G/r)\int d^3y\,\varepsilon$ at large r, is an example. Approaching flat space thus does not fix all asymptotic data. The powers in Eq. (2.19) admit a nonzero energy and momentum and exclude tails that would make them infinite. The parity conditions make the symplectic form and the charges of rotations and boosts finite; without them, these quantities would receive divergent contributions from large r [34, 35].

Together with the constraints (2.18) and regularity in the interior, these conditions define the space Γ of admissible initial data and solutions. A point of Γ is a complete set of gravitational and matter data on the initial slice

that belongs to the specified problem; through the evolution equations it determines a solution. Γ is the classical phase space of the experiment. Its tangent vectors are the admissible variations, which solve the linearized constraints and the linearized boundary conditions. Once the gauge redundancies identified below are removed, quantizing Γ gives the Hilbert space of Sec. II C. The prepared state is a superposition over configurations in Γ, each branch of the matter carrying the gravitational field that it sources.

Which transformations are redundancies. General relativity is diffeomorphism covariant, and the active and passive descriptions of a diffeomorphism remain equivalent. Covariance by itself, however, does not decide which transformations are allowed, or which are gauge redundancies, in a problem with prescribed asymptotic data. Three operations must be kept apart:

1. transporting the entire problem, in which the fields, the prescribed asymptotic data, the preparation, and the detector are transformed together, so that the problem is mapped onto an equivalent one;

2. transforming the fields within the problem while its prescribed data are held fixed, which is possible only if the transformation preserves those data;

3. a gauge equivalence, which identifies configurations that represent the same physical state.

The difference between the first two operations is not that between active and passive descriptions. It is the difference between transforming all the data and transforming the fields relative to prescribed data, and it arises whenever a problem contains fixed background data; the boundary does not introduce a failure of covariance. Predictions are computed with the asymptotic reference and the detector held fixed, so the second and third operations are the ones at issue.

Consider the infinitesimal diffeomorphism generated by the vector field $\kappa_g \xi^\mu(t, \mathbf{x})$; we call ξ^μ its *descriptor*. In the pullback convention, $\delta_\xi g_{\mu\nu} = \mathcal{L}_{\kappa_g \xi} g_{\mu\nu}$, the metric perturbation and the matter fields change at fixed coordinate values by

$$\delta_\xi h_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu + O(\kappa_g),$$
$$\delta_\xi \phi_\alpha = \kappa_g \, \xi^\mu \partial_\mu \phi_\alpha.$$

This local formula is the same for every descriptor. What the transformation does to the problem is decided far away, as three examples show.

1. *A descriptor with compact spatial support* changes nothing near infinity. It may depend on time and may re-slice the interior or change its spatial coordinates, while the prescribed asymptotic data and every quantity referred to them are untouched. It preserves Γ, and it is the standard example of a proper gauge transformation.

2. *A translation, rotation, boost, or asymptotic time translation* also preserves Γ. For a constant translation, $\xi^i = a^i$, the flat metric is unchanged and the perturbation is merely carried along, but the matter fields are displaced by a distance $\kappa_g |\mathbf{a}|$ relative to the laboratory axes. The overlap of each particle with its selected outgoing modes therefore changes. Rotations and boosts are Killing vectors of Minkowski space and likewise leave the flat metric unchanged. The time translation advances the system relative to the asymptotic clock: it is the evolution. These transformations act within the problem but change its state when the detector and the asymptotic reference are held fixed. Transforming the detector as well would restore the original probabilities, but that is the first operation, an expression of covariance, not a gauge redundancy. They are asymptotic symmetries.

3. *A dilation*, $\xi_i = x_i$, gives $\delta_\xi h_{ij} = 2\delta_{ij}$ at all distances. The transformed spatial metric no longer approaches δ_{ij} in the laboratory coordinates: the distant rulers are rescaled. The dilation transports the problem to one with rescaled prescribed data, but it does not preserve Γ, however small κ_g is. The transformed geometry is still asymptotically flat, since that property is diffeomorphism invariant; what fails is the prescribed condition $\gamma_{ij} \to \delta_{ij}$ in the fixed laboratory coordinates. As a passive change of coordinates of the whole problem, the dilation is harmless. A dilation confined to a bounded region belongs to the first example; Sec. II E uses one to illustrate the limits of the perturbative patch.

Preservation is not yet gauge equivalence. Let G_{allowed} consist of the transformations that preserve Γ, and let G_{proper} be its normal subgroup of gauge redundancies. The physical phase space and the group acting on it are the quotients

$$\Gamma_{\text{phys}} = \Gamma / G_{\text{proper}},$$
$$G_{\text{asym}} = G_{\text{allowed}} / G_{\text{proper}}.$$

An asymptotic symmetry is thus an equivalence class of allowed transformations, not one particular extension into the interior. Under the Regge–Teitelboim conditions, G_{asym} is the Poincaré group [34].

The infinitesimal phase-space criterion. Whether an allowed transformation is proper is decided by its generator. A transformation with normal and tangential components $(\xi^\perp, \xi^i)$ on a slice is generated canonically by

$$\mathscr{G}_\Sigma[\xi] = \int_\Sigma d^3x \, \left(\xi^\perp \mathcal{H} + \xi^i \mathcal{H}_i \right) + B_{\partial\Sigma}[\xi]. \tag{2.20}$$

Here $\mathcal{H}$ and $\mathcal{H}_i$ are the Hamiltonian and momentum constraints in canonical form (Appendix A, Eq. (A.3)); they vanish on solutions. The surface term $B_{\partial\Sigma}[\xi]$ is

evaluated at spatial infinity. It is fixed by requiring the generator to have well-defined variations for the falloffs (2.19) [34]. On solutions only the surface term survives, the gravitational counterpart of Gauss's law, so boundary contributions distinguish an asymptotic symmetry from a proper gauge transformation. A descriptor with compact support gives zero. The asymptotic time translation gives the ADM energy [25],

$$E_{\mathrm{ADM}} = \frac{1}{16\pi G} \lim_{r\to\infty} \oint_{S_r} dS^i \left(\partial_j \gamma_{ij} - \partial_i \gamma_{jj}\right),$$

which measures the $1/r$ tail of the spatial metric. In the gauge of Sec. II C the TT part does not contribute, and the scalar potential gives $E_{\mathrm{ADM}} = \int d^3x\, \varepsilon$ at linear order (Appendix A A). Translations, rotations, and boosts give the ADM momentum, the angular momentum, and the boost charge.

The precise criterion uses the presymplectic form. The action supplies a symplectic current, bilinear and antisymmetric in two variations. Integrated over Σ, with the boundary treatment appropriate to the problem, it defines a form on admissible variations, which in the convention of Eq. (A.2) is

$$\mathbf{\Omega}_\Sigma(\delta_1, \delta_2) = \int_\Sigma d^3x \left(\delta_1\pi^{ij}\, \delta_2\gamma_{ij} - \delta_2\pi^{ij}\, \delta_1\gamma_{ij}\right) + (\text{matter terms}).$$

The form is presymplectic: it can have nonzero vectors that pair to zero with every admissible variation. A proper gauge direction is such a vector [23, 36]:

$$\mathbf{\Omega}_\Sigma(\delta, \delta_\xi) = 0 \quad \text{for every admissible variation } \delta, \quad (2.21)$$

throughout Γ. Where a Hamiltonian generator exists, $\delta\mathscr{G}_\Sigma[\xi] = \mathbf{\Omega}_\Sigma(\delta, \delta_\xi)$, and the surface term in Eq. (2.20) is what makes this relation hold. Equation (2.21) then says that the generator is constant on each connected component of Γ, ordinarily normalized to zero. A charge that vanishes on one particular solution is not enough. The normalized energy vanishes on Minkowski spacetime, yet the asymptotic time translation is a nontrivial symmetry of the full phase space: it generates the evolution, and the vanishing of the bulk constraints on physical states does not remove this dynamics (Appendix A A). Likewise, the mean total momentum of the benchmark state of Sec. IV B vanishes, yet a translation still moves both particles relative to the detector modes. Equation (2.21) is a criterion for infinitesimal, connected gauge directions.

A physical observable is invariant under proper gauge transformations, and in the quantum theory so are its expectation values. Asymptotic symmetries preserve the problem but can act on its states. Observable values can therefore change when an asymptotic symmetry acts on the state while the observable and the asymptotic reference are held fixed, and no state need be invariant under them. Nor is preserving a coordinate gauge the same as being proper gauge: translations preserve the reference condition $Q_a = 0$ of Sec. II C but are asymptotic symmetries. The operations that arise in practice fall into four classes:

Why "vanishing at infinity" is only a shorthand. Smooth descriptors $\xi^\mu(t, \mathbf{x})$ with compact spatial support are proper, since their surface term vanishes. They may be nonzero at the initial and final times of the experiment. The representatives of the preparation and of the detector must then be transformed with them (Secs. II D and II E); the physical preparation and detector are unchanged. This class suffices for everything below. For descriptors without compact support, vanishing at infinity and acting trivially on the phase space are different statements: decay rates, derivatives, and surface terms matter. Decaying descriptors can also be proper, but admitting them requires checking the falloffs, the domain of the inverse operators of Appendix A C, and the surface term. The inverse operators show how the boundary condition enters the reduction itself. Solving the constraints (Sec. II C) and defining the gauge coordinates (Appendix A C) use inverses, such as ∇^{-2} and M^{-1}, that select the solution decaying at infinity, and these undo only transformations that decay. The decaying inverse of the vector operator M_{ij} of Appendix A C gives $M^{-1}M\xi = \xi$ for a compactly supported ξ_i, but zero for a constant translation $\xi_i = a_i$, for which $M\xi = 0$. The reduction therefore does not mistake a translation, which carries the ADM momentum, for a redundancy. Conversely, under the Regge–Teitelboim conditions some descriptors that do not vanish at infinity, for example odd-parity $O(1)$ ones, also have vanishing charges. Admitting them would require the same domain analysis, which the argument below does not need.

Where BMS fits. Spatial infinity, where our slices end, is a different limiting regime from null infinity. The former is reached at large distance on a fixed slice; the latter is reached along outgoing light rays, $r \to \infty$ at fixed retarded time $u = t - r$. At null infinity, Bondi, van der Burg, and Metzner [37] and Sachs [38] found that the transformations preserving asymptotic flatness form an infinite-dimensional group, the Bondi–Metzner–Sachs

Operation	Example	Status within the specified problem
Preserves Γ and satisfies Eq. (2.21)	compactly supported descriptor; re-slicing of the interior at fixed asymptotic clock	Proper gauge redundancy: predictions must not depend on it
Preserves Γ but has a nonvanishing surface generator	translation, rotation, boost, asymptotic time translation	Physical asymptotic symmetry: acts nontrivially on states
Fails to preserve Γ (changes the prescribed asymptotic data or violates the falloffs)	dilation of the laboratory coordinates	Not an allowed transformation of this phase space; it transports the problem to another one
Changes the reference that defines the measured subsystems or the preparation, or acts on the state but not on the detector	packet positions defined in a different spatial gauge (Sec. IV C)	Describes a different experiment: not a symmetry statement

(BMS) group. In a conformal completion it is the group that preserves a prescribed universal structure, a degenerate metric and a field of null generators on null infinity, defined up to linked rescalings that are constant along the generators [39]. Besides the Lorentz transformations, BMS contains the *supertranslations*, direction-dependent shifts of retarded time, $u \to u + f(\mathbf{n})$, where $\mathbf{n}$ labels the generator; the group is their semidirect product. The $\ell \leq 1$ harmonics of f are the ordinary spacetime translations. The Bondi shear, by contrast, is solution-dependent data. A supertranslation shifts it by an amount independent of retarded time even when the news vanishes, so it changes the asymptotic state without creating radiation; this change of vacuum is related to the gravitational memory effect [40]. BMS transformations thus preserve the prescribed asymptotic problem but can act nontrivially on its physical states. They are not additional redundancies to be quotiented out, and physical observables need not be invariant under them. Which asymptotic group is realized at spatial infinity depends on the boundary and parity conditions there. Under the Regge–Teitelboim conditions the supertranslation charges vanish and the group is the Poincaré group, while twisted parity conditions realize BMS at spatial infinity [34, 35]. The gauge-independence result below needs only compactly supported descriptors, which are proper in either completion, so it does not depend on this choice. It does not construct a BMS action, soft sectors, memory, or scattering data at null infinity, and a finite-time experiment specified on Cauchy slices does not need them. The TT Fock vacuum fixes the radiative preparation; it does not specify every soft boundary datum of an enlarged asymptotic phase space, and it does not make the asymptotic charges vanish. Gravitons emitted during the interval are traced out at t_f. Their later propagation toward null infinity does not change the matter state; predicting radiation, memory, or charge balance there would require additional observables.

C. Constrained Cauchy data and physical observables

Gauge and perturbative form. Three steps should be kept apart: the 3+1 decomposition of Sec. II B, the choice of a gauge section, made here, and the elimination of the constrained variables by solving the constraints, below. We use asymptotically Cartesian spatial coordinates, in which $\gamma_{ij} \to \delta_{ij}$ far away, and *maximal slicing*, $K = 0$ [41]. Maximal slicing fixes the time coordinate; it is the analogue of a Coulomb-type gauge condition for the slicing. The spatial coordinates are chosen so that the spatial-metric perturbation has only a conformally flat scalar part and a TT part (Appendix A C, gauge $Q_a = 0$). To linear order we write

$$\begin{aligned} N &= 1 + n, \\ & N_i \text{ (first order)}, \\ & \gamma_{ij} = \delta_{ij} + 2\psi\delta_{ij} + \kappa_g h^{\mathrm{TT}}_{ij}, \\ & K_{ij} = -\tfrac{1}{2}\mathcal{L}_n \gamma_{ij}, \\ & K = 0. \end{aligned} \tag{2.22}$$

- n is the lapse perturbation. In the Newtonian limit it equals the Newtonian potential Φ_N.
- ψ is the **scalar metric potential**, the conformal factor of the spatial metric. For a static source $\psi = -\Phi_N > 0$, so $\gamma_{ij} \simeq (1 - 2\Phi_N)\delta_{ij}$, the familiar weak-field spatial metric.
- h^{TT}_{ij} is transverse ($\partial_i h^{\mathrm{TT}}_{ij} = 0$) and traceless ($h^{\mathrm{TT}}_{ii} = 0$). It contains the two graviton polarizations. The factor κ_g makes h^{TT} a canonically normalized field.

Counting degrees of freedom. The spatial metric has six components. Three are removed by the choice of spatial coordinates, leaving ψ and the two TT components. The Hamiltonian constraint determines ψ. For the extrinsic curvature, any symmetric traceless tensor can be split (the York decomposition [33, 42]) into a *longitudinal* part built from a vector W_i and a TT part:

$$K_{ij} = (\mathbb{L}W)_{ij} + K_{ij}^{\mathrm{TT}},$$
$$(\mathbb{L}W)_{ij} = \partial_i W_j + \partial_j W_i - \tfrac{2}{3}\delta_{ij}\partial_l W_l.$$

The trace vanishes by maximal slicing, and the momentum constraint determines W_i. What remains free is h_{ij}^{TT} and its conjugate momentum ($\propto K_{ij}^{\mathrm{TT}}$): two polarizations, i.e. the gravitons [25]. The same conclusion appears in the gauge-invariant scalar–vector–tensor decomposition of linearized gravity, in which only the TT part obeys a wave equation and the other parts obey Poisson-type equations (Sec. 2.2 of Ref. [43]).

To linear order in G, the metric (2.22) has ${}^{(3)}R = -4\nabla^2\psi$ (the TT part does not contribute to the linearized scalar curvature), and the terms quadratic in K_{ij} are second order. Using $\partial_j(\mathbb{L}W)_{ij} = \nabla^2 W_i + \frac{1}{3}\partial_i\partial_j W_j$, the constraints (2.18) become, in position and in Fourier space ($\nabla \to i\mathbf{k}$),

$$\begin{aligned} \nabla^2\psi &= -4\pi G\varepsilon, \\ \psi(\mathbf{k}) &= \frac{4\pi G}{k^2}\varepsilon(\mathbf{k}), \\ (\nabla^2\delta_{ij} + \tfrac{1}{3}\partial_i\partial_j)W_j &= 8\pi G j_i, \\ W_i(\mathbf{k}) &= -\frac{8\pi G}{k^2}\left(\delta_{ij} - \frac{k_i k_j}{4k^2}\right) j_j(\mathbf{k}). \end{aligned} \tag{2.23}$$

The first line is Poisson's equation: ψ is minus the Newtonian potential, $\psi(\mathbf{x}) = G\int d^3y\,\varepsilon(\mathbf{y})/|\mathbf{x}-\mathbf{y}|$. The second is its gravitomagnetic analogue, sourced by the momentum density. At this order ε and j_i reduce to T^{00} and T^{0i}. The fields are required to decay at spatial infinity, which fixes the homogeneous solutions of these elliptic equations. This is the boundary condition (2.19) at work: a homogeneous solution that does not decay, such as a constant or a linear term added to ψ, would change the prescribed asymptotic data.

The sources are operators. Crucially, ε and j_i here are operators on the matter Hilbert space, not numbers. Solving Eq. (2.23) with operator-valued sources makes the constrained field ψ itself a matter operator. Acting on the branch superposition, each branch comes with its own Newtonian field ("the field follows the branch"), and matrix elements between different packet components are retained. Replacing ε by its expectation value would give a semiclassical (mean-field) model. That is a different theory, and it cannot generate entanglement.

For $k = |\mathbf{k}| > 0$ we need two projectors. $P^{\perp}$ projects vectors onto the plane transverse to $\mathbf{k}$. Π^{TT} projects symmetric tensors onto their transverse-traceless part, the two-dimensional space of graviton polarizations for that $\mathbf{k}$:

$$\begin{aligned} P^{\perp}_{ij} &= \delta_{ij} - \frac{k_i k_j}{k^2}, \\ \Pi^{\mathrm{TT}}_{ij,kl} &= \frac{1}{2}(P^{\perp}_{ik}P^{\perp}_{jl} + P^{\perp}_{il}P^{\perp}_{jk} - P^{\perp}_{ij}P^{\perp}_{kl}). \end{aligned} \tag{2.24}$$

A bare product state fails the sourced constraint. Before choosing a quantum state, consider what the constraints require of it. For one nonrelativistic particle of mass m, the leading Hamiltonian constraint smeared with a real test function f reads, in the canonical variables of Appendix A C (where P_0 is the variable that fixes the scalar curvature of the slice),

$$\begin{aligned} C[f] &= P[f]\otimes 1 - \frac{\kappa_g m}{2}\, 1\otimes f(\hat{\mathbf{x}}), \\ P[f] &= \int d^3x\, f(\mathbf{x}) P_0(\mathbf{x}). \end{aligned} \tag{2.25}$$

Suppose a bare product $|\Phi\rangle_{\mathrm{g}} \otimes |\psi\rangle_{\mathrm{m}}$ of a gravitational state and a matter state satisfied $C[f]\,|\Phi\rangle|\psi\rangle = 0$ for every f. Projecting the matter factor onto the complement of $|\psi\rangle$ removes the first term and leaves

$$\big\|(1 - |\psi\rangle\langle\psi|)\, f(\hat{\mathbf{x}})|\psi\rangle\big\|^2 = \mathrm{Var}_\psi\, f(\hat{\mathbf{x}}) = 0 \quad \text{for every } f. \tag{2.26}$$

A finite-width packet violates this. A gravitational state that matches the mean source cannot match its fluctuations: the constrained field must be an operator function of the matter position. This is an elementary obstruction in the leading model, not a complete nonlinear constraint solution. It does not exclude a product of reduced matter variables and independent TT variables, which is what we use below. Nor does it exclude a spin degree of freedom whose gravitational source is initially spin independent, or auxiliary product labels that are later projected onto physical states.

Initial state. The reduced initial state is the matter state of Eq. (2.11) times the TT vacuum:

$$\varrho_{\mathrm{red}}(0) = \rho_0 \otimes |0_{\mathrm{TT}}\rangle\langle 0_{\mathrm{TT}}|. \tag{2.27}$$

The constrained fields ψ and W_i do not appear as separate tensor factors: they are functions of the matter operators, fixed by Eq. (2.23). The independent TT oscillators have the quadratic action $\frac{1}{2}\int[(\dot{h}^{\mathrm{TT}})^2 - (\nabla h^{\mathrm{TT}})^2]$, where $(\dot{h}^{\mathrm{TT}})^2$ means $\dot{h}^{\mathrm{TT}}_{ij}\dot{h}^{\mathrm{TT}}_{ij}$. Each Fourier mode is therefore a harmonic oscillator of frequency $k = |\mathbf{k}|$. With

$\pi^{\rm TT}_{ij} = \dot{h}^{\rm TT}_{ij}$ the conjugate momentum, the vacuum covariances are the familiar $1/(2k)$ and $k/2$ per polarization:

$$\begin{aligned}\langle h^{\rm TT}_{ij}(\mathbf{k}) h^{\rm TT}_{kl}(\mathbf{k}')\rangle_0 &= (2\pi)^3\delta^3(\mathbf{k}+\mathbf{k}')\frac{\Pi^{\rm TT}_{ij,kl}}{2k},\\ \langle \pi^{\rm TT}_{ij}(\mathbf{k}) \pi^{\rm TT}_{kl}(\mathbf{k}')\rangle_0 &= (2\pi)^3\delta^3(\mathbf{k}+\mathbf{k}')\frac{k\Pi^{\rm TT}_{ij,kl}}{2}.\end{aligned} \tag{2.28}$$

Their means and symmetrized position–momentum covariance vanish. After the constraints have been solved and the coordinates fixed (reduced phase-space quantization [23]), the physical Hilbert space is simply the tensor product of the matter Fock space and the TT Fock space, with the ordinary inner product. Appendix A derives this and writes the normalized wavefunctional. There is no leftover "longitudinal graviton" Hilbert space to trace over: the non-TT parts of the gravitational field are not independent degrees of freedom.

A sudden preparation. Equation (2.27) declares the state on the initial Cauchy slice. We do not claim that it can be manufactured there instantaneously by spatially local operations; the calculation concerns its subsequent evolution. It is a sudden preparation in one respect: a uniformly moving mass carries a TT near field of order Gmv^2, which the TT vacuum does not contain. This mismatch is the origin of the finite-time TT gain and loss computed in Sec. III B. A different gravitational preparation is a different experiment (Sec. III C).

Physical observables and the declared reference. Gauge-invariant observables in gravity cannot be strictly local; they must be anchored to a reference [16, 17]. We make the anchoring explicit. Let π be the map that follows the proper gauge orbit of a configuration X on the constraint surface to its unique representative on the reference section $Q_a = 0$, i.e. maximal slicing with the spatial gauge of Appendix A C. Section II E constructs it. For any function f of the reduced variables, define its physical extension

$$\begin{aligned}&F_f(X) = f\big(\pi(X)\big),\\ &\quad \pi(gX) = \pi(X) \ \text{ for proper } g,\\ &\quad \text{hence}\\ &\quad F_f(gX) = F_f(X).\end{aligned} \tag{2.29}$$

F_f is a complete (relational) observable in the sense of Ref. [44]: it is the value of f when the configuration is read in the reference section. The selected modes of Sec. II A, the projector $P_{\rm sel}$, the one-particle number restriction, and the trace over TT modes all refer to these physical variables. The subsystem definition is thereby part of the gauge-invariant construction, rather than a choice made separately in each gauge.

The reference has a physical meaning. The detection time is the asymptotic time t_f, carried into the interior by maximal slices, and positions are those of the conformally flat spatial gauge, anchored to the asymptotic rest frame. This is a declaration of which experiment is described. Another reference section defines other observables, equally gauge invariant, whose order-G entanglement can differ. That is a different experiment, not a gauge transformation (last row of the table in Sec. II B). Three questions must therefore be kept apart: a change of computational gauge, with the preparation and the detector transported, which describes the same experiment; a change of the physical reference or of the declared preparation, which can describe a different experiment; and numerical precision and omitted higher orders, which concern the accuracy of the prediction for a given experiment.

The A and B mode algebras commute. In the reduced description they are generated by different matter fields, and the dressing map of Sec. II D carries the algebra as a whole, so they remain commuting in any other description. This is what makes "local" measurements on the two selected mode spaces well defined. Here "local" means acting on one species' selected modes; it is not a claim of compact spacetime support for a gravitationally dressed observable. Commutation is a property of the declared construction, not of every gravitational dressing [16].

D. Dressing and the two descriptions

The reduced description eliminates the constrained variables before quantization. It is useful to compare it with a description that keeps them, for two reasons: covariant calculations keep them (Sec. III C), and the comparison shows how the preparation and the detector carry the gravitational field.

In Appendix A C the scalar and longitudinal parts of the spatial metric and of its momentum are organized into four canonical pairs (Q_a, P_a), $a = 0, \dots, 3$, with $\{Q_a(\mathbf{x}), P_b(\mathbf{y})\} = \delta_{ab}\delta^3(\mathbf{x}-\mathbf{y})$. The coordinate $Q_0 = \frac{1}{2}\nabla^{-2}p^i{}_i$ is an inverse-Laplacian smearing of the trace of the extrinsic curvature, and Q_i measures the longitudinal ("pure coordinate") part of h_{ij}. The reference section is $Q_a = 0$. Keeping only the matter sources at first order, the constraints read

$$C_a = P_a - \kappa_g J_a, \qquad J_0 = \frac{1}{2}\varepsilon, \qquad J_i = j_i. \tag{2.30}$$

This is the matter-sourced part of the perturbative construction. At the same order the full constraints also contain terms quadratic in the gravitational variables; Sec. II E keeps them.

Dirac states and the first-order dressing. If Q_a is kept as a variable, physical states are wavefunctionals $\Psi[Q]$ obeying $C_a\Psi = 0$ with $P_a = -i\delta/\delta Q_a$. To first order the solution is a Q-dependent phase times the reduced state:

$$D[Q] = \int d^3x \left(\frac{1}{2}Q_0\varepsilon + Q_i j_i\right),$$
$$\Psi_{\rm phys}[Q] = \left(1 + i\kappa_g D - \frac{\kappa_g^2}{2}D^2 + i\kappa_g^2 D_2 + \cdots\right)\Psi_{\rm red}. \tag{2.31}$$

Since $\delta D/\delta Q_a = J_a$, acting with $-i\,\delta/\delta Q_a$ on $(1 + i\kappa_g D)\Psi_{\rm red}$ gives $\kappa_g J_a \Psi_{\rm red}$, which solves the constraint at this order. The phase $e^{i\kappa_g D}$ is the gravitational analogue of Dirac's dressing of a charge by its Coulomb field [15–17]. It attaches to each matter configuration the gravitational field that the constraints require.

Dressed observables. Since $[P_a, D] = -iJ_a$, the dressing $E = 1 + i\kappa_g D + \cdots$ intertwines the constraints with the gauge momenta, $E^\dagger C_a E = P_a + O(\kappa_g^2)$. A reduced observable commutes with P_a, so its dressed form commutes with the constraints at this order: $[C_a, EO^{\rm red}E^\dagger] = E[P_a, O^{\rm red}]E^\dagger + O(\kappa_g^2) = O(\kappa_g^2)$. The constraint generates the gauge transformation of the gravitational and of the matter variables together, and the dressing cancels its action on the matter; transforming only the gravitational variables would leave the matter term uncanceled.

The second-order coefficient D_2 cannot be obtained by exponentiating the first-order phase, because the sources do not commute:

$$[C_a(\mathbf{x}), C_b(\mathbf{y})] = \kappa_g^2[J_a(\mathbf{x}), J_b(\mathbf{y})] \neq 0. \tag{2.32}$$

For example, energy density and momentum density at different points do not commute. Consistency at second order therefore requires the nonlinear terms of the full ADM constraints, which restore the non-Abelian constraint algebra. Section II E and Appendix B show how to organize them.

The dictionary. Write E_t for the completed dressing map at time t, with E_i at the initial time $t = 0$ and E_f at the detection time t_f. The same experiment is represented in the two descriptions by

$$\rho_i^{\rm cov} = E_i\,\varrho_{\rm red}(0)\,E_i^\dagger,$$
$$O^{\rm cov} = E_f\,O^{\rm red}E_f^\dagger, \tag{2.33}$$
$$U_{\rm cov} = E_f\,U_{\rm red}E_i^\dagger.$$

In words: the initial state is dressed at $t = 0$, the detector observable at t_f, and the evolution maps one to the other. Here "cov" denotes the matched covariant description of the same experiment; a covariant calculation that starts from a bare product of matter and a freshly chosen gravitational vacuum does not automatically describe this state. The map E_t relates two descriptions; it is neither a spacetime diffeomorphism nor an asymptotic-symmetry operator. Preparation, evolution, and detector must be transported together; transforming the state without the detector describes a different experiment. The three maps must use the same physical pairing; they are not an arbitrary unitary on a fixed auxiliary Fock space. Constraint solutions such as Eq. (2.31) are not normalizable in an inner product that integrates over the gauge coordinates. Probabilities use the physical inner product inherited from the reduction, $\langle E\psi_1, E\psi_2\rangle_{\rm phys} = \langle\psi_1, \psi_2\rangle_{\rm red}$, which counts each gauge orbit once. The traces in these probabilities run over the complete physical system, TT modes included; the matter channel follows by tracing out the physical TT modes afterwards, not by a separately defined matter-only trace. Equation (2.33) is the operator form of the reduction proved in Sec. II E. It does not follow from assuming that the probabilities agree and cycling a trace. Where the dressing depends on time, the generators of evolution are related by

$$H_{\rm red} = E_t^\dagger H_{\rm cov} E_t - iE_t^\dagger \dot{E}_t. \tag{2.34}$$

The Newtonian interaction arises through this relation, i.e. through the constrained dynamics. Conjugation by E_t shifts P_a to $P_a + \kappa_g J_a$. The part of the gravitational energy that is quadratic in the constrained momenta then becomes the instantaneous constraint interaction given in Sec. III A and derived in Appendix A E, including its A–B cross term. This parallels electrodynamics, where conjugating the longitudinal electric energy by Dirac's dressing produces the Coulomb energy. The interaction is not an equal-time phase between the dressings of A and B: their generators D_A and D_B commute.

E. Gauge independence of the predictions

The first-order dressing shows the mechanism, but it does not by itself prove that probabilities through order $\kappa_g^2 = 32\pi G$ are independent of the gauge. We now give the proof in five steps. Steps 1–3 are a finite-time version of the gauge-independence theorem for constrained systems [19–23]. The theorem requires gauge-invariant (BRST-invariant) boundary conditions [22, 23]. The content specific to this paper is the identification of these physical boundary kernels for the experiment: the constrained Cauchy state, the physically extended detector modes, and the environmental trace. Step 4 checks the endpoint cancellation explicitly at order G, and step 5 extracts the order-G consequences.

Domain. We compare descriptions related by admitted proper transformations (Sec. II B), with compact spatial support. In the weak-field scaling

$$\begin{aligned}\delta x^\mu &= \kappa_g \xi^\mu,\\ \xi^\mu &= O(1),\\ \delta h_{\mu\nu} &= \partial_\mu \xi_\nu + \partial_\nu \xi_\mu + O(\kappa_g) = O(1),\end{aligned}$$

so that $\delta g_{\mu\nu} = O(\kappa_g)$. Each descriptor and its relevant norms are held fixed as $\kappa_g \to 0$, and finite paths of transformations must stay inside the perturbative patch. Within that patch the reference condition $Q_a = 0$ meets each nearby proper orbit once, its linear gauge-fixing operator is invertible on the admitted domain, and every section compared covers the same orbits with the same multiplicity. These conditions matter even in BRST language: a formal change of gauge-fixing fermion can change orbit coverage, and such changes are outside the theorem [45]. The patch is a genuine restriction even in flat space. A compactly supported vector field equal to x^i near the origin generates a proper flow that dilates the interior by e^s. For $s = O(1)$ it produces $h'_{ij} = (e^{2s} - 1)\delta_{ij}/\kappa_g$ there, outside the perturbative expansion, although the geometry is still flat.

Step 1: constrained physical variables. Expand the full ADM constraints, normalized so that their leading term is P_a:

$$C_a = P_a + \kappa_g c_a^{(1)}(Q, P, z) + \kappa_g^2 c_a^{(2)}(Q, P, z) + \cdots , \tag{2.35}$$

where z collects the matter and TT variables. The coefficient $c_a^{(1)}$ contains the matter sources $-J_a$ and also terms quadratic in the gravitational variables; Eq. (2.30) keeps only the former. Since $\partial C_a/\partial P_b = \delta_a{}^b + O(\kappa_g)$, the constraint surface can be solved locally for the gauge momenta, as a formal series:

$$\begin{aligned}C = 0 \iff P_a &= A_a(Q, z) = \kappa_g A_a^{(1)} + \kappa_g^2 A_a^{(2)} + \cdots ,\\ \bar{C}_a &\equiv P_a - A_a(Q, z),\\ C_a &= K_a{}^b \bar{C}_b,\end{aligned} \tag{2.36}$$

with K invertible. Appendix B A gives the recursion for A_a. The $\bar{C}_a$ have the same zero set as the C_a. Replacing one set by the other is a change of constraint basis, not a unitary conjugation. Classically, the $\bar{C}_a$ commute strongly in this local chart. Writing ∂_a for the derivative with respect to Q_a (spatial labels understood), their bracket

$$\{\bar{C}_a, \bar{C}_b\} = \partial_a A_b - \partial_b A_a + \{A_a, A_b\}_z \tag{2.37}$$

does not depend on P. First-classness makes it vanish on the constraint surface $P = A$; since it has no P dependence, it vanishes identically. The return map π of Eq. (2.29) follows the orbit generated by the $\bar{C}_a$ to $Q = 0$. On the constraint surface, the coordinates of $\pi(X)$ are the reduced variables $z_{\rm phys}$ that define the preparation and the detector. No single conjugation by a dressing can achieve the same for the original ADM constraints, whose algebra is non-Abelian (Appendix B B).

Step 2: the measure. In a canonical gauge section $\chi(Q, P, z) = 0$, the phase-space path-integral measure is [19]

$$d\mu_\chi = dQ\, dP\, dz\; \delta[C]\, \delta[\chi]\; |\det\{\chi, C\}| \,. \tag{2.38}$$

On the constraint surface,

$$\begin{aligned}\delta[C] &= \frac{\delta[\bar{C}]}{|\det K|},\\ |\det\{\chi, C\}| &= |\det K|\, \left|\det\{\chi, \bar{C}\}\right| ,\end{aligned} \tag{2.39}$$

so the change of constraint basis cancels between the two factors. The non-Abelian form of the original constraints therefore does not obstruct the reduction. Integrating over P with $\delta[\bar{C}]$ sets $P = A$. Now introduce orbit coordinates η at fixed $z_{\rm phys} = \pi(X)$. The gauge flow translates Q and acts on z by a canonical transformation, so $dQ\, dz = dz_{\rm phys}\, d\eta$ with unit Jacobian, and $\{\chi, \bar{C}\}$ is the derivative of χ along the orbit. For each physical point, provided the section meets the orbit once,

$$\int d\eta\; \delta[\chi(z_{\rm phys}, \eta)] \left|\det \frac{\partial \chi}{\partial \eta}\right| = 1. \tag{2.40}$$

The section has dropped out of the measure, and no physical data have been integrated out. For $\chi = Q$ this reduces to Eq. (A.8).

Step 3: the action and the boundary kernels. Let θ be the canonical one-form. The proper orbit directions are null directions of the presymplectic form on the constraint surface, Eq. (2.21), so locally

$$\theta\big|_{C=0} = \pi^*\theta_{\rm red} + dB, \qquad S\big|_{C=0} = S_{\rm red}[z_{\rm phys}] + B_f - B_i. \tag{2.41}$$

With the asymptotic clock fixed, the Hamiltonian on the constraint surface is the ADM surface energy, which is the same function of $z_{\rm phys}$ on every proper orbit. The endpoint functions B_f and B_i, with the accompanying changes of polarization and measure, are absorbed when the prepared state and the detector are expressed in the same physical pairing. This is the origin of the dressing maps of Eq. (2.33). For a pure endpoint phase it is the familiar cancellation of $e^{iB_f - iB_i}$ against the boundary wavefunctions. On the forward (+) and backward (−) branches of the closed time path, the endpoint transformations accompany the initial density kernel, the final detector kernel, and the final TT trace. After the orbit integrations, the probability functional is the same on every admitted section:

$$\mathcal{Z}_\chi[\rho_i, O] = \int \mathcal{D}z_+ \mathcal{D}z_-\ \mathcal{B}_{\rho_i,O}[z_+, z_-]\, e^{iS_{\rm red}[z_+] - iS_{\rm red}[z_-]} = \mathcal{Z}_{\rm red}[\rho_i, O]. \tag{2.42}$$

Here $\mathcal{B}_{\rho_i,O}$ denotes the common physical boundary kernels and environmental gluing, specified once on the reduced space. Equation (2.42) is derived from the reduction of the measure and the action; equality of propagators in different gauges is not assumed.

The quantum prescription. Equations (2.38)–(2.42) are formal perturbative expressions. They hold in the quantum theory only with one regulated measure, operator ordering, and counterterm prescription that maintains the corresponding Ward identities and is used for all representations [22, 23, 46]. In a covariant treatment this includes the determinant (ghost) sector and the necessary contact terms. The classical Jacobian argument does not fix operator ordering or composite counterterms, and these cannot be inferred from the cyclicity of the trace; an uncanceled measure or composite-operator anomaly would violate the requirement. The reduced calculation of Sec. III uses on-shell one-body mass renormalization, symmetric ordering of same-species products, and the one-particle number restriction, all defined in the reduced representation. Other descriptions must transport these choices through π, rather than make fresh coordinate-defined ones in each gauge. We do not construct an explicit regulator, every boundary composite counterterm of a covariant implementation, or the complete gravitational dressing D_2. A covariant calculation that chooses these ingredients independently still requires a matching check. For the leading negativity these omissions matter little. The local terms that these choices affect drop out of the leading entanglement block (Sec. IV A), and the remaining ingredients are controlled by the explicit argument of step 4.

The general argument applies to canonical sections $\chi(Q, P, z) = 0$. Covariant gauges such as de Donder constrain the histories of the lapse and shift and their time derivatives, so they are not of this form. Their equivalence with canonical gauges follows from the Fradkin–Vilkovisky construction, in which the multipliers become canonical variables [21, 47]. At the order needed here we use instead the history-space Gaussian argument of Appendix B F, which admits any linear section of the off-shell histories, de Donder included.

Step 4: the explicit endpoint identity. At order G the cancellation can be checked directly, for prescribed, field-independent descriptors $\xi^\mu(t, \mathbf{x})$ in the admitted domain. (Comparisons of field-dependent sections rest on steps 1–3.) Under a linearized proper transformation, $\delta_\xi h_{\mu\nu} = \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$. Let S_1 be the coefficient of the linear matter–gravity coupling. For a conserved stress matrix element ($\partial_\mu T^{\mu\nu} = 0$, which holds for the on-shell vertex of Eq. (3.3)), integration by parts without a spatial surface term gives

$$S_1 = \frac{1}{2}\int_{t_i}^{t_f} dt\, d^3x\ h_{\mu\nu}T^{\mu\nu}, \qquad \delta_\xi S_1 = b_\xi(t_f) - b_\xi(t_i), \qquad b_\xi(t) = \int d^3x\, \xi_\nu T^{0\nu}. \tag{2.43}$$

The bulk variation does not vanish; it is a pure endpoint term. To see what cancels it, express the linearized ADM momentum in metric variables,

$$p^{ij} = \dot h_{ij} - \partial_i h_{0j} - \partial_j h_{0i} - \delta_{ij}\left(\dot h_{kk} - 2\partial_k h_{0k}\right). \tag{2.44}$$

On the admitted domain, where $\nabla^{-2}\nabla^2$ and $M^{-1}M$ (Appendix A C) act as the identity,

$$\delta_\xi p^{ij} = 2(\delta_{ij}\nabla^2 - \partial_i\partial_j)\xi_0, \qquad \delta_\xi Q_0 = 2\xi_0, \qquad \delta_\xi Q_i = \xi_i, \qquad \delta_\xi D = \int d^3x\,(\xi_0\varepsilon + \xi_i j_i) = b_\xi, \tag{2.45}$$

with $\xi_0 = -\xi^0$, $J_0 = \varepsilon/2$, and $J_i = j_i$. Hence the completed linear coupling is invariant:

$$\boxed{\delta_\xi\left[S_1 - D(t_f) + D(t_i)\right] = 0.} \tag{2.46}$$

The signs are those of pulling the bare covariant evolution back to the reduced states, $E_f^\dagger U_{\rm cov} E_i$. The identity tests every admitted temporal profile ξ_0, i.e. changes of the lapse, as well as spatial profiles ξ_i. In particular, a re-slicing of the interior at fixed asymptotic clock is a compactly supported ξ_0, and the dressing of the preparation and of the detector absorbs it. Changing the clock, or the slices that define the detection event, changes the observable instead.

Equivalently, $S_1 - D(t_f) + D(t_i) = \frac{1}{2}\langle h, \mathcal{T}\rangle$ for a source $\mathcal{T}^{\mu\nu}$ that consists of the bulk stress in the window together with distributions at the two endpoints, and $\partial_\mu \mathcal{T}^{\mu\nu} = 0$ holds including those distributions. This is the finite-time replacement for conservation of the windowed bulk stress, which fails at the endpoints, Eq. (3.17).

Inserted into the free gravitational Gaussian integral, the completed vertices make the mutual A–B exchange and the first-order emission amplitudes independent of the section. This includes the equal-time contact terms generated by the momenta contained in D (Appendix B F). These are exactly the ingredients of the leading negativity (Sec. IV A).

Step 5: consequences at order G. Expanding the complete probability functional, i.e. the action, measure, preparation, and detector together,

$$\begin{aligned} p_\chi &= \sum_{n=0}^{2} \kappa_g^n p_\chi^{(n)} + O(\kappa_g^3), \\ p_\chi^{(n)} &= p_{Q=0}^{(n)} \quad (n = 0, 1, 2). \end{aligned} \tag{2.47}$$

The second-order terms must be expanded consistently: one must not truncate E at first order while keeping the bulk probability to second order. For the declared state the remainder is smaller still. The action depends on $h_{\mu\nu}$ only through $\kappa_g h_{\mu\nu}$. The TT vacuum and the completed dressing $E(\kappa_g Q)$ are invariant under $(\kappa_g, h, p) \to (-\kappa_g, -h, -p)$, and the detector observables are matter operators. Every detector probability is therefore even in κ_g:

$$p = p^{(0)} + \kappa_g^2 p^{(2)} + O(\kappa_g^4). \tag{2.48}$$

Gauge conditions and counterterms written in terms of the full metric respect the same symmetry. The argument uses a parity-symmetric gravitational state, which includes any zero-mean Gaussian TT state; a coherent TT state of fixed amplitude breaks it.

Applying Eqs. (2.47) and (2.48) to a complete set of detector matrix units makes the unnormalized selected matrix and the selection probability agree between sections through order G. The matrices are compared after the detector basis is transported back to the same reference modes; a bare coordinate matrix in an independently chosen basis need not have the same entries. Since $p_{\rm sel} = 1 + O(G)$ for the declared input, conditioning preserves the agreement. Continuity of the trace norm and of the partial transpose on the finite-dimensional mode space then gives

$$\begin{aligned} \rho_{{\rm det},\chi} - \rho_{{\rm det},Q=0} &= O(G^2), \\ \mathcal{N}_\chi - \mathcal{N}_{Q=0} &= O(G^2). \end{aligned} \tag{2.49}$$

Result. For the physical preparation and detector defined above, the order-G detector state and its leading negativity are independent of the proper-gauge section used to compute them. This includes the case in which the order-G coefficient happens to vanish. The statement identifies which coefficients are equal. It does not say that the finite-coupling remainder vanishes, and no relative bound follows from dividing by a coefficient that may be zero. The statement covers admitted descriptors with controlled norms in the local perturbative patch, with the same boundary problem and the same physical inputs. It does not identify changed preparations, changed references, or asymptotic symmetries with gauge transformations, and it says nothing about large transformations that leave the fixed-background expansion. Nor does it turn a missing endpoint term, an omitted contact term, or an inconsistent approximation into a gauge-invariant result.

III. FINITE-TIME GRAVITATIONAL DYNAMICS

This section derives the reduced dynamics of the declared experiment on the finite interval $[0, t_f]$. Subsection A gives the interaction, B the reduced channel and the selection, C its covariant counterpart and the role of the endpoint data, and D the Newtonian limit, which explains the mechanism.

A. Stress vertex, constraint interaction, and TT coupling

Stress tensor of a one-particle state. Gravity couples to the stress–energy tensor. For the free scalar field of species α it is

$$T_\alpha^{\mu\nu} = {:}\partial^\mu\phi_\alpha\partial^\nu\phi_\alpha - \tfrac{1}{2}\eta^{\mu\nu}\left(\partial_\rho\phi_\alpha\partial^\rho\phi_\alpha + m_\alpha^2\phi_\alpha^2\right){:}\ . \tag{3.1}$$

By Eq. (2.4), its number-preserving part on one-particle states is fixed by its matrix elements between momentum eigenstates. Inserting the mode expansion (2.3) and keeping only the $a^\dagger a$ terms gives

$$\langle \mathbf{p}'|T^{\mu\nu}_\alpha(t,\mathbf{x})|\mathbf{p}\rangle_\alpha = \tau^{\mu\nu}_\alpha(p',p)\, e^{i(E_{\alpha,\mathbf{p}'}-E_{\alpha,\mathbf{p}})t - i(\mathbf{p}'-\mathbf{p})\cdot\mathbf{x}}, \tag{3.2}$$

where $p^\mu = (E_{\alpha,\mathbf{p}},\mathbf{p})$ and $p'^\mu = (E_{\alpha,\mathbf{p}'},\mathbf{p}')$ are on-shell four-momenta ($p\cdot p = -m_\alpha^2$), and the coefficient is

$$\tau^{\mu\nu}_\alpha(p',p) = \frac{p'^\mu p^\nu + p'^\nu p^\mu - \eta^{\mu\nu}(p'\cdot p + m_\alpha^2)}{2\sqrt{E_{\alpha,\mathbf{p}'}E_{\alpha,\mathbf{p}}}}, \qquad (p'-p)_\mu \tau^{\mu\nu}_\alpha = 0. \tag{3.3}$$

We call $\tau^{\mu\nu}_\alpha(p',p)$ the **one-particle stress vertex**: it is the momentum-space kernel of the operator $T^{\mu\nu}_\alpha(0,\mathbf{0})$ on the one-particle sector. *Derivation:* a derivative ∂^μ acting on the annihilation part of ϕ (which contains $e^{ip\cdot x}$) gives ip^μ, and acting on the creation part (which contains $e^{-ip'\cdot x}$) gives $-ip'^\mu$. The two orderings of $\partial^\mu\phi\partial^\nu\phi$ give $p'^\mu p^\nu + p'^\nu p^\mu$. The Lagrangian term gives $-\eta^{\mu\nu}(p'\cdot p + m_\alpha^2)$. The denominator comes from the $1/\sqrt{2E}$ in the field normalization. *Checks:* for $\mathbf{p}'=\mathbf{p}$ one finds $\tau^{00} = E_{\mathbf{p}}$, $\tau^{0i} = p^i$, and $\tau^{ij} = p^i p^j/E_{\mathbf{p}}$, which are the energy, momentum, and momentum flux of one particle. The second relation in Eq. (3.3) is $\partial_\mu T^{\mu\nu} = 0$ in momentum space. It holds because both momenta are on shell.

The operators that source gravity: ε, $\mathbf{j}$, T. The gravitational constraints and couplings involve the components of $T^{\mu\nu}_\alpha$ in the 3+1 split. We give them names:

$$\varepsilon_\alpha \equiv T^{00}_\alpha \quad \text{(energy density)}, \qquad j^i_\alpha \equiv T^{0i}_\alpha \quad \text{(momentum density)}, \qquad \mathsf{T}^{ij}_\alpha \equiv T^{ij}_\alpha \quad \text{(spatial stress)}.$$

At the order we work, these coincide with the normal-frame quantities ε and j_i of Eq. (2.18). All three are *operators*. Their one-particle kernels are the corresponding components of the vertex (3.3): τ^{00}_α, τ^{0i}_α, and τ^{ij}_α. Concretely, by Eq. (3.2), on the one-particle sector

$$T^{\mu\nu}_\alpha(t,\mathbf{x}) = \int_{\mathbf{p},\mathbf{p}'} \tau^{\mu\nu}_\alpha(p',p)\, e^{i(E_{\alpha,\mathbf{p}'}-E_{\alpha,\mathbf{p}})t - i(\mathbf{p}'-\mathbf{p})\cdot\mathbf{x}}\, |\mathbf{p}'\rangle_\alpha\langle\mathbf{p}|.$$

We use the Fourier convention in which the component with wavevector $\mathbf{k}$ *raises the particle's momentum by* $\mathbf{k}$. Since $\int d^3x\, e^{i\mathbf{k}\cdot\mathbf{x}} e^{-i(\mathbf{p}'-\mathbf{p})\cdot\mathbf{x}} = (2\pi)^3\delta^3(\mathbf{p}'-\mathbf{p}-\mathbf{k})$,

$$T^{\mu\nu}_\alpha(t,\mathbf{k}) \equiv \int d^3x\, e^{i\mathbf{k}\cdot\mathbf{x}}\, T^{\mu\nu}_\alpha(t,\mathbf{x}) = \int_{\mathbf{p}} \tau^{\mu\nu}_\alpha(p+\mathbf{k},p)\, e^{i(E_{\alpha,\mathbf{p}+\mathbf{k}}-E_{\alpha,\mathbf{p}})t}\, |\mathbf{p}+\mathbf{k}\rangle_\alpha\langle\mathbf{p}|, \tag{3.4}$$

so that $\varepsilon_\alpha(t,\mathbf{k}) = T^{00}_\alpha(t,\mathbf{k})$, $j^i_\alpha(t,\mathbf{k}) = T^{0i}_\alpha(t,\mathbf{k})$, and $\mathsf{T}^{ij}_\alpha(t,\mathbf{k}) = T^{ij}_\alpha(t,\mathbf{k})$. To summarize the notation: $T^{\mu\nu}$ is the operator; $\tau^{\mu\nu}$ is its matrix element between momentum eigenstates; ε, $\mathbf{j}$, T are names for the components of $T^{\mu\nu}$, with kernels τ^{00}, τ^{0i}, τ^{ij}. Here $\tau(p+\mathbf{k},p)$ means the vertex (3.3) with outgoing momentum $\mathbf{p}+\mathbf{k}$ and incoming momentum $\mathbf{p}$, both on shell. Each term of Eq. (3.4) does three things: it moves the particle from $\mathbf{p}$ to $\mathbf{p}+\mathbf{k}$, multiplies the amplitude by the vertex, and supplies the phase $e^{i\omega t}$. Here $\omega = E_{\mathbf{p}+\mathbf{k}} - E_{\mathbf{p}}$ is the recoil energy of the particle when it absorbs momentum $\mathbf{k}$. Over a finite time interval, ω need not equal the frequency $|\mathbf{k}|$ of any graviton.

The constraint interaction: the idea. The mechanism is the gravitational version of the Coulomb interaction in Coulomb-gauge electrodynamics. There, the scalar potential A_0 enters the action without a time derivative. Its field equation is Gauss's law, $\nabla^2 A_0 = -\rho$, which fixes A_0 at each instant in terms of the charges *on the same time slice*. Substituting the solution back into the action leaves the transverse photons plus an instantaneous Coulomb energy $\frac{1}{2}\int d^3x\, d^3y\, \rho(\mathbf{x})\rho(\mathbf{y})/(4\pi|\mathbf{x}-\mathbf{y}|)$. In our gauge, gravity works the same way. The lapse pertur-

bation n, the shift N_i, and the scalar potential ψ obey elliptic (Poisson-type) equations sourced by the matter on the same slice (Eq. (2.23) and Appendix A E). They can therefore be eliminated, which leaves the TT gravitons plus an instantaneous interaction between the sources. The comparison is spelled out in the table of Appendix A B. Appendix A E carries out the elimination: it solves the constraint equations for the constrained fields, evaluates the action on the solution, and passes to the Hamiltonian. Throughout, each source α is described by its three operators $(\varepsilon_\alpha, \mathbf{j}_\alpha, \mathsf{T}_\alpha)$. For a Fourier component with wavevector $\mathbf{k}$, let $\mathbf{n} = \mathbf{k}/k$ and split each current into its longitudinal and transverse parts, $\mathbf{j}_L = \mathbf{n}(\mathbf{n}\cdot\mathbf{j})$ and $\mathbf{j}_T = \mathbf{j} - \mathbf{j}_L$.

The constraint kernel. The mutual part of the resulting interaction is $H_C^{AB} = -(\kappa_g^2/4)\int_{\mathbf{k}} \mathcal{C}_{AB}$, with the Fourier-space kernel

$$\mathcal{C}_{AB}(\mathbf{k}) = \frac{1}{k^2}\left[\frac{1}{2}\varepsilon_A\varepsilon_B - 2\mathbf{j}_{TA}\cdot\mathbf{j}_{TB} - \frac{3}{2}\mathbf{j}_{LA}\cdot\mathbf{j}_{LB} + \frac{1}{2}(\varepsilon_A \operatorname{tr}\mathsf{T}_B + \operatorname{tr}\mathsf{T}_A\varepsilon_B)\right]. \tag{3.5}$$

Here ε_A stands for $\varepsilon_A(\mathbf{k})$ and ε_B for $\varepsilon_B(-\mathbf{k})$, and likewise for $\mathbf{j}$ and T. Thus A gains momentum $\mathbf{k}$ and B gains $-\mathbf{k}$. We write $\mathcal{C}_{AB}(t;\mathbf{k},-\mathbf{k})$ when this assignment, and the time argument of the interaction-picture operators, need to be explicit. $\mathcal{C}_{AB}$ is symmetric under $A \leftrightarrow B$ (with $\mathbf{k}\to-\mathbf{k}$). For a single total source T, the bilinear $\mathcal{C}[T,T] = \mathcal{C}_{AA} + \mathcal{C}_{BB} + 2\,\mathcal{C}_{AB}$ is Eq. (3.5) with both arguments equal to T.

Reading the kernel. Each term in Eq. (3.5) is an instantaneous potential in Fourier space ($1/k^2$ is the Fourier transform of $1/(4\pi r)$). The $\varepsilon_A\varepsilon_B$ term is Newtonian attraction. The $\mathbf{j}\cdot\mathbf{j}$ terms are gravitomagnetic, velocity-dependent interactions; the transverse and longitudinal parts of the current enter with different weights, because the shift responds differently to them. The $\varepsilon\operatorname{tr}\mathsf{T}$ terms are corrections from pressure and kinetic stress, through the lapse and ψ.

The constraint Hamiltonian. In the quantum theory, ε, $\mathbf{j}$, and T are the one-particle operators of Eq. (3.4), and products of operators of the same species are symmetrically ordered. With $T = T_A + T_B$, the bilinear $\mathcal{C}[T,T]$ contains the self-interaction of each particle and the cross terms $\mathcal{C}_{AB} + \mathcal{C}_{BA} = 2\mathcal{C}_{AB}$. The mutual term therefore has twice the prefactor:

$$\begin{aligned} H_C[T] &= -\frac{\kappa_g^2}{8}\int_{\mathbf{k}} \mathcal{C}[T,T](\mathbf{k}), \\ H_C^{AB}(t) &= -\frac{\kappa_g^2}{4}\int_{\mathbf{k}} \mathcal{C}_{AB}(t;\mathbf{k},-\mathbf{k}). \end{aligned} \tag{3.6}$$

Newtonian limit. For slow particles, $|\mathbf{j}| \sim \varepsilon v$ and $\operatorname{tr}\mathsf{T} \sim \varepsilon v^2$, so the energy-density term dominates, and the momentum and spatial-stress terms are its velocity-dependent completion. Using $\kappa_g^2 = 32\pi G$ and $\int_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}/k^2 = 1/(4\pi r)$, one recovers the Newtonian interaction:

$$H_C^{AB} \longrightarrow -\frac{\kappa_g^2}{8}\int_{\mathbf{k}} \frac{\varepsilon_A(\mathbf{k})\varepsilon_B(-\mathbf{k})}{k^2} = -G\int d^3x\, d^3y\, \frac{\varepsilon_A(\mathbf{x})\varepsilon_B(\mathbf{y})}{|\mathbf{x}-\mathbf{y}|}. \tag{3.7}$$

Two checks. (i) *Post-Newtonian limit.* For two classical point particles, $\varepsilon_\alpha = m_\alpha\gamma_\alpha\delta^3(\mathbf{x}-\mathbf{x}_\alpha)$, $\mathbf{j}_\alpha = \varepsilon_\alpha\mathbf{v}_\alpha$, and $\mathsf{T}_\alpha = \varepsilon_\alpha\mathbf{v}_\alpha\mathbf{v}_\alpha$, where $\gamma_\alpha = (1-v_\alpha^2)^{-1/2}$. Transforming Eq. (3.5) back to position space with $\int_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}/k^2 = 1/(4\pi r)$ and $\int_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}k_ik_j/k^4 = (\delta_{ij} - \hat r_i\hat r_j)/(8\pi r)$ gives

$$H_C^{AB} = -\frac{Gm_Am_B\gamma_A\gamma_B}{r}\left[1 + v_A^2 + v_B^2 - \frac{7}{2}\mathbf{v}_A\cdot\mathbf{v}_B - \frac{1}{2}(\mathbf{v}_A\cdot\hat{\mathbf{r}})(\mathbf{v}_B\cdot\hat{\mathbf{r}})\right],$$

which is the integrand of the point-history phase (C.17). Expanded to order v^2, the bracket times $\gamma_A\gamma_B$ becomes $1 + \frac{3}{2}(v_A^2+v_B^2) - \frac{7}{2}\mathbf{v}_A\cdot\mathbf{v}_B - \frac{1}{2}(\mathbf{v}_A\cdot\hat{\mathbf{r}})(\mathbf{v}_B\cdot\hat{\mathbf{r}})$. This is exactly the velocity-dependent two-body term of the first

post-Newtonian (Einstein–Infeld–Hoffmann) Lagrangian [48, 49]. TT exchange is suppressed by further powers of v^2, consistent with the benchmark numbers of Sec. IV B. (ii) *Covariant limit.* The same kernel appears when the covariant one-graviton exchange amplitude is split into its TT pole part and an instantaneous remainder, Eq. (A.16). This interaction therefore comes from the same gravitational action as the radiative exchange below.

Coupling to gravitons. Expanding the matter action to first order in the metric [10], $\delta S = \frac{1}{2}\int \delta g_{\mu\nu} T^{\mu\nu}$ with $\delta g_{ij} = \kappa_g h_{ij}^{\rm TT}$, gives the TT coupling $-\frac{\kappa_g}{2}\int h_{ij}^{\rm TT} T^{ij}$ in the Hamiltonian. The total reduced interaction is

$$H_I(t) = H_C(t) - \frac{\kappa_g}{2}\int d^3x\, h_{ij}^{\rm TT}(t,\mathbf{x}) T^{ij}(t,\mathbf{x}) + H_{\rm ct}^{\rm ren}(t). \tag{3.8}$$

The counterterm Hamiltonian $H_{\rm ct}^{\rm ren}$ contains the local "seagull" and mass/field-renormalization terms of the chosen scalar effective theory. We use on-shell one-body renormalization: the stationary gravitational self-energy of each particle is absorbed into the physical dispersion $E_{\alpha,\mathbf{p}}$ already used for free propagation. Finite-time self-energy transients and the full gain/loss channel remain; their explicit momentum integrals are given in Appendix E.

B. The selected finite-time channel

What we compute. The goal is the matter state at t_f after the gravitons have been traced out,

$$\rho_{\rm matter}(t_f) = {\rm Tr}_{\rm TT}\left[U\left(\rho_0 \otimes |0_{\rm TT}\rangle\langle 0_{\rm TT}|\right)U^\dagger\right],$$

to first order in G, followed by the projection onto the selected modes (2.14). Here U is the interaction-picture evolution operator from 0 to t_f generated by H_I of Eq. (3.8). The coupling orders are: H_C and $H_{\rm ct}^{\rm ren}$ are $O(\kappa_g^2) = O(G)$, while $H_{\rm TT}$ is $O(\kappa_g)$. The Dyson series, truncated consistently at $O(\kappa_g^2)$, is therefore

$$U = 1 - i\int_0^{t_f} dt\left[H_C(t) + H_{\rm ct}^{\rm ren}(t) + H_{\rm TT}(t)\right] - \int_0^{t_f} dt \int_0^t dt'\, H_{\rm TT}(t) H_{\rm TT}(t') + O(\kappa_g^3). \tag{3.9}$$

Terms such as H_C^2 or $H_C H_{\rm TT}$ are of order κ_g^4 or κ_g^3 and are dropped. The rest of this subsection evaluates Eq. (3.9) term by term.

Graviton-number sectors. $H_{\rm TT} = -\frac{\kappa_g}{2}\int h_{ij}^{\rm TT} T^{ij}$ is linear in the graviton field. In the interaction picture the TT field is

$$\begin{aligned} h_{ij}^{\rm TT}(t,\mathbf{x}) &= \sum_{\lambda=1,2}\int_{\mathbf{k}} \frac{1}{\sqrt{2k}}\left[e_{ij}^{\lambda}(\mathbf{k})\,\beta_{\lambda\mathbf{k}}\, e^{-ikt+i\mathbf{k}\cdot\mathbf{x}} + e_{ij}^{\lambda *}(\mathbf{k})\,\beta_{\lambda\mathbf{k}}^\dagger\, e^{ikt-i\mathbf{k}\cdot\mathbf{x}}\right], \\ [\beta_{\lambda\mathbf{k}}, \beta_{\lambda'\mathbf{k}'}^\dagger] &= \delta_{\lambda\lambda'}(2\pi)^3\delta^3(\mathbf{k}-\mathbf{k}'), \end{aligned} \tag{3.10}$$

where $\beta_{\lambda\mathbf{k}}$ annihilates a graviton of polarization λ and momentum $\mathbf{k}$. The polarization tensors are normalized by $\sum_\lambda e_{ij}^\lambda e_{kl}^{\lambda *} = \Pi_{ij,kl}^{\rm TT}$, which reproduces Eq. (2.28). Inserting Eq. (3.10) puts $H_{\rm TT}$ in the form

$$H_{\rm TT}(t) = \sum_\lambda \int_{\mathbf{k}} \left[\mathsf{v}_{\lambda\mathbf{k}}(t)\,\beta_{\lambda\mathbf{k}}^\dagger + \mathsf{v}_{\lambda\mathbf{k}}^\dagger(t)\,\beta_{\lambda\mathbf{k}}\right],$$

which defines the matter operator $\mathsf{v}_{\lambda\mathbf{k}}(t)$, the coefficient of the graviton creation operator. Acting on the graviton vacuum, U then produces a zero-graviton and a one-graviton component. (There is also a two-graviton component, but its amplitude is $O(\kappa_g^2)$, so its probability is $O(G^2)$.)

$$U\,|\Psi\rangle|0_{\rm TT}\rangle = \mathcal{U}_0|\Psi\rangle\,|0_{\rm TT}\rangle + \sum_\lambda \int_{\mathbf{k}} L_{\lambda\mathbf{k}}|\Psi\rangle\,\beta^\dagger_{\lambda\mathbf{k}}|0_{\rm TT}\rangle + \cdots,$$
$$\mathcal{U}_0 \equiv \langle 0_{\rm TT}|U|0_{\rm TT}\rangle.$$

Here $\mathcal{U}_0$ and $L_{\lambda\mathbf{k}}$ are operators on the matter Hilbert space. The trace over gravitons removes all interference between different graviton numbers. Hence

$$\rho_{\rm matter}(t_f) = \mathcal{U}_0\,\rho_0\,\mathcal{U}_0^\dagger + \sum_\lambda \int_{\mathbf{k}} L_{\lambda\mathbf{k}}\,\rho_0\,L^\dagger_{\lambda\mathbf{k}} + O(G^2).$$

This is a Kraus decomposition. The first term describes histories in which no graviton is present at t_f, and the second describes histories with one final graviton $(\lambda,\mathbf{k})$. We now compute $L_{\lambda\mathbf{k}}$ and $\mathcal{U}_0$.

One-graviton sector: emission. The one-graviton component comes from the first-order term of Eq. (3.9): $L_{\lambda\mathbf{k}} = -i\int_0^{t_f} \mathsf{v}_{\lambda\mathbf{k}}(t)\,dt$. When particle α with momentum $\mathbf{p}$ emits a graviton of momentum $\mathbf{k}$, it recoils to $\mathbf{p}-\mathbf{k}$. The energy mismatch of this process, and the finite-time integral that multiplies every amplitude, are

$$\Delta_\alpha(\mathbf{p},\mathbf{k}) = E_{\alpha,\mathbf{p}-\mathbf{k}} + k - E_{\alpha,\mathbf{p}} > 0,$$
$$\mathcal{A}_{t_f}(z) = \int_0^{t_f} e^{izt}\,dt = t_f e^{izt_f/2}\,\mathrm{sinc}(zt_f/2). \tag{3.11}$$

Here $\mathrm{sinc}\,x = \sin x/x$, with its continuous value at zero. The mismatch is strictly positive because a massive particle moves slower than light, $|\nabla_{\mathbf{p}}E| < 1$: a free massive particle cannot emit an on-shell graviton while conserving energy and momentum. This excludes steady emission by inertial motion in the infinite-time limit. Over a finite interval, however, $\mathcal{A}_{t_f}(\Delta) \neq 0$, and the emission amplitude depends on the declared preparation and observation interval. Taking the coefficient of $\beta^\dagger_{\lambda\mathbf{k}}$ in $-\frac{\kappa_g}{2}\int d^3x\, h^{\rm TT}_{ij}T^{ij}$ and using Eq. (3.4) at wavevector $-\mathbf{k}$, the emission vertex and its time integral are

$$\mathsf{v}_{\lambda\mathbf{k}}(t) = -\frac{\kappa_g}{2\sqrt{2k}}\sum_{\alpha=A,B}\int_{\mathbf{p}} e^{\lambda *}_{ij}\tau^{ij}_\alpha(p-\mathbf{k},p)e^{i\Delta_\alpha t}|\mathbf{p}-\mathbf{k}\rangle\langle\mathbf{p}|_\alpha,$$
$$L_{\lambda\mathbf{k}} = -i\int_0^{t_f}\mathsf{v}_{\lambda\mathbf{k}}(t)\,dt, \qquad L = L_A\otimes I + I\otimes L_B. \tag{3.12}$$

The operator $\mathsf{v}_{\lambda\mathbf{k}}$ moves particle α from $\mathbf{p}$ to $\mathbf{p}-\mathbf{k}$ with amplitude given by the spatial stress vertex, projected on the graviton polarization. The factor $e^{i\Delta_\alpha t}$ combines the particle's recoil phase with the graviton's phase e^{ikt}. $L_{\lambda\mathbf{k}}$ is the **Kraus operator** [50] for "the matter evolves while emitting a graviton $(\lambda,\mathbf{k})$ that is still present at t_f". Because either particle can emit into the same mode, $L = L_A\otimes I + I\otimes L_B$ is a *sum of one-body operators*. Its physical meaning is fixed by expectation values: $\sum_\lambda\int_{\mathbf{k}}\mathrm{Tr}(L_{\lambda\mathbf{k}}\rho_0 L^\dagger_{\lambda\mathbf{k}})$ is the total probability that a graviton is emitted, and the squared amplitude contains local terms ($L^\dagger_A L_A$, $L^\dagger_B L_B$) and interference (cross) terms ($L^\dagger_A L_B$ and its conjugate).

Zero-graviton sector: exchange and loss. In the no-graviton amplitude $\mathcal{U}_0$, the first-order terms $-i\int(H_C + H^{\rm ren}_{\rm ct})$ contribute directly. The $H_{\rm TT}$ term does not, since $\langle 0|\beta|0\rangle = 0$. The second-order $H_{\rm TT}H_{\rm TT}$ term contributes through $\langle 0_{\rm TT}|H_{\rm TT}(t)H_{\rm TT}(t')|0_{\rm TT}\rangle = \sum_\lambda\int_{\mathbf{k}}\mathsf{v}^\dagger_{\lambda\mathbf{k}}(t)\mathsf{v}_{\lambda\mathbf{k}}(t')$, where only $\beta\beta^\dagger$ survives: a graviton is emitted at t' and reabsorbed at $t > t'$, by the same or the other particle. Hence

$$\mathcal{U}_0 = 1 - i\mathsf{K}_C - i\mathsf{K}_{\rm ren} - \mathsf{V}_{\rm TT} + O(G^2),$$

where $\mathsf{K}_C = \int_0^{t_f} H_C dt$ and $\mathsf{K}_{\rm ren} = \int_0^{t_f} H^{\rm ren}_{\rm ct}dt$ are integrated generators. (Throughout, H denotes a Hamilto-

nian and K the dimensionless time integral of one.) The remaining operator is the ordered double integral

$$\mathsf{V}_{\rm TT} = \sum_\lambda \int_{\mathbf{k}} \int_0^{t_f} dt \int_0^t dt'\, \mathsf{v}^\dagger_{\lambda\mathbf{k}}(t)\mathsf{v}_{\lambda\mathbf{k}}(t'),$$
$$\Gamma_{\rm TT} = \mathsf{V}_{\rm TT} + \mathsf{V}^\dagger_{\rm TT} = \sum_\lambda \int_{\mathbf{k}} L^\dagger_{\lambda\mathbf{k}} L_{\lambda\mathbf{k}}, \tag{3.13}$$
$$\mathsf{K}_{\rm TT} = \frac{\mathsf{V}_{\rm TT} - \mathsf{V}^\dagger_{\rm TT}}{2i}.$$

Split $\mathsf{V}_{\rm TT} = \frac{1}{2}\Gamma_{\rm TT} + i\mathsf{K}_{\rm TT}$ into Hermitian and anti-Hermitian parts, as in the second line. The two parts have different physical roles:

- $\mathsf{K}_{\rm TT}$ enters $\mathcal{U}_0$ exactly like a Hamiltonian, $\mathcal{U}_0 \supset -i\mathsf{K}_{\rm TT}$. It is the effective generator of *virtual* graviton exchange. Its cross-species part is TT exchange between A and B, and its same-species part is a finite-time self-energy.

- $\Gamma_{\rm TT}$ reduces the norm of the no-graviton amplitude: to $O(G)$, $\mathcal{U}_0^\dagger\mathcal{U}_0 = 1 - \Gamma_{\rm TT}$. The second line of Eq. (3.13) states that $\Gamma_{\rm TT} = \sum_\lambda \int_{\mathbf{k}} L^\dagger_{\lambda\mathbf{k}} L_{\lambda\mathbf{k}}$. It follows by adding the two time orderings: $\mathsf{V} + \mathsf{V}^\dagger$ is the double integral over the full square $[0, t_f]^2$, which factorizes into $(\int \mathsf{v})^\dagger(\int \mathsf{v})$. So $\mathrm{Tr}(\Gamma_{\rm TT}\rho_0)$ is the probability that a graviton is emitted, and the norm lost by the no-graviton branch equals exactly the norm gained by the one-graviton branch. This is unitarity at finite time. No Markov approximation and no infinite-time energy delta function are involved.

The total integrated generator is $\mathsf{K} = \mathsf{K}_C + \mathsf{K}_{\rm TT} + \mathsf{K}_{\rm ren}$.

The reduced density matrix. Inserting $\mathcal{U}_0 = 1 - i\mathsf{K} - \frac{1}{2}\Gamma_{\rm TT}$ into the Kraus form and keeping first order,

$$\rho_{\rm matter}(t_f) = \rho_0 - i[\mathsf{K}, \rho_0] + \sum_\lambda \int_{\mathbf{k}} \Big[L_{\lambda\mathbf{k}}\rho_0 L^\dagger_{\lambda\mathbf{k}} - \tfrac{1}{2}\{L^\dagger_{\lambda\mathbf{k}} L_{\lambda\mathbf{k}}, \rho_0\}\Big] + O(G^2).$$

The three first-order terms are: the commutator (conservative evolution: mutual constraint exchange, Hermitian TT exchange, and local self terms); the *gain* term $L\rho_0 L^\dagger$ (matter outcomes accompanied by a final graviton); and the *loss* anticommutator (the corresponding depletion of the no-graviton branch). The trace is preserved: $\mathrm{Tr}(L\rho_0 L^\dagger) = \mathrm{Tr}(L^\dagger L\rho_0)$ cancels the loss. This has the familiar Lindblad (GKSL) form [50–52], but it is the exact first-order *finite-time* channel, not a Markovian master equation: no rate, Born–Markov, or secular approximation is used. Projecting with $P_{\rm sel}$ (Eq. (2.14)) gives the selected block and, by normalizing it, the conditional state:

$$\widetilde{\rho}_{\rm sel} = P_{\rm sel}\left\{\rho_0 - i[\mathsf{K}, \rho_0] + \sum_\lambda \int_{\mathbf{k}} \left[L_{\lambda\mathbf{k}}\rho_0 L^\dagger_{\lambda\mathbf{k}} - \frac{1}{2}\{L^\dagger_{\lambda\mathbf{k}} L_{\lambda\mathbf{k}}, \rho_0\}\right]\right\} P_{\rm sel} + O(G^2),$$
$$\rho_{\rm det} = \rho_0 + \delta\widetilde{\rho}_{\rm sel} - \rho_0\,\mathrm{Tr}\,\delta\widetilde{\rho}_{\rm sel} + O(G^2). \tag{3.14}$$

Here $\delta\widetilde{\rho}_{\rm sel} = \widetilde{\rho}_{\rm sel} - \rho_0$ is the first-order part of the selected block.

What the detector keeps: selection probability. The trace of $\widetilde{\rho}_{\rm sel}$ is the probability $p_{\rm sel}$ that both particles are found in their selected modes. Using $P_{\rm sel}\rho_0 = \rho_0 P_{\rm sel} = \rho_0$, the three first-order terms contribute as follows.

- *Conservative term.* $\mathrm{Tr}(P_{\rm sel}[\mathsf{K}, \rho_0]P_{\rm sel}) = \mathrm{Tr}(\mathsf{K}\rho_0) - \mathrm{Tr}(\rho_0\mathsf{K}) = 0$. Hermitian evolution moves no prob-

ability out of the selected space at first order; its leakage probability, $\|Q_{\rm sel}\mathsf{K}|\Psi\rangle\|^2$ for an initial state $|\Psi\rangle$, is $O(G^2)$.

- *Loss term.* $-\sum_\lambda \int_{\mathbf{k}} \mathrm{Tr}(L^\dagger L\rho_0) = -\,\mathrm{Tr}(\Gamma_{\rm TT}\rho_0)$, minus the total emission probability.
- *Gain term.* $+\sum_\lambda \int_{\mathbf{k}} \mathrm{Tr}(P_{\rm sel}L\rho_0 L^\dagger P_{\rm sel})$, the probability of emitting a graviton *and* still finding both particles in the selected modes.

The difference between the last two is the probability of emitting a graviton with a recoil large enough to kick at least one particle out of its selected modes. To make this explicit, split the emission amplitude with $Q_{\rm sel} = I - P_{\rm sel}$:

$$\begin{aligned} M_{\lambda\mathbf{k}} &= P_{\rm sel}L_{\lambda\mathbf{k}}P_{\rm sel}, \\ B_{\lambda\mathbf{k}} &= P_{\rm sel}L^\dagger_{\lambda\mathbf{k}}Q_{\rm sel}L_{\lambda\mathbf{k}}P_{\rm sel} \geq 0, \\ P_{\rm sel}L^\dagger L P_{\rm sel} &= M^\dagger M + B. \end{aligned} \tag{3.15}$$

The operators have direct probabilistic meanings in the initial state:

- $M_{\lambda\mathbf{k}} = P_{\rm sel}L_{\lambda\mathbf{k}}P_{\rm sel}$ is the amplitude to emit graviton $(\lambda,\mathbf{k})$ while staying inside the selected space. $\sum_\lambda \int_{\mathbf{k}} \mathrm{Tr}(M\rho_0 M^\dagger)$ is the *selected gain*, the probability of "emission, and both particles still detected in their modes".
- $B_{\lambda\mathbf{k}} = P_{\rm sel}L^\dagger_{\lambda\mathbf{k}}Q_{\rm sel}L_{\lambda\mathbf{k}}P_{\rm sel} \geq 0$ is a probability operator. $\mathrm{Tr}(B_{\lambda\mathbf{k}}\rho_0) = \mathrm{Tr}(Q_{\rm sel}L_{\lambda\mathbf{k}}\rho_0 L^\dagger_{\lambda\mathbf{k}}Q_{\rm sel})$ is the probability of emitting $(\lambda,\mathbf{k})$ *and not finding* the pair in the selected modes.
- The identity $P_{\rm sel}L^\dagger L P_{\rm sel} = M^\dagger M + B$ splits the total emission probability into these two alternatives.

Because $\rho_0 = P_{\rm sel}\rho_0 P_{\rm sel}$, the selected dissipator becomes $M\rho_0 M^\dagger - \{M^\dagger M,\rho_0\}/2 - \{B,\rho_0\}/2$. This is a standard dissipator with jump operator M, plus an extra loss $-\{B,\rho_0\}/2$ that removes exactly the probability that leaks out. The selection probability is therefore

$$p_{\rm sel} = 1 - \sum_\lambda \int_{\mathbf{k}} \mathrm{Tr}(B_{\lambda\mathbf{k}}\rho_0) + O(G^2). \tag{3.16}$$

In the benchmark of Sec. IV B, for example, the total emission probability is $\mathrm{Tr}(\Gamma_{\rm TT}\rho_0) = 2.11203\times10^{-10}$ and the selected gain is 1.34927×10^{-10}. Their difference, 7.62766×10^{-11}, is $1-p_{\rm sel}$ (Appendix E, Eq. (E.22)).

Emission amplitudes are order $\sqrt{G}$, so their leakage probability can be order G. Conservative evolution changes amplitudes at order G and first leaks probability from an initially selected subspace at order G^2. The latter distinction does not by itself justify applying a witness to unselected counts, because interference with unselected amplitudes can already enter at order G.

For same-species contributions, the two vertices compose over the full intermediate momentum space. A projected mutual formula cannot be converted to a self term merely by equating species labels. On-shell subtraction fixes the stationary one-body dispersion, while the residual self Hamiltonian acts locally. Such local Hermitian terms cannot create first-order interspecies entanglement from the initial product, although they remain part of the selected state. The gain terms must still be kept in the entanglement test. Equations (3.12)–(3.16) specify this channel in the stated number-preserving sector; retaining additional scalar Fock sectors would enlarge its transition operators.

C. Covariant description and endpoint data

Closed-time-path formulation. The same density matrix can be computed covariantly, without first solving the constraints. In the closed-time-path (CTP, or Schwinger–Keldysh) formalism [53–55], the ket is evolved forward and the bra backward in time, and the metric perturbation is integrated out in a covariant gauge (de Donder) [10]. The graviton then enters through two kernels. The *retarded* Green function describes how the field responds causally to a source. The *symmetric* (Hadamard) function describes vacuum fluctuations and is responsible for noise and decoherence. We use this formulation as a cross-check and to clarify the role of the initial gravitational field.

At finite times there is a subtlety. The sources act only inside the observation window $\chi(t) = \Theta(t)\Theta(t_f - t)$, where Θ here denotes the Heaviside step function. A conserved stress multiplied by this window is no longer conserved at the endpoints:

$$\partial_\mu(\chi T^{\mu\nu}) = \delta(t)T^{0\nu}(0,\mathbf{x}) - \delta(t-t_f)T^{0\nu}(t_f,\mathbf{x}). \tag{3.17}$$

The windowed source appears to be switched on at $t=0$ and off at t_f. A covariant calculation that includes only the windowed source is therefore not gauge invariant by itself. The missing pieces are the gravitational field already present on the initial slice, i.e. the constrained field of our state, and the dressing of the detector at t_f. Conservation at each scalar vertex, Eq. (3.3), is fully compatible with these temporal fluxes. Under a linearized gauge change the variation of the bulk interaction is an endpoint term, and the completed vertex of Eq. (2.46) shows how the dressed state and detector cancel it.

Appendix C carries out the covariant slab calculation. It identifies the endpoint terms that the slab needs in order to reproduce the reduced channel (3.14), including the cancellation of the non-TT noise. That the dressed preparation and detector supply exactly these terms follows from the section-independence argument of Sec. II E with matched physical boundary data. The comparison is therefore a consistency condition, not an independent evaluation of the endpoint kernels. A Ward identity alone

would not fix those kernels, since a gauge-invariant boundary functional can be added without violating it.

Static example. Consider two static masses at distance R during a time t_f. If the gravitational field were produced only by the windowed sources through the retarded Green function, the influence of B would reach A only after the light-travel time R, giving the phase $\Phi_{\rm slab} = g(t_f - R)_+/R$. In our state, however, the Newtonian field is already present at $t = 0$, since it solves the initial constraint. It contributes the remaining phase Φ_∂, and the sum is the full Newtonian phase:

$$\begin{aligned} \Phi_{\rm slab} &= \frac{g(t_f - R)_+}{R}, \\ \Phi_\partial &= \frac{g\min(t_f, R)}{R}, \\ \Phi_{\rm slab} + \Phi_\partial &= \frac{g t_f}{R}, \quad x_+ = \max(x, 0). \end{aligned} \tag{3.18}$$

For static sources the initial-edge term can be checked directly. Each component of the de Donder field of mass B obeys the free wave equation, with the static profile $\Phi_0 \propto 1/r_B$ as initial data and zero initial time derivative, where $r_B = |\mathbf{y} - \mathbf{x}_B|$. By the spherical-mean solution of the wave equation and the mean-value property of $1/r_B$,

$$\begin{aligned} \Phi(t, \mathbf{x}) &= \partial_t\big(t\, M_t[\Phi_0](\mathbf{x})\big), \\ M_t\left[\frac{1}{r_B}\right](\mathbf{x}) &= \begin{cases} 1/R, & t < R, \\ 1/t, & t > R, \end{cases} \end{aligned} \tag{3.19}$$

where M_t is the mean over the sphere of radius t about $\mathbf{x}$ and $R = |\mathbf{x} - \mathbf{x}_B|$. The freely evolved initial field at the position of A therefore equals its Newtonian value for $t < R$ and vanishes afterwards, when the retarded field of the source inside the slab takes over. Its contribution to the phase of A is exactly Φ_∂ in Eq. (3.18). This checks the initial-edge term for static sources only; it does not replace the general quantum endpoint calculation.

The static field is already present on the initial slice. Its contribution changes the finite-time phase but does not change the causal response to operations performed after that slice. In general, the field is the sum of the solution determined by the initial data and the retarded response to sources inside the interval. The exact onset of the latter for point histories is analyzed in Appendix D.

A different gravitational preparation describes a different experiment. A coherent TT displacement produces a mean-field drive, potentially already at order κ_g when its amplitude is fixed. Thermal or squeezed states change the symmetric covariance; the free TT commutator remains state independent. A stationary moving dressing also differs from the finite-slice vacuum used here. For the declared state, constraint interaction and TT response are both included in (3.14), and their Newtonian limit provides the physical interpretation developed next.

D. Newtonian limit

Before evaluating the full Gaussian channel, it helps to see how the particles become entangled in the simplest limit of the constraint interaction. At low speed, the energy-density term dominates the momentum and spatial-stress terms. If the packets are also narrow compared with their separation and transitions between branches can be neglected, then on each joint branch (s_A, s_B) the interaction is just the Newtonian potential evaluated along the two center histories. Let $r_{s_A s_B}(t)$ be the distance between the center of branch s_A of A and the center of branch s_B of B. The branch Hamiltonian and the phase accumulated on that joint branch are

$$\begin{aligned} H_N^{s_A s_B}(t) &= -\frac{g}{r_{s_A s_B}(t)}, \\ \Phi_N^{s_A s_B} &= -\mathsf{K}_N^{s_A s_B} = g\int_0^T \frac{dt}{r_{s_A s_B}(t)}. \end{aligned} \tag{3.20}$$

The state on joint branch (s_A, s_B) is multiplied by $e^{-i\mathsf{K}} = e^{i\Phi}$. Since the interaction is attractive, $\mathsf{K} < 0$ and the phase Φ is positive. Here $g = Gm_A m_B$, and in this section the observation ends at the designed crossing time, $t_f = T$. The initial constraint field is already included in this limit of the reduced Hamiltonian, so Eq. (3.20) is the conservative phase for the declared Cauchy preparation. A retarded source switched on only inside the time slab has an additional onset restriction and describes a different comparison (Sec. III C and Appendix D).

Branch phases in the planar geometry. Every branch phase has the form $\Phi_N^{s_A s_B} = gT\, F(\delta_{s_A s_B})$, where $\delta_{s_A s_B} = (s_A d_A - s_B d_B)/2$ is the relative arm displacement of Eq. (2.15) and F is a geometric integral over the remaining fraction $u = 1 - t/T$ of the flight. Appendix D A gives F in closed form.

Which part of the phase entangles. Four branch phases can always be decomposed into a part independent of both labels, a part depending on one label only, and a part depending on the product of both: $\Phi_{s_A s_B} = \phi_0 + s_A\phi_A^{\rm loc} + s_B\phi_B^{\rm loc} + s_A s_B\Theta$. With the joint branches ordered as $(++, +-, -+, --)$, this is a linear map with a 4×4 Hadamard matrix, which is its own inverse up to a factor 4:

$$\begin{pmatrix}\Phi_{++}\\ \Phi_{+-}\\ \Phi_{-+}\\ \Phi_{--}\end{pmatrix} = \mathsf{H}_4 \begin{pmatrix}\phi_0\\ \phi_A^{\mathrm{loc}}\\ \phi_B^{\mathrm{loc}}\\ \Theta\end{pmatrix},$$
$$\begin{pmatrix}\phi_0\\ \phi_A^{\mathrm{loc}}\\ \phi_B^{\mathrm{loc}}\\ \Theta\end{pmatrix} = \frac{1}{4}\,\mathsf{H}_4 \begin{pmatrix}\Phi_{++}\\ \Phi_{+-}\\ \Phi_{-+}\\ \Phi_{--}\end{pmatrix}, \qquad (3.21)$$
$$\mathsf{H}_4 = \begin{pmatrix}1 & 1 & 1 & 1\\ 1 & 1 & -1 & -1\\ 1 & -1 & 1 & -1\\ 1 & -1 & -1 & 1\end{pmatrix}.$$

Each row of H_4 lists $(1, s_A, s_B, s_A s_B)$ for one joint branch. In particular, the last row of the inverse gives the entangling phase as a signed sum, $\Theta = \frac{1}{4}(\Phi_{++} - \Phi_{+-} - \Phi_{-+} + \Phi_{--})$. Because the evolution on joint branch (s_A, s_B) is multiplication by $e^{i\Phi_{s_A s_B}}$, the evolution operator is diagonal in the branch basis. Since s_α is the eigenvalue of Z_α (Eq. (2.10)), it factorizes as

$$\mathrm{diag}\big(e^{i\Phi_{++}}, e^{i\Phi_{+-}}, e^{i\Phi_{-+}}, e^{i\Phi_{--}}\big) = e^{i\phi_0}\; e^{i\phi_A^{\mathrm{loc}} Z_A}\; e^{i\phi_B^{\mathrm{loc}} Z_B}\; e^{i\Theta Z_A Z_B}.$$

This is a common phase, a local Z rotation on each particle, and the joint gate $e^{i\Theta Z_A Z_B}$. Local unitaries cannot change entanglement, so only the last factor can entangle the initial product state. The signed sum defining Θ removes every term that depends on neither branch label or on only one of them. Appendix D A gives the resulting Θ_N in closed form for the planar geometry.

After the local phases are removed, and using $Z|+_x\rangle = |-_x\rangle$ together with $e^{i\Theta Z_A Z_B} = \cos\Theta + i\sin\Theta\, Z_A Z_B$, the equal input cats become

$$|\psi_\Theta\rangle = e^{i\Theta Z_A Z_B}|+_x, +_x\rangle = \cos\Theta\,|+_x, +_x\rangle + i\sin\Theta\,|-_x, -_x\rangle. \qquad (3.22)$$

Entanglement test. To quantify the resulting entanglement we use the *partial transpose*, which transposes only particle B's indices, and the *negativity* [56]:

$$(\rho^{T_B})_{r_A r_B, s_A s_B} = \rho_{r_A s_B, s_A r_B},$$
$$\mathcal{N}(\rho) = \frac{\|\rho^{T_B}\|_1 - 1}{2}. \qquad (3.23)$$

Here $\|M\|_1 = \mathrm{Tr}\sqrt{M^\dagger M}$ is the trace norm, so $\mathcal{N}$ equals the sum of the magnitudes of the negative eigenvalues of ρ^{T_B}. For two qubits, the Peres–Horodecki criterion [57, 58] states that ρ is entangled if and only if ρ^{T_B} has a negative eigenvalue. The two product components in Eq. (3.22) are orthogonal on each particle, so its Schmidt coefficients are $|\cos\Theta|$ and $|\sin\Theta|$. Consequently

$$\text{eigenvalues of } \rho_\Theta^{T_B}: \quad \cos^2\Theta, \quad \sin^2\Theta, \quad +\frac{|\sin 2\Theta|}{2}, \quad -\frac{|\sin 2\Theta|}{2};$$
$$\mathcal{N}(\rho_\Theta) = \frac{|\sin 2\Theta|}{2}. \qquad (3.24)$$

For small Θ the leading negative eigenvalue is $-|\sin 2\Theta|/2 \simeq -|\Theta|$. A sign reversal of the phase therefore changes the joint coherence without changing the

amount of entanglement. In Section IV we apply the same criterion to a mixed selected state that also contains transitions, TT gain, and loss.

Cancellation during the incoming flight. The entangling phase is a signed combination of four positive branch integrals, so it can be small even when their common part is appreciable. For small arms it is governed by the mixed second derivative (tidal tensor) of the Newtonian potential, contracted with the two branch separations. In the planar geometry its angular factor changes sign when the longitudinal separation $2Lu$ passes through $\sqrt{2}\,b$, so early and late portions of the incoming flight contribute with opposite signs. The leading small-arm coefficient therefore vanishes at $2L/b = 2.9198\ldots$. Appendix D A derives these statements and gives the limiting forms. This cancellation is a property of the leading small-arm Newtonian phase only; Section IV C tests the full selected constraint amplitude in the same planar family.

IV. ENTANGLEMENT OF THE SELECTED STATE

Section III D used the Newtonian branch phases, which ignore the finite momentum width of the packets. In reality the interaction can connect *different* selected modes, and graviton emission can leave the matter inside or outside their span. We therefore evaluate the full channel (3.14) in the orthonormal Gaussian modes and extract the entanglement from the conditional density matrix. Only five of its matrix elements enter the leading entanglement test (Sec. IV A).

A. Matrix elements and the leading partial-transpose block

Matrix elements. Every operator in Eq. (3.14) is built from the one-particle operators (3.4). Its matrix elements between the selected mode states are therefore integrals of packet amplitudes times vertices: for a Fourier component $O = \int_{\mathbf{p}} o(\mathbf{p}+\mathbf{k},\mathbf{p})\,|\mathbf{p}+\mathbf{k}\rangle\langle\mathbf{p}|$, one has $\langle r|O|s\rangle_\alpha = \int_{\mathbf{p}} w^*_{\alpha r}(\mathbf{p}+\mathbf{k})\,o(\mathbf{p}+\mathbf{k},\mathbf{p})\,w_{\alpha s}(\mathbf{p})$. Appendix E A writes out the mutual constraint generator, the emission amplitudes, and the ordered TT exchange in this form. They retain the full stress vertex, the exact square-root energies and recoil, and transitions between the two selected modes, which a four-phase description omits. In the Newtonian limit with narrow packets, the diagonal elements of the constraint generator reduce to $\mathsf{K}_N = -\Phi_N$ of Eq. (3.20).

The leading partial-transpose block. In the local X basis, $|0\rangle = |+_x\rangle$ and $|1\rangle = |-_x\rangle$ for each particle, the input is the pure product $\rho_0 = |00\rangle\langle 00|$. Its partial transpose has eigenvalue one on $|00\rangle$ and a threefold-degenerate eigenvalue zero on the span of $|01\rangle$, $|10\rangle$, and $|11\rangle$. Entanglement at first order means that one of these zero eigenvalues becomes negative. By degenerate perturbation theory, their first-order values are the eigenvalues of a 3×3 matrix D: the first-order correction to $\rho_{\rm det}^{T_B}$, including the conditioning on selection, restricted to that null subspace. Because the input is a product and every graviton jump operator is a sum of one-body operators, D is fixed by five numbers (Appendix F A). They are the Hermitian double-flip amplitudes (constraint and TT) and their sum, the correlated loss, and the local gains:

$$\begin{aligned}
\mathcal{E}_C &= \langle 11|\mathsf{K}_C^{AB}|00\rangle,\\
\mathcal{E}_{\rm TT} &= \langle 11|\mathsf{K}_{\rm TT}^{AB}|00\rangle,\\
\mathcal{E} &= \mathcal{E}_C + \mathcal{E}_{\rm TT},\\
\gamma_{\rm cross} &= \langle 11|\Gamma_{\rm cross}|00\rangle,\\
\nu_\alpha &= \sum_\lambda \int_{\mathbf{k}} |\langle 1|L_{\alpha,\lambda\mathbf{k}}|0\rangle|^2.
\end{aligned} \tag{4.1}$$

Here $\mathsf{K}_C^{AB} = \int_0^{t_f} H_C^{AB}\,dt$ and $\mathsf{K}_{\rm TT}^{AB}$ are the A–B parts of K_C and $\mathsf{K}_{\rm TT}$, and $\Gamma_{\rm cross} = \sum_\lambda \int_{\mathbf{k}} (L_A^\dagger L_B + L_B^\dagger L_A)$ is the cross-species part of $\Gamma_{\rm TT}$. The block below is written with $\delta\rho \equiv \delta\rho_{\rm det} = \delta\widetilde{\rho}_{\rm sel} - \rho_0\,{\rm Tr}\,\delta\widetilde{\rho}_{\rm sel}$, the first-order change of the conditional state. For a diagonal, branch-preserving generator $(\mathsf{K}_C^{AB})_{s_As_B,s_As_B} = -\Phi_{s_As_B}$, the double-flip amplitude is $\mathcal{E}_C = -\Theta$, the entangling phase of Eq. (3.21) (Appendix F A).

The block. In the ordered basis $(|01\rangle, |10\rangle, |11\rangle)$, with the double-flip coherence $\langle 00|\delta\rho|11\rangle = i\mathcal{E}^* - \gamma^*_{\rm cross}/2$ placed at position $(01,10)$ by the partial transpose, the block is

$$\mathsf{D} = \begin{pmatrix} \nu_B & i\mathcal{E}^* - \gamma^*_{\rm cross}/2 & 0\\ -i\mathcal{E} - \gamma_{\rm cross}/2 & \nu_A & 0\\ 0 & 0 & 0 \end{pmatrix}. \tag{4.2}$$

No branch-preservation assumption enters this reduction. The two nonzero eigenvalues and the resulting leading negativity are

$$\begin{aligned}
\lambda_\pm &= \frac{\nu_A+\nu_B}{2} \pm \sqrt{\frac{(\nu_A-\nu_B)^2}{4} + |-i\mathcal{E} - \gamma_{\rm cross}/2|^2}, \qquad \lambda_0 = 0,\\
\mathcal{N}(\rho_{\rm det}) &= [-\lambda_-]_+ + O(G^2), \qquad [x]_+ = \max(x,0).
\end{aligned} \tag{4.3}$$

All entries of the block are first order in G. The products of entries inside the square root arise from diagonalizing that first-order matrix; they do not supply second-order density-matrix information. With only Hermitian evolution ($\nu_\alpha = \gamma_{\rm cross} = 0$), Eq. (4.3) reduces to $\lambda_- = -|\mathcal{E}|$, which matches $-|\Theta|$ of Sec. III D. Local gain ($\nu_\alpha > 0$) pushes the eigenvalues up and competes with entanglement. Appendix F B gives the complementary diagonal-noise derivation and its positivity conditions. For a complex prepared cat, the local input basis must be transformed consistently before the partial transpose is taken, as is done in Section IV C.

A compact form. With $c = -i\mathcal{E} - \gamma_{\rm cross}/2$, Eq. (4.3) gives

$$\mathcal{N}^{[2]} = \frac{1}{2}\Big[\sqrt{(\nu_A - \nu_B)^2 + 4|c|^2} - (\nu_A + \nu_B)\Big]_+, \quad (4.4)$$

where the superscript [2] denotes the result through κ_g^2. Coherent exchange alone does not decide the answer. For nonnegative gains, the selected state is entangled at leading order if and only if $|c|^2 > \nu_A\nu_B$.

Why the leading block is gauge independent. For the product input $\rho_0 = |00\rangle\langle 00|$, let $P_\perp = 1 - \rho_0^{T_B}$ project onto the initially empty subspace. Any correction that has the input on one side and local operators on the other, $X\rho_0 + \rho_0 Y$ with $X = X_A \otimes 1 + 1 \otimes X_B$ and likewise for Y, satisfies

$$P_\perp \big(X\rho_0 + \rho_0 Y\big)^{T_B} P_\perp = 0. \quad (4.5)$$

Appendix B E gives the two-line proof. Equation (4.5) covers local Hamiltonians, local losses and their conditioning correction, same-species contact terms, the independent second-order dressing D_2 (which contains at most one species; Appendix B D), gravitational seagulls, and the renormalization counterterms. Such terms change local coherences and the selection probability, but not the leading block. What remains is built from first-order amplitudes. $\mathcal{E}$ and $\gamma_{\rm cross}$ involve one vertex of each species, and ν_α is the square of a first-order emission amplitude. The endpoint identity (2.46), inserted into the free gravitational Gaussian integral with its canonical contacts, makes each of them section independent (Appendix B F). Hence the order-G block is the same in every section. With the parity of the declared state, Eq. (2.48), the difference is $\delta_\xi \mathsf{D} = O(G^2)$, and the leading negativity is gauge independent. This explicit argument needs only the declared endpoint dressing and the free Gaussian. It does not depend on how the local counterterms are chosen.

The TT and constraint parts are separately gauge invariant. In linearized gravity the metric perturbation decomposes into gauge-invariant scalar, vector, and tensor parts [43]. In particular, for every admitted descriptor,

$$\begin{aligned} \delta_\xi h_{ij}^{\rm TT} &= \Pi_{ij,kl}^{\rm TT}\,(\partial_k\xi_l + \partial_l\xi_k) = 0, \\ \mathcal{E} &= \mathcal{E}_{\rm TT} + \mathcal{E}_C, \end{aligned} \quad (4.6)$$

where the projector acts in Fourier space. The dressing D contains only the Q_a, which have no TT component. The TT part of each completed vertex is therefore the bulk coupling $\frac{1}{2}\int h_{ij}^{\rm TT} T^{ij}$ alone, and the free action has no TT–constraint cross term. The TT exchange amplitude $\mathcal{E}_{\rm TT}$ is consequently the same in every gauge, and so is the remainder $\mathcal{E}_C$, the *completed* constraint contribution. In the reduced description $\mathcal{E}_C$ is computed from H_C alone, because $D = 0$ on the reference section; in another gauge it also contains the endpoint terms that the dressing supplies. The split refers to the laboratory frame, which defines the 3+1 decomposition, and it holds at order G: beyond linear order $h^{\rm TT}$ is not gauge invariant.

B. A benchmark for the full Gaussian channel

Consider the planar input with the dimensionless parameters

$$L = 2, \quad b = 1, \quad d_A = d_B = 0.8, \quad T = t_f = 100, \quad m_A = m_B = 5000, \quad g = 10^{-4}, \quad \ell_A = \ell_B = 0.1. \quad (4.7)$$

All quantities are in units with $\hbar = c = 1$ and an arbitrary unit of length; masses are in the corresponding inverse-length units and $g = Gm_Am_B$ is dimensionless. Its nominal branch speed is about 0.0204, the nonrelativistic final width is $\sigma_x(T) = 0.141421$, and the exact raw overlap is $S = e^{-16} = 1.12535 \times 10^{-7}$. (The two terms in the exponent of Eq. (2.7) each equal 8.) The closest center separation is $r_{\rm min} = 4/\sqrt{16.64} = 0.980581$, while the conservative relative-displacement bound is approximately 4.2×10^{-4}. These scales describe a weak, well-separated example rather than a proposed laboratory parameter set.

The exact channel uses the momentum preparation $\mathbf{p}_{\alpha s} = m_\alpha \mathbf{v}_{\alpha s}$,

$$\mathbf{p}_{A,\pm} = (100, \mp 20, 0), \qquad \mathbf{p}_{B,\pm} = (-100, \mp 20, 0). \quad (4.8)$$

The momenta are not retuned when the square-root dispersion is used. For example, the exact mean velocity of the $A,+$ packet is

$(0.0199957913414, -0.00399915826828, 0)$, compared with its nominal $(0.02, -0.004, 0)$. A comparison with the prescribed point histories therefore includes this center shift as well as finite width and recoil.

Table I lists the leading partial-transpose eigenvalue and its contributions, rounded to four significant figures; Appendix E G gives full-precision values. In the additive rows, the constraint result is followed by the Hermitian TT shift and then by the conditional TT dissipative shift. All eigenvalues refer to the first-order projected spectrum, i.e. to the smallest eigenvalue of D, at the single observation time of Eq. (4.7).

The constraint interaction provides almost all of the entanglement. Its difference from the Newtonian comparison comes from the velocity-dependent constraint terms and the exact packet evolution. The TT exchange and the evaluated TT gain and loss are relative corrections of order 10^{-5} and 10^{-6}. They are meaningful as order-G coefficients; whether they could be resolved at $g = 10^{-4}$ depends on the unknown $O(G^2)$ terms. The entries of the block (4.2) are $\mathcal{E} \simeq -5.158 \times 10^{-6}$ (with imaginary part $\simeq 1.0 \times 10^{-11}$), $\nu_A \simeq 6.496 \times 10^{-12}$, $\nu_B \simeq 6.493 \times 10^{-12}$, and $\gamma_{\rm cross} \simeq -6.683 \times 10^{-12}$. Some entries of the full selected matrix are of order 10^{-4}, because local phases need not be as small as the entangling amplitude. They do not enter the leading block (Appendix E E).

The eigenvalues of D and the leading negativity are

$$\begin{aligned} &\text{eigenvalues of } \mathsf{D}: \quad -5.158 \times 10^{-6}, \quad 0, \quad +5.158 \times 10^{-6}; \\ &\mathcal{N}(\rho_{\rm det}) = 5.158 \times 10^{-6} + O(G^2). \end{aligned} \tag{4.9}$$

The numerical integration uncertainty is about 10^{-12} (Appendix E F); it is not an estimate of the omitted $O(G^2)$ terms. A matched point-history calculation, which omits finite width and recoil, gives -5.157×10^{-6} (Appendix E G). The result is for the number-preserving scalar channel and the finite-slice TT vacuum of Sec. II C, and it refers to the declared preparation and detector. By Sec. II E it does not depend on the proper-gauge section used to compute it.

C. TT-sector entanglement at a zero of the completed constraint amplitude

In the benchmark the constraint interaction generates nearly all the entanglement; TT exchange is a relative correction of order 10^{-5}. Since $\mathcal{E}_{\rm TT}$ and $\mathcal{E}_C$ are separately gauge invariant at this order (Eq. (4.6)), a sharper question can be asked: does the TT sector entangle the selected modes when the completed constraint part of the leading amplitude is tuned to zero? This question has a concrete alternative to compare against. Consider a model in which the constraint interaction is quantum but the TT field is a classical mean field, sourced by $\langle T^{\rm TT}\rangle$ and producing only local matter evolution. Such a model gives no leading-order entanglement at a zero of $\mathcal{E}_C$, whereas perturbative quantum gravity predicts a negative partial transpose there.

The cancellation condition and a suitable input. Let $|\psi_A\rangle$ and $|\psi_B\rangle$ be normalized states in the two local selected spaces, with orthogonal complements $|\psi_A^\perp\rangle$ and $|\psi_B^\perp\rangle$. For the product input $|\psi\rangle$, the matrix element of the constraint generator that flips both factors to their complements is

$$\begin{aligned} |\psi\rangle &= |\psi_A, \psi_B\rangle, \\ |\chi\rangle &= |\psi_A^\perp, \psi_B^\perp\rangle, \\ \rho_0' &= |\psi\rangle\langle\psi|, \\ \mathcal{E}_C &= \langle\chi|P_{\rm sel}\mathsf{K}_C P_{\rm sel}|\psi\rangle. \end{aligned} \tag{4.10}$$

This is the constraint part of the double-flip amplitude $\mathcal{E}$ of Eq. (4.1), now for a general product input. For constraint evolution alone, the block (4.2) has $\nu_\alpha = \gamma_{\rm cross} = 0$, so the three initially zero partial-transpose eigenvalues acquire the first-order values $-|\mathcal{E}_C|, 0, +|\mathcal{E}_C|$. Local Hamiltonians cannot contribute to this matrix element, because it would require an unchanged local overlap such as $\langle\psi_B^\perp|\psi_B\rangle$, which is zero. The full mutual constraint matrix, Eq. (E.5) of Appendix E A, therefore contains all the constraint terms relevant to this test, including velocity-dependent stress, finite width, recoil, and transitions between the selected modes. The goal is to make $\mathcal{E}_C = 0$.

The condition $\mathcal{E}_C = 0$ is generally complex and imposes two real equations. A zero of the Newtonian signed branch phase does not suffice. For example, with the original real even cats and $T = 100$, tuning L near 1.4299482 removes the real part while leaving an imaginary part of about 2.214×10^{-11}. This finite-width transition contribution is comparable to the TT effect we seek to resolve.

A symmetry that makes the condition real. For equal masses, widths, and arm lengths in the planar geometry, the setup is invariant under spatial inversion $\mathbf{x} \to -\mathbf{x}$ followed by exchange of the two species. Inversion maps branch s of A onto branch $-s$ of B (compare the table in Sec. II A), and the exchange maps it back to A. This operation preserves the free evolution, the constraint interaction, and the TT-vacuum channel. On the selected two-qubit space it acts as a branch flip on

TABLE I. Benchmark at leading order in G.

Calculation or contribution	Value
Newtonian finite-width comparison	-5.460×10^{-6}
Full exact-recoil constraint interaction	-5.158×10^{-6}
Hermitian TT shift	$+7.401 \times 10^{-11}$
Conditional TT dissipative shift	$+6.494 \times 10^{-12}$
Full conditional eigenvalue λ_-	-5.158×10^{-6}
Selection probability $p_{\rm sel}$	$1 - 7.628 \times 10^{-11}$

each particle ($X \otimes X$) followed by a SWAP. In the basis $(++, +-, -+, --)$,

$$\begin{gathered} U_{\rm IX} = \mathrm{SWAP}(X \otimes X), \qquad U_{\rm IX}^\dagger = U_{\rm IX}, \qquad U_{\rm IX}^2 = I, \\ U_{\rm IX}|++\rangle = |--\rangle, \qquad U_{\rm IX}|+-\rangle = |+-\rangle, \qquad U_{\rm IX}|-+\rangle = |-+\rangle, \\ [U_{\rm IX}, P_{\rm sel}\mathsf{K}_C P_{\rm sel}] = 0. \end{gathered} \tag{4.11}$$

Now choose the prepared relative phases in the orthonormal packet basis so that

$$\begin{aligned} |\psi_A\rangle &= \frac{c_{A+}^\dagger + e^{i\theta_A^{\rm in}} c_{A-}^\dagger}{\sqrt{2}}|0_A\rangle, \\ |\psi_A^\perp\rangle &= \frac{c_{A+}^\dagger - e^{i\theta_A^{\rm in}} c_{A-}^\dagger}{\sqrt{2}}|0_A\rangle, \\ |\psi_B\rangle &= \frac{c_{B+}^\dagger + e^{i\theta_B^{\rm in}} c_{B-}^\dagger}{\sqrt{2}}|0_B\rangle, \\ |\psi_B^\perp\rangle &= \frac{c_{B+}^\dagger - e^{i\theta_B^{\rm in}} c_{B-}^\dagger}{\sqrt{2}}|0_B\rangle, \\ \theta_A^{\rm in} + \theta_B^{\rm in} &= \pi \pmod{2\pi}. \end{aligned} \tag{4.12}$$

The symmetry then gives $U_{\rm IX}|\psi\rangle = -|\chi\rangle$. To check this, expand $|\psi\rangle = \frac{1}{2}\big(|++\rangle + e^{i\theta_B^{\rm in}}|+-\rangle + e^{i\theta_A^{\rm in}}|-+\rangle + e^{i(\theta_A^{\rm in}+\theta_B^{\rm in})}|--\rangle\big)$. Applying Eq. (4.11) swaps the coefficients of $|++\rangle$ and $|--\rangle$, and with $e^{i(\theta_A^{\rm in}+\theta_B^{\rm in})} = -1$ the result is $\frac{1}{2}\big(-|++\rangle + e^{i\theta_B^{\rm in}}|+-\rangle + e^{i\theta_A^{\rm in}}|-+\rangle + |--\rangle\big) = -|\chi\rangle$. Since $U_{\rm IX}$ is Hermitian and commutes with the Hermitian operator $P_{\rm sel}\mathsf{K}_C P_{\rm sel}$, their product is Hermitian, and therefore

$$\mathcal{E}_C = -\langle\psi|U_{\rm IX}(P_{\rm sel}\mathsf{K}_C P_{\rm sel})|\psi\rangle \in \mathbb{R}. \tag{4.13}$$

This argument uses the full selected operator and survives finite-width transitions. It also makes the Hermitian TT double-flip amplitude and the cross-loss matrix element real. It removes one of the two real conditions from the root search without forcing either entangler to vanish.

Both examples below use $\theta_A^{\rm in} = \theta_B^{\rm in} = \pi/2$. These are changes to the *prepared* state, made before any gravitational evolution. In terms of the overlapping raw modes, the same local state has, up to an overall phase, relative phase $\varphi_\alpha = \arccos(-S_\alpha)$ and normalization $\sqrt{2(1-S_\alpha^2)}$. Assigning a raw relative phase $\pi/2$ would describe a different state. The alternative $\theta_A^{\rm in} = 0$, $\theta_B^{\rm in} = \pi$ is also allowed by the symmetry, but it requires its own root calculation.

Simple and stationary zeros.

Keep $b = 1$, $d_A = d_B = 0.8$, $m_A = m_B = 5000$, $\ell_A = \ell_B = 0.1$, and $g = 10^{-4}$. At each geometry, choose the central momenta according to the free-flight prescription $\mathbf{p}_{\alpha s} = m_\alpha \mathbf{v}_{\alpha s}(L, T)$ of Eq. (2.2), and evaluate the channel up to $t_f = T$ with the exact square-root dispersion. We look for two kinds of zeros of the (now real) function $\mathcal{E}_C(L, T)$. A *simple* zero is found by tuning L at fixed $T = 100$. A *stationary* zero additionally has $\partial_L \mathcal{E}_C = 0$, so that a small error in L enters only quadratically:

$$\begin{aligned} &\mathcal{E}_C(L_*, 100) = 0, \\ &\mathcal{E}_C(L_\dagger, T_\dagger) = 0, \qquad \partial_L \mathcal{E}_C(L, T)|_{(L_\dagger, T_\dagger)} = 0. \end{aligned} \tag{4.14}$$

The derivative in the second line changes positions and momenta together according to the prescribed geometry. An initial-position error at fixed momentum is a different variation.

At a constraint zero, the general entangling block of Section IV A reduces to a TT block. For each particle use the local labels $|0\rangle_A = |\psi_A\rangle$, $|1\rangle_A = |\psi_A^\perp\rangle$, and similarly for B, and order the initially empty subspace as $(|01\rangle, |10\rangle, |11\rangle)$. With $\mathcal{E}_{\rm TT} = \langle\chi|\mathsf{K}_{\rm TT}|\psi\rangle$, $\gamma_{\rm cross} = \langle\chi|\Gamma_{\rm cross}|\psi\rangle$, and ν_α the integrated TT gain into the orthogonal local mode,

$$\mathsf{D}_{\mathrm{TT}} = \begin{pmatrix} \nu_B & -\gamma_{\mathrm{cross}}/2 + i\mathcal{E}_{\mathrm{TT}} & 0 \\ -\gamma_{\mathrm{cross}}/2 - i\mathcal{E}_{\mathrm{TT}} & \nu_A & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \lambda_-^{\mathrm{TT}} = \frac{\nu_A + \nu_B}{2} - \sqrt{\frac{(\nu_A - \nu_B)^2}{4} + \mathcal{E}_{\mathrm{TT}}^2 + \frac{\gamma_{\mathrm{cross}}^2}{4}}. \tag{4.15}$$

The displayed real form follows from the symmetry. The numerical assembly retains complex sampling residuals and uses the general complex formula (4.3). For the prepared complex cats, the density matrix must first be expressed in this local input basis before its partial transpose is formed.

The table uses the changed input of Eq. (4.12), exact Gaussian constraint and TT kernels, unrestricted loss, and selection normalization. The quoted uncertainties concern numerical integration. Appendix G gives the stored root representatives, convergence, and parameter sensitivity.

The negative eigenvalues come from Hermitian exchange through the TT modes. At both roots, the dissipative contribution alone has a nonnegative leading block. At the stationary root, setting $\mathcal{E}_{\mathrm{TT}} = 0$ gives the two nonzero eigenvalues $(2.3683, 4.9379) \times 10^{-9}$, both positive. Vacuum gain and loss therefore *reduce* the entanglement generated by exchange in these examples. At the stationary root their effect is too large to justify expanding Eq. (4.15) in the noise-to-exchange ratio.

The surviving exchange couples to the packets' kinetic stress through the physical TT modes. It belongs to the Hermitian part of the finite-time, ordered, no-final-graviton amplitude, i.e. to K_{TT} in Eq. (3.13). The accompanying gain and loss are finite-slice preparation effects, consistent with the strictly positive emission mismatch for an inertial massive particle. They are not a steady radiation rate.

Neither root is a zero of the Newtonian contribution. At both roots the finite-width Newtonian entangling phase is canceled by the remaining constraint terms, and the point-Newtonian roots differ from the full selected root (Appendix G B).

The simple root requires L to be tuned to about one part in 10^7. At the stationary root a coordinated change of positions and momenta enters only quadratically, but an isolated position error at fixed momentum does not. Appendix G C gives the tolerances; variations of T, b, the arms, and the prepared phases must be controlled as well.

A symmetry of the whole setup can also cancel the constraint entangler, but it may remove the TT entangler with it. For crossed splitting directions it does (Appendix G D), so that construction cannot replace the tuned cancellation; other symmetric preparations are not excluded.

For the roots in Eq. (4.14), the established statement is

$$\begin{aligned} &\mathcal{E}_C = 0, \\ &\quad \lambda_-^{\mathrm{TT}} < 0, \\ &\quad \lambda_{\min}(\rho_{\mathrm{det}}^{T_B}) = \lambda_-^{\mathrm{TT}} + O(G^2) \quad \text{for the specified input and selected modes.} \end{aligned} \tag{4.16}$$

Because $\mathcal{E}_C$ and $\mathcal{E}_{\mathrm{TT}}$ are separately gauge invariant at this order, Eq. (4.16) is a gauge-invariant statement about the declared preparation and detector. It is not a statement about a coordinate split of the interaction.

What the result does and does not show. At the roots, the leading entanglement is produced by the TT sector. This distinguishes the prediction from the mean-field model described above, in which the leading-order entangling amplitude vanishes at the roots. It does not establish that every conceivable classical-TT model predicts a separable state, and the absence of leading-order entanglement in that model is not an all-orders separability statement. Five further qualifications apply.

1. **It is not a detection of outgoing radiation.** For inertial point sources, the TT response of Eq. (D.21) consists of two parts. One depends only on the instantaneous separation: it is the TT near field of a uniformly moving mass. The other is an initial-slice transient that acts only until the light front from the initial slice arrives, i.e. for $t \lesssim 3$ of the 100 time units at the simple zero. This point-source structure does not by itself show that all finite-propagation effects in the full packet calculation are preparation transients. Emitted gravitons enter through the gains and losses, which reduce the entanglement; the Hermitian TT exchange is the no-final-graviton part of the amplitude. To the extent that the exchange is the bound near field, instantaneous operator-valued potentials with the corresponding velocity-dependent terms could mimic it.

TABLE II. Leading-order data at the two zeros, rounded to four significant figures (full values in Appendix G B).

Quantity	Simple zero	Stationary zero
L	1.42994819774	2.3975949
$T = t_f$	100	22.5036458
Absolute tolerance on $\lvert\mathcal{E}_C\rvert$	10^{-15}	10^{-13}
$\mathcal{E}_{\mathrm{TT}}$	$+4.461 \times 10^{-11}$	$+7.918 \times 10^{-9}$
ν_A	2.781×10^{-12}	3.653×10^{-9}
ν_B	2.781×10^{-12}	3.653×10^{-9}
γ_{cross}	-3.522×10^{-12}	-2.570×10^{-9}
Conditional λ_-	-4.186×10^{-11}	-4.368×10^{-9}
$1 - p_{\mathrm{sel}}$ through order G	2.237×10^{-11}	5.537×10^{-8}
Numerical uncertainty in λ_-	10^{-13}	5×10^{-11}

2. **The zero is sensitive to the preparation.** The roots are defined for the declared preparation. An order-G change in how the product preparation is defined, for example positions specified in a different spatial gauge, adds a one-time entangling phase at the preparation. A rough scaling estimate puts that phase at order $g\,v$, whereas TT exchange accumulates only as about $g\,v^4T/r$, so the zero condition is parametrically sensitive to the definition. This is a sensitivity to the physical preparation, not an uncertainty in the prediction for the declared one. Quantifying it would require a packet-level calculation.

3. **Higher orders are not controlled.** The $O(G^2)$ constraint contribution and corrections from gravitational deflection have not been bounded below the surviving TT signal at $g = 10^{-4}$, and small common and local phases alone do not establish such a bound. Lowering g at fixed dimensionless packet geometry scales every first-order eigenvalue and loss by $g/10^{-4}$, while higher orders carry further powers. For example, the stationary-root leading eigenvalue becomes approximately -4.37×10^{-13} at $g = 10^{-8}$. The clean regime is small g.

4. **The noise depends on the preparation.** At the stationary root the TT gains are comparable to $\mathcal{E}_{\mathrm{TT}}$ and reduce the entanglement. They depend on the sudden TT-vacuum preparation of Sec. II C.

5. **The cancellation is input-specific.** $\mathcal{E}_C = 0$ removes the constraint contribution to one entangling matrix element for one input. The selected constraint operator does not vanish: at the simple root its nonlocal part, after the identity and one-body terms are subtracted, has norm about 8×10^{-11}, larger than $\mathcal{E}_{\mathrm{TT}}$. Another input or entangling direction would generally receive a constraint contribution, and correlations outside the selected modes are not constrained.

V. OPERATIONAL MEANING

The entanglement computed above belongs to physical mode subsystems: the selected outgoing modes of A and of B, defined through the reference section (Sec. II C). This section shows that it can be certified, in principle, by local measurements on those modes. It is not an experimental proposal. The ability to analyze the selected modes is an explicit measurement assumption, not an engineering demonstration.

Local mode observables. On the selected space of particle α, the Pauli operators of Eq. (2.10) are the mode matrix units $\mathcal{O}^{\alpha}_{rs}$ of Eq. (2.13) arranged into Hermitian combinations. Measuring Z_α asks which outgoing mode the particle occupies. Measuring X_α or Y_α asks for the relative phase of the two modes, i.e. an interferometric mode analysis. Here "local" means acting on one species' selected modes. The two algebras commute (Sec. II C), so joint statistics of A and B observables are well defined.

A witness for the calculated state. Let $|v_-\rangle$ be an eigenvector of $\rho_{\mathrm{det}}^{T_B}$ with negative eigenvalue λ_-, and define

$$W = \big(|v_-\rangle\langle v_-|\big)^{T_B}, \qquad \mathrm{Tr}(W\rho_{\mathrm{det}}) = \langle v_-|\rho_{\mathrm{det}}^{T_B}|v_-\rangle = \lambda_- < 0. \tag{5.1}$$

For every separable state ρ_{sep}, $\rho_{\mathrm{sep}}^{T_B} \geq 0$, hence $\mathrm{Tr}(W\rho_{\mathrm{sep}}) \geq 0$. A negative expectation value of W therefore certifies entanglement. At leading order $|v_-\rangle$ lies in the null subspace of Sec. IV A. In the input-adapted basis $|00\rangle = |\psi_A, \psi_B\rangle$ of Appendix F, the eigenvector of the 2×2 block of Eq. (4.2) is $|v_-\rangle = (|01\rangle - e^{i\varphi}|10\rangle)/\sqrt{2}$ for $\nu_A = \nu_B$, with $\varphi = \arg c$ and $c = -i\mathcal{E} - \gamma_{\mathrm{cross}}/2$. Then

$$W = \frac{1}{2}\Big(|01\rangle\langle 01| + |10\rangle\langle 10| - e^{-i\varphi}|00\rangle\langle 11| - e^{i\varphi}|11\rangle\langle 00|\Big), \tag{5.2}$$

which is a combination of products of local operators. Using the first-order eigenvector costs nothing at the working accuracy: for any unit vector $|v\rangle$ in the null subspace, $\langle v|\rho_{\rm det}^{T_B}|v\rangle$ differs from the corresponding entry of D only at $O(G^2)$.

For the real even cats the input basis is $|0\rangle = |+_x\rangle$, $|1\rangle = |-_x\rangle$ for each particle, and the benchmark has $\varphi \simeq \pi/2$. Rewriting the matrix units of Eq. (5.2) with the branch-basis Pauli operators of Eq. (2.10) then gives

$$W \simeq \frac{1}{4}\Big[1 - X_A X_B + \big(Y_A Z_B + Z_A Y_B\big)\Big], \qquad \langle W\rangle = \lambda_- \simeq -5.16\times 10^{-6}. \tag{5.3}$$

For the ideal phase state of Eq. (3.22) the same operator gives $\langle W\rangle = -\frac{1}{2}\sin 2\Theta \simeq -\Theta$. Its value requires the correlations $\langle X_A X_B\rangle$, $\langle Y_A Z_B\rangle$, and $\langle Z_A Y_B\rangle$, which local mode analysis can obtain in principle on repeated preparations. One practical point matters at second order. The predicted one-body phases are much larger than the entangling amplitude (Sec. IV B). They do not affect the leading block, but they would enter the witness value at second order if the local analysis axes were not rotated to absorb them. The witness should therefore be applied in local bases rotated by the predicted one-body phases. The magnitude in Eq. (5.3) shows that the certificate is in principle; statistical feasibility is outside the scope of this paper.

Selection and conditioning. The witness is evaluated on $\rho_{\rm det}$, the state conditioned on finding both particles in their selected modes, which happens with probability $p_{\rm sel} \simeq 1 - 7.6\times 10^{-11}$ in the benchmark. Conditioning is implemented by product local filters $P_A \otimes P_B$, and product local filters cannot create entanglement from a separable state. A negative value of W on the selected state therefore certifies entanglement of the matter modes. The certificate assumes that the specified filters and mode observables are implemented. A position click is not automatically such a filter: other motional modes, and their interference with the selected amplitudes, can contribute to the same position distribution.

Why overlap fringes alone do not certify. In free flight, the natural readout is the interference pattern at the crossing time, when the two branches of each particle overlap in space. This readout is not sufficient. At exact overlap the two local envelopes have equal magnitude, so position statistics carry no information about the branch populations. They are sensitive only to products of 1, X, and Y on each particle. The ideal phase state then has a separable mimic with the same joint position distribution (Appendix H). This is why the witness above uses mode observables. Detection away from overlap restores the population information, but certification from positions then requires calibrated local mode support or an independent bound on contamination from other modes.

VI. DISCUSSION AND CONCLUSIONS

We have calculated the gravitationally induced entanglement of two freely propagating quantum particles as a constrained Cauchy problem in the effective quantum field theory of general relativity. The constraints are solved with operator-valued matter sources, so the constrained part of the gravitational field follows each branch of the matter state. The TT modes are the only independent gravitational degrees of freedom; they start in their vacuum and are traced out. A bare product of a gravitational state and a finite-width matter packet would fail the sourced constraint, so the physical preparation is a dressed state, not such a product.

The predictions refer to physical mode observables, defined by returning each configuration to a reference section anchored to the asymptotic frame and clock. That frame and clock are the prescribed asymptotic data of the problem. They define its solution space and, with it, the distinction between proper gauge transformations, which act trivially on physical states, and asymptotic symmetries, which preserve the problem but act on its states. For this preparation and detector they are independent of the proper gauge used to compute them. Structurally, the Faddeev measure and the action reduce to the same probability functional on every admitted section once the finite-time boundary kernels are expressed in a common physical pairing. Explicitly, the completed linear coupling $S_1 - D(t_f) + D(t_i)$ is invariant under changes of the lapse and of the shift, and inserted into the free Gaussian it fixes the mutual exchange and emission amplitudes on which the leading negativity depends. For the declared, parity-symmetric gravitational state, the detector state and the leading negativity agree between sections up to $O(G^2)$. The result does not cover changes of the physical preparation or of the reference that defines the subsystems: those are different experiments. Nor does it cover asymptotic symmetries, which act on states rather than being redundancies.

The finite-time channel contains the instantaneous constraint exchange, the virtual TT exchange, and TT gain and loss, followed by conditioning on the selected outgoing modes. Its leading negativity depends only on the double-flip amplitude, the cross loss, and the local gains. In the benchmark the selected state is entangled, with smallest partial-transpose eigenvalue $\simeq -5.16\times 10^{-6}$ at leading order, almost entirely from the constraint interaction. A witness built from local mode observables certifies this state in principle, provided the selected modes can be analyzed. Position statistics at exact overlap cannot, because the ideal phase state has a separable mimic there.

At order G the TT part and the completed constraint part of the entangling amplitude are separately gauge invariant. For tuned preparations the constraint part vanishes and the TT sector still produces a negative partial transpose. This distinguishes the prediction from a model with quantum constraint interactions and a classical TT mean field. It is not, however, a detection of outgoing

radiation: for inertial point sources the TT response consists of the bound near field of the moving masses and an initial-slice transient. The zero is specific to the declared input and selected modes and is sensitive to how the preparation is defined; the $O(G^2)$ terms have not been bounded below the TT signal at the benchmark coupling.

The scope of the calculation is perturbative and effective. It retains the number-preserving one-particle scalar sector, on-shell one-body renormalization, the finite-slice TT vacuum, and terms through first order in G. It uses compactly supported proper descriptors and Regge–Teitelboim asymptotics; it does not construct BMS charges, soft sectors, or scattering data. Two extensions would strengthen it. An independent covariant evaluation of the endpoint kernels would turn the consistency condition of Appendix C into a second calculation. A packet-level calculation of how the leading entanglement depends on the definition of the prepared product state would quantify the sensitivity identified in Sec. IV C.

Appendix A: Canonical formulation of gravity, constraints, and the constrained initial state

This appendix is a concise review of the Hamiltonian (ADM) formulation of general relativity and of how its constraints are handled, specialized to the linearized setting of the main text. It is written for a reader who knows general relativity from a standard course but has not used the canonical formalism. Sections A and B review standard material; fuller treatments are in Refs. [23, 25–28, 30–32]. Sections C–E apply that material to justify three statements used in Secs. II and III:

1. The scalar and longitudinal parts of the gravitational field are fixed by the matter; they are not independent degrees of freedom and do not form an environment.
2. The physical Hilbert space is the tensor product of the matter and TT Fock spaces with the ordinary inner product, so the initial state (2.27) is a legitimate physical state.
3. Eliminating the constrained fields produces the instantaneous constraint Hamiltonian H_C of Eq. (3.6).

A. The ADM Hamiltonian formulation

Take the 3+1 split of Sec. II B: lapse N, shift N^i, spatial metric γ_{ij}, and extrinsic curvature $K_{ij} = -\frac{1}{2N}(\dot\gamma_{ij} - D_iN_j - D_jN_i)$. With $\sqrt{-g} = N\sqrt{\gamma}$ and the Gauss–Codazzi relations, which express the four-dimensional curvature through the intrinsic curvature ${}^{(3)}R$ of the slice and its extrinsic curvature, the Einstein–Hilbert action becomes [25, 28, 32]

$$\begin{aligned} S_{\rm EH} &= \frac{1}{16\pi G}\int d^4x\,\sqrt{-g}\,{}^{(4)}R = \int dt\, L_{\rm ADM},\\ L_{\rm ADM} &= \frac{1}{16\pi G}\int d^3x\, N\sqrt{\gamma}\left({}^{(3)}R + K_{ij}K^{ij} - K^2\right) + (\text{surface terms}). \end{aligned} \tag{A.1}$$

Only γ_{ij} appears with a time derivative, through K_{ij}. The lapse and shift appear without time derivatives. The momentum conjugate to γ_{ij}, and the canonical bracket, are

$$\pi^{ij} = \frac{\partial L_{\rm ADM}}{\partial\dot\gamma_{ij}} = \frac{\sqrt{\gamma}}{16\pi G}\left(K\gamma^{ij} - K^{ij}\right), \qquad \{\gamma_{ij}(\mathbf{x}), \pi^{kl}(\mathbf{y})\} = \delta^k_{(i}\delta^l_{j)}\,\delta^3(\mathbf{x}-\mathbf{y}). \tag{A.2}$$

A Legendre transform gives the Hamiltonian, including the matter contributions:

$$\begin{aligned} H &= \int d^3x\left(N\mathcal{H} + N^i\mathcal{H}_i\right) + H_\partial,\\ \mathcal{H} &= \frac{16\pi G}{\sqrt{\gamma}}\left(\pi_{ij}\pi^{ij} - \tfrac{1}{2}\pi^2\right) - \frac{\sqrt{\gamma}}{16\pi G}\,{}^{(3)}R + \sqrt{\gamma}\,\varepsilon, \qquad \mathcal{H}_i = -2\gamma_{ij}D_k\pi^{jk} - \sqrt{\gamma}\,j_i. \end{aligned} \tag{A.3}$$

Here $\pi = \gamma_{ij}\pi^{ij}$, and H_∂ is a surface term at spatial infinity which, for asymptotically flat spacetimes, is the ADM energy. Three features of Eq. (A.3) matter for us.

- **Lapse and shift are Lagrange multipliers.** Varying N and N^i gives $\mathcal{H} = 0$ and $\mathcal{H}_i = 0$. Using Eq. (A.2), the gravitational part of $\mathcal{H}$ equals $\frac{\sqrt{\gamma}}{16\pi G}(K_{ij}K^{ij} - K^2 - {}^{(3)}R)$, and the gravitational part of $\mathcal{H}_i$ equals $\frac{\sqrt{\gamma}}{8\pi G}D_j(K^j{}_i - \delta^j{}_iK)$. So these two conditions are exactly the Hamiltonian and momentum constraints (2.18).
- **The Hamiltonian is a sum of constraints plus a surface term.** On solutions, the bulk part of H vanishes, and the energy of an isolated system is carried entirely by H_∂. Physical time evolution is therefore evolution relative to the asymptotic region. This is why the detector modes are anchored to the asymptotic reference (Secs. II B and II C). At linear order, integrating the Hamiltonian constraint over the slice gives the leading, matter-sourced relation $H_\partial = \int d^3x\,\varepsilon$: the constraint relates the surface energy to the matter energy and does not set it to zero. The quadratic terms of the constraint add the gravitational binding energy at the next order. Once the constraints are solved and the gauge is fixed, H_∂ becomes the reduced Hamiltonian, which then contains H_C and the TT coupling; no separate surface term is added to it.

- **Choosing N and N^i is choosing coordinates.** The lapse and shift determine how the slices and the spatial coordinates advance in time. They are fixed by imposing coordinate (gauge) conditions, here maximal slicing and the spatial gauge of Sec. C below.

B. Constraints and gauge: Dirac's classification and two routes to quantization

First-class constraints. The constraints $\mathcal{H} \approx 0$ and $\mathcal{H}_i \approx 0$ are *first class* in Dirac's terminology [23, 26, 27]: their Poisson brackets with one another vanish whenever the constraints hold. (They close into the algebra of deformations of a hypersurface.) First-class constraints play two roles. They restrict the allowed initial data. And, used as generators, they produce gauge transformations. For example, $\int d^3x\, \xi^i \mathcal{H}_i$ generates the change of (γ_{ij}, π^{ij}) and of the matter fields under an infinitesimal spatial diffeomorphism, and $\int d^3x\, \xi\, \mathcal{H}$ moves the slice along its normal (on solutions). Phase-space points related by proper gauge transformations of this kind, for example with ξ of compact support, describe the same physical state. A transformation that preserves the boundary conditions but has a nonvanishing boundary contribution to its generator, Eq. (2.20), such as a translation, is an asymptotic symmetry instead (Sec. II B).

Counting. At each point, (γ_{ij}, π^{ij}) comprise 12 phase-space variables. Each first-class constraint removes two: one through the constraint itself and one through the gauge condition that fixes the transformation it generates. This leaves $12 - 2 \times 4 = 4$ phase-space variables, i.e. two configuration degrees of freedom: the two graviton polarizations.

Two routes to quantization.

- *Dirac quantization.* Keep all variables and impose the constraints as operator conditions on states, $\hat{C}_a|\Psi\rangle = 0$.
- *Reduced phase-space quantization.* Choose gauge conditions $\chi_a = 0$ for which the matrix $\{\chi_a, C_b\}$ is invertible. The combined set $\Phi_\alpha = (\chi_a, C_a)$ is then *second class*. Solve all of them classically and quantize the remaining variables, using the Dirac bracket

$$\{A, B\}_D = \{A, B\} - \{A, \Phi_\alpha\}\,(\mathcal{M}^{-1})^{\alpha\beta}\,\{\Phi_\beta, B\}, \qquad \mathcal{M}_{\alpha\beta} = \{\Phi_\alpha, \Phi_\beta\}. \tag{A.4}$$

The Dirac bracket is the bracket induced on the surface where all Φ_α vanish. The main text uses the second route, carried out in Secs. C–E below; Sec. II D of the main text relates it to the first.

The electrodynamics analogy. Every step has a familiar counterpart in Coulomb-gauge electrodynamics:

Electrodynamics, Coulomb gauge	Linearized gravity, this paper
A_0: no time derivative, Lagrange multiplier	lapse n and shift N_i
Gauss's law $\nabla \cdot \mathbf{E} = \rho$	Hamiltonian and momentum constraints, Eq. (2.18)
gauge condition $\nabla \cdot \mathbf{A} = 0$	$Q_a = 0$: maximal slicing plus spatial gauge (Sec. C)
longitudinal $\mathbf{E}_L = -\nabla\nabla^{-2}\rho$, fixed by the charges	ψ and W_i (longitudinal K_{ij}), fixed by ε and j_i
two transverse photon polarizations	two TT graviton polarizations
instantaneous Coulomb energy $\frac{1}{2}\int \rho\,\rho/(4\pi r)$	instantaneous constraint Hamiltonian H_C, Eq. (3.6)
Dirac's dressed charged states [15]	dressed matter states, Sec. II D

There are three differences. Like charges attract in gravity, which accounts for the overall minus sign of H_C. The sources include momentum and stress, not only a charge density. And the constraint algebra is non-Abelian: the source operators do not commute with one another. The last point matters only at order κ_g^2 (Sec. II D and Appendix B).

C. Linearized canonical variables and the gauge $Q_a = 0$

We now expand about flat space. Write $\gamma_{ij} = \delta_{ij} + \kappa_g h_{ij}$, where h_{ij} now denotes the *full* perturbation (not only its TT part), and $p^{ij} = \kappa_g \pi^{ij}$, with π^{ij} the ADM momentum of Eq. (A.2). Because (γ_{ij}, π^{ij}) are canonical, so are (h_{ij}, p^{ij}). The six components of h_{ij} and of p^{ij} are reorganized into four "scalar and vector" canonical pairs (Q_a, P_a), $a = 0, 1, 2, 3$, plus the two TT pairs. All variables below are fields on the same spatial slice, with spatial indices raised and lowered using δ_{ij}. In position space, define

$$
\begin{aligned}
P_0 &= \partial_i\partial_j h_{ij} - \nabla^2 h^i{}_i, \qquad Q_0 = \frac{1}{2}\nabla^{-2} p^i{}_i, \\
F_i &= \partial_j h_{ij} - \frac{1}{3}\partial_i h^j{}_j, \qquad M_{ij} = \nabla^2\delta_{ij} + \frac{1}{3}\partial_i\partial_j, \\
Q_i &= (M^{-1})_{ij} F_j, \qquad\qquad P_i = -2\partial_j p^{ij}.
\end{aligned}
\tag{A.5}
$$

Here ∇^{-2} is the inverse spatial Laplacian with the boundary condition that its solution vanishes at spatial infinity:

$$
(\nabla^{-2} f)(\mathbf{x}) = -\int d^3y\, \frac{f(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|}.
$$

The inverse of M_{ij} uses the same boundary condition. Thus Q_0 and Q_i remain nonlocal functions of the canonical fields, as required to separate the constrained and TT variables.

What these variables are.

- $P_0 = \partial_i\partial_j h_{ij} - \nabla^2 h^i{}_i$ determines the linearized scalar curvature of the slice: ${}^{(3)}R = \kappa_g P_0$.
- $Q_0 = \frac{1}{2}\nabla^{-2} p^i{}_i$ is its conjugate coordinate. With the convention of Eq. (A.2), $p^i{}_i = 4K/\kappa_g$ at linear order, so $Q_0 = (2/\kappa_g)\nabla^{-2}K$. It is an inverse-Laplacian smearing of the trace of the extrinsic curvature; setting $Q_0 = 0$ imposes $K = 0$.
- F_i measures the part of h_{ij} that a change of spatial coordinates $x^i \to x^i + \xi^i$ would produce, namely $\delta h_{ij} \propto \partial_i\xi_j + \partial_j\xi_i$. Inverting M_{ij} extracts the generating vector Q_i.
- $P_i = -2\partial_j p^{ij}$ is the combination that appears in the linearized momentum constraint.

Brackets. From $\{h_{ij}(\mathbf{x}), p^{kl}(\mathbf{y})\} = \delta^k_{(i}\delta^l_{j)}\delta^3(\mathbf{x}-\mathbf{y})$, contraction with the momentum trace gives

$$
\{P_0(\mathbf{x}), Q_0(\mathbf{y})\} = -\nabla^{-2}_{\mathbf{y}}\nabla^2_{\mathbf{x}}\delta^3(\mathbf{x}-\mathbf{y}) = -\delta^3(\mathbf{x}-\mathbf{y}).
$$

Similarly, $\{F_j(\mathbf{x}), P_l(\mathbf{y})\} = M^{(\mathbf{x})}_{jl}\delta^3(\mathbf{x}-\mathbf{y})$. Hence $\{Q_a(\mathbf{x}), P_b(\mathbf{y})\} = \delta_{ab}\delta^3(\mathbf{x}-\mathbf{y})$ for $a, b = 0, 1, 2, 3$. The brackets among the Q variables and among the P variables vanish, and all of them commute with the TT pairs.

Linearized constraints and gauge conditions. At linear order the K^2 terms of the Hamiltonian constraint are negligible, so it reads ${}^{(3)}R = 16\pi G\varepsilon$, i.e. $\kappa_g P_0 = 16\pi G\varepsilon$. With $16\pi G = \kappa_g^2/2$ this is $P_0 = \frac{\kappa_g}{2}\varepsilon$. The momentum constraint $\mathcal{H}_i = 0$ linearizes to $-2\partial_j\pi^{ij} = j_i$, i.e. $P_i = \kappa_g j_i$. For the gauge conditions we choose $Q_a = 0$:

- $Q_0 = 0$ means $p^i{}_i = 0$, i.e. $K = 0$: **maximal slicing**.
- $Q_i = 0$ means $\partial_j h_{ij} = \frac{1}{3}\partial_i h_{kk}$: the metric perturbation has **no longitudinal ("pure coordinate") part**.

The constraints $C_a = 0$ and the reconstruction of the metric perturbation then read

$$
\begin{aligned}
C_0 = P_0 - \frac{\kappa_g}{2}\varepsilon = 0, \qquad C_i = P_i - \kappa_g j_i = 0, \\
h_{ij} = h^{\mathrm{TT}}_{ij} - \frac{1}{2}\delta_{ij}\nabla^{-2}P_0 + \partial_i Q_j + \partial_j Q_i.
\end{aligned}
\tag{A.6}
$$

The second line is the general decomposition of a symmetric tensor into TT, trace, and longitudinal pieces ($2+1+3=6$ components). In the gauge $Q_a = 0$ only the first two survive. In particular, $\psi = -\frac{\kappa_g}{4}\nabla^{-2}P_0 = -\frac{\kappa_g^2}{8}\nabla^{-2}\varepsilon$, so $\nabla^2\psi = -4\pi G\varepsilon$, in agreement with the Hamiltonian constraint of Section II C ($\kappa_g^2/8 = 4\pi G$).

Why matter keeps its ordinary brackets. The gauge conditions and constraints $\Phi = (Q_a, C_a)$ form a second-class set. Since the sources J_a ($J_0 = \varepsilon/2$, $J_i = j_i$) are matter operators, $\{Q_a, C_b\} = \{Q_a, P_b\} = \delta_{ab}$. In the ordering (Q, C), the matrix of Eq. (A.4) and its inverse are

$$
\mathcal{M} = \begin{pmatrix} 0 & I \\ -I & X \end{pmatrix}, \qquad \mathcal{M}^{-1} = \begin{pmatrix} X & -I \\ I & 0 \end{pmatrix}, \qquad X = \{C, C\} = \kappa_g^2\{J, J\}.
$$

For matter or TT variables A and B, $\{A, Q_a\} = \{Q_a, B\} = 0$. In the correction term of Eq. (A.4), $\{A, \Phi_\alpha\}$ can therefore be nonzero only for $\Phi_\alpha = C$, and the row of $\mathcal{M}^{-1}$ belonging to C is nonzero only in the Q column, where $\{Q, B\} = 0$. The correction vanishes, and **matter and TT variables keep their ordinary Poisson brackets**. The reduced phase space is simply matter $\oplus$ TT.

The same conclusion can be read off from the canonical one-form, $\Theta = \int p\,\delta q$, whose exterior derivative is the symplectic form:

$$\begin{aligned}\int p^{ij}\delta h_{ij} &= \int p^{ij}_{\rm TT}\delta h^{\rm TT}_{ij} + P_i\delta Q_i - Q_0\delta P_0 \\ &= \int p^{ij}_{\rm TT}\delta h^{\rm TT}_{ij} + P_a\delta Q_a - \delta(P_0Q_0), \\ \Theta_{\rm red} &= \sum_{\alpha=A,B}\int \pi_\alpha\delta\phi_\alpha + \int p^{ij}_{\rm TT}\delta h^{\rm TT}_{ij}.\end{aligned} \tag{A.7}$$

The total variation $-\delta(P_0Q_0)$ in the second line does not affect the symplectic form. On the surface $Q_a = 0$, $C_a = 0$, the terms $P_a\delta Q_a$ vanish, leaving the matter variables $(\phi_\alpha, \pi_\alpha)$ and the TT variables as independent canonical coordinates. The phase-space path-integral measure reduces in the same way. This is the standard gauge-fixed path integral for a constrained system, in which delta functions impose the constraints and gauge conditions and a determinant (the Faddeev–Popov factor) makes the result independent of the gauge choice [23]:

$$\mathcal{D}z\,\mathcal{D}Q\,\mathcal{D}P\,\delta[Q]\delta[C(Q,P,z)]\,|\det\{Q_a, C_b\}| = \mathcal{D}z. \tag{A.8}$$

Here z denotes all matter and TT phase-space variables. The constraint delta functional integrates out the momenta conjugate to the fixed gauge coordinates, and its Jacobian cancels the determinant. Equation (A.8) is a *reduction*, not a trace over longitudinal oscillators: there is nothing left to trace over.

D. The reduced state and operator-valued constrained fields

The reduced wavefunctional. Quantizing the reduced phase space gives the tensor product of the matter Fock space and the TT Fock space. Let $h_\lambda(\mathbf{k})$ be real TT configuration coordinates, one per polarization λ, normalized so that every mode is an oscillator of frequency k. The normalized reduced wavefunctional, i.e. the state of Eq. (2.27) in the mixed momentum/field representation, is

$$\Psi_{\rm red}(\mathbf{p}_A, \mathbf{p}_B, h_\lambda; 0) = f_A(\mathbf{p}_A) f_B(\mathbf{p}_B)\,\mathcal{N}\exp\left[-\frac{1}{2}\sum_{\lambda=1}^{2}\int_{\mathbf{k}} k\,h_\lambda(\mathbf{k})h_\lambda(-\mathbf{k})\right]. \tag{A.9}$$

The Gaussian in h_λ is the harmonic-oscillator ground state of each TT mode, with normalization constant $\mathcal{N}$ (not to be confused with the negativity). The factor $f_A(\mathbf{p}_A)f_B(\mathbf{p}_B)$ is the momentum coefficient of the two-species Fock state restricted to the $(N_A, N_B) = (1, 1)$ sector. Section IV C replaces only this matter coefficient by that of the phase-shifted product state in Eq. (4.12); the TT vacuum Gaussian is unchanged. The inner product is the ordinary integral over the two matter momenta and the TT configurations.

The constrained field is an operator. Nothing in Eq. (A.9) refers to ψ or W_i. They are recovered as operators built from matter, by solving the constraints as operator equations. For example, using Eq. (3.4),

$$\widehat{\psi}(\mathbf{k}) = \frac{\kappa_g^2}{8k^2}\widehat{\varepsilon}(\mathbf{k}), \qquad \widehat{\varepsilon}(\mathbf{k}) = \sum_\alpha\int_{\mathbf{p}} \tau^{00}_\alpha(p+\mathbf{k}, p)|\mathbf{p}+\mathbf{k}\rangle\langle\mathbf{p}|_\alpha. \tag{A.10}$$

Acting on Eq. (A.9), this operator retains transitions between packet components. Replacing it with its expectation value would discard those transitions and define a semiclassical model different from the one evaluated here.

The relation between this reduced state and the Dirac description, in which the gauge coordinates Q_a are kept, is the dressing of Eq. (2.31); the dictionary that transports preparation, evolution, and detector together is Eq. (2.33).

E. Eliminating the constrained fields: the constraint Hamiltonian

This section derives the constraint Hamiltonian H_C quoted in Sec. III A.

Quadratic action. Insert the gauge-fixed metric (2.22) into the ADM Lagrangian (A.1) and keep terms of second order. The linearized extrinsic curvature is $K_{ij} = -\dot\psi\,\delta_{ij} + \frac{1}{2}(\partial_i N_j + \partial_j N_i) - \frac{\kappa_g}{2}\dot h_{ij}^{\rm TT}$. In the kinetic combination $K_{ij}K^{ij} - K^2$, the longitudinal part of the shift drops out after integration by parts. The terms involving $h^{\rm TT}$ decouple from the scalar–vector sector at this order, because they are transverse and traceless. For a conformally flat metric, $N\sqrt{\gamma}\,{}^{(3)}\!R$ contributes $2(\nabla\psi)^2 - 4n\nabla^2\psi$ at second order. Splitting the shift into transverse and longitudinal parts, $N_i = N_i^T + N_i^L$, the scalar–vector part of the gravitational Lagrangian is

$$L_{g,\rm SV} = \frac{1}{16\pi G}\int d^3x\left[-6\dot\psi^2 + 2(\nabla\psi)^2 - 4n\nabla^2\psi + 4\dot\psi\,\partial_i N_i + \frac{1}{2}(\partial_i N_j^T)^2\right]. \tag{A.11}$$

Matter coupling. To first order in the metric perturbation, the matter action changes by $\frac{1}{2}\int \delta g_{\mu\nu}T^{\mu\nu}$. With $\delta g_{00} = -2n$, $\delta g_{0i} = N_i$, and $\delta g_{ij} = 2\psi\delta_{ij}$, this is $\int d^3x(-n\varepsilon + N_i j_i + \psi\,\mathrm{tr}\,\mathsf{T})$.

Field equations. The lapse and shift act as Lagrange multipliers. Varying n gives the Hamiltonian constraint. Varying the transverse and longitudinal shift gives the two parts of the momentum constraint. The linearized maximal-slicing condition is $K = -3\dot\psi + \partial_i N_i = 0$. Varying ψ, and using that condition to cancel the $\ddot\psi$ terms, gives an elliptic equation for n. Altogether:

$$\begin{aligned}
\nabla^2\psi &= -4\pi G\varepsilon, & N_i^T(\mathbf{k}) &= -\frac{16\pi G}{k^2}j_i^T(\mathbf{k}),\\
\partial_i\dot\psi &= 4\pi G j_i^L, & \partial_i N_i &= 3\dot\psi,\\
N_i^L(\mathbf{k}) &= -\frac{12\pi G}{k^2}j_i^L(\mathbf{k}), & n(\mathbf{k}) &= -\frac{4\pi G}{k^2}[\varepsilon(\mathbf{k}) + \mathrm{tr}\,\mathsf{T}(\mathbf{k})].
\end{aligned} \tag{A.12}$$

Although ψ appears with a kinetic term $\dot\psi^2$ in Eq. (A.11), it is not an independent degree of freedom. Its value is fixed by the Hamiltonian constraint, and its time derivative by the longitudinal momentum constraint. In canonical language, $\psi = -\frac{\kappa_g}{4}\nabla^{-2}P_0$ is determined by $C_0 = 0$, while the coordinate Q_0 conjugate to P_0 is fixed by the gauge condition. All four fields in Eq. (A.12) are thus fixed instantaneously by the matter.

The field of each source. In the matter coupling above, ε, $\mathbf{j}$, and T are the *total* (A plus B) operators. The field equations (A.12) contain no propagation, and because they are linear, each field is a sum of the fields produced by the two sources separately, $\psi = \psi_A + \psi_B$ and so on, with

$$\psi_\alpha = \frac{4\pi G}{k^2}\,\varepsilon_\alpha, \qquad n_\alpha = -\frac{4\pi G}{k^2}\,(\varepsilon_\alpha + \mathrm{tr}\,\mathsf{T}_\alpha), \qquad \mathbf{N}_\alpha^T = -\frac{16\pi G}{k^2}\,\mathbf{j}_{T\alpha}, \qquad \mathbf{N}_\alpha^L = -\frac{12\pi G}{k^2}\,\mathbf{j}_{L\alpha}. \tag{A.13}$$

Each field at time t is determined by the sources at the same time t. The first is the Newtonian potential of source α (with $\psi = -\Phi_N$). The lapse is sourced by $\varepsilon + \mathrm{tr}\,\mathsf{T}$, i.e. by the "active gravitational mass density" $\rho + 3p$ familiar from cosmology. The shift is the gravitomagnetic potential sourced by the momentum density.

Evaluating the action on the solution. The action is quadratic in the constrained fields $\phi = (\psi, n, \mathbf{N})$ and linear in the sources. For any Lagrangian of the form $L = \frac{1}{2}\phi\cdot\mathcal{K}\phi + \phi\cdot J$, the solution $\phi_* = -\mathcal{K}^{-1}J$ gives $\frac{1}{2}\phi_*\cdot\mathcal{K}\phi_* = -\frac{1}{2}\phi_*\cdot J$. The on-shell Lagrangian is therefore *half* the source coupling evaluated on the solution, $L_* = \frac{1}{2}\phi_*\cdot J$. (Here this holds up to total time derivatives, which are the endpoint terms treated in Sec. II E and Appendix C.) Writing the source coupling as a sum over which source produces the field (β) and which source feels it (α),

$$L_{\rm int} = \frac{1}{2}\sum_{\alpha,\beta=A,B}\ell_{\alpha\beta}, \qquad \ell_{\alpha\beta} \equiv \int d^3x\,(-n_\beta\,\varepsilon_\alpha + \mathbf{N}_\beta\cdot\mathbf{j}_\alpha + \psi_\beta\,\mathrm{tr}\,\mathsf{T}_\alpha). \tag{A.14}$$

The terms with $\alpha = \beta$ are gravitational self-energies (see the renormalization remark after Eq. (3.8)). The terms with $\alpha \neq \beta$ are the mutual interaction, $L_{\rm int}^{AB} = \frac{1}{2}(\ell_{AB} + \ell_{BA})$. To evaluate ℓ_{AB}, use Parseval's relation $\int d^3x\, f(\mathbf{x})g(\mathbf{x}) = \int_{\mathbf{k}} f(\mathbf{k})g(-\mathbf{k})$, placing A's operators at $+\mathbf{k}$ and B's fields at $-\mathbf{k}$, and insert Eq. (A.13). Since $\mathbf{N}_B^T\cdot\mathbf{j}_A = \mathbf{N}_B^T\cdot\mathbf{j}_{TA}$ and $\mathbf{N}_B^L\cdot\mathbf{j}_A = \mathbf{N}_B^L\cdot\mathbf{j}_{LA}$,

$$\ell_{AB} = \int_{\mathbf{k}}\frac{4\pi G}{k^2}\Big[(\varepsilon_B + \mathrm{tr}\,\mathsf{T}_B)\,\varepsilon_A - 4\,\mathbf{j}_{TB}\cdot\mathbf{j}_{TA} - 3\,\mathbf{j}_{LB}\cdot\mathbf{j}_{LA} + \varepsilon_B\,\mathrm{tr}\,\mathsf{T}_A\Big].$$

ℓ_{BA} is the same with $A \leftrightarrow B$. Relabeling $\mathbf{k} \to -\mathbf{k}$ in it keeps A at $+\mathbf{k}$. Adding the two and multiplying by $\frac{1}{2}$,

$$L_{\rm int}^{AB} = \int_{\mathbf{k}} \frac{4\pi G}{k^2} \Big[\varepsilon_A \varepsilon_B - 4\, \mathbf{j}_{TA} \cdot \mathbf{j}_{TB} - 3\, \mathbf{j}_{LA} \cdot \mathbf{j}_{LB} + \varepsilon_A \,\mathrm{tr}\, \mathsf{T}_B + \mathrm{tr}\, \mathsf{T}_A\, \varepsilon_B \Big] \equiv \frac{\kappa_g^2}{4} \int_{\mathbf{k}} \mathcal{C}_{AB}(\mathbf{k}). \tag{A.15}$$

The last equality defines the constraint kernel $\mathcal{C}_{AB}$; with $4\pi G = \kappa_g^2/8$ it is Eq. (3.5).

Passing to the Hamiltonian. For a small interaction, the change of the Hamiltonian at fixed canonical variables is minus the change of the Lagrangian at fixed velocities, $H_{\rm int}(q,p) = -L_{\rm int}(q,\dot q) + O(L_{\rm int}^2)$. (In $H = p\dot q - L$, the first-order change of $\dot q$ at fixed p cancels because $p = \partial L/\partial \dot q$.) At first order in G, therefore, $H_C = -L_{\rm int}$. For a single total source the interaction Lagrangian is $L_{\rm int} = (\kappa_g^2/8) \int_{\mathbf{k}} \mathcal{C}[T,T]$, which gives Eq. (3.6).

Covariant cross-check. The same kernel appears in a covariant calculation. Let

$$\mathcal{P}^{\rm dD}_{\mu\nu,\rho\sigma} = \tfrac{1}{2} \left(\eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho} - \eta_{\mu\nu}\eta_{\rho\sigma} \right)$$

be the numerator of the graviton propagator in de Donder (harmonic) gauge [10], and write $T_A \mathcal{P}^{\rm dD} T_B = T_A^{\mu\nu} \mathcal{P}^{\rm dD}_{\mu\nu,\rho\sigma} T_B^{\rho\sigma} = T_A^{\mu\nu} T_{B\,\mu\nu} - \frac{1}{2} T_A{}^{\mu}{}_{\mu} T_B{}^{\nu}{}_{\nu}$. The one-graviton exchange amplitude between two conserved sources is proportional to $T_A \mathcal{P}^{\rm dD} T_B/(k^2 - \omega^2)$. Using conservation, $\omega\varepsilon = \mathbf{k}\cdot\mathbf{j}$ and $\omega \mathbf{j} = \mathsf{T}\mathbf{k}$ up to the sign conventions of Eq. (C.4), to eliminate the longitudinal and temporal components, one finds for sources with frequency ω and wavevector $\mathbf{k}$

$$\frac{T_A \mathcal{P}^{\rm dD} T_B}{k^2 - (\omega + i0)^2} = \frac{\mathsf{T}_A \Pi^{\rm TT} \mathsf{T}_B}{k^2 - (\omega + i0)^2} + \mathcal{C}_{AB}(\mathbf{k}), \tag{A.16}$$

where $\mathcal{C}_{AB}$ is Eq. (3.5) and $\mathsf{T}_A \Pi^{\rm TT} \mathsf{T}_B = \mathsf{T}_A^{ij} \Pi^{\rm TT}_{ij,kl} \mathsf{T}_B^{kl}$. The first term on the right contains the radiative pole at $\omega = \pm k$ and only the two physical polarizations. The second contains $1/k^2$ but no radiative frequency pole; it is the instantaneous constraint interaction. So the covariant propagator "knows" the same split into radiative TT exchange and instantaneous constraint interaction that the canonical reduction makes explicit. (Check: for static masses only T^{00} is nonzero, the TT term vanishes, and $T_A \mathcal{P}^{\rm dD} T_B = \frac{1}{2} \varepsilon_A \varepsilon_B$, which matches the Newtonian term of Eq. (3.5).) For sharply windowed sources, conservation includes the temporal boundary terms of Eq. (3.17); Appendix C shows how those terms turn Eq. (A.16) into the finite-time channel.

Appendix B: Constraint completion, second-order dressing, and the explicit order-G argument

This appendix supplies the technical steps behind Sec. II E and the gauge-independence argument for the leading block in Sec. IV A. Section A solves the completed constraints for the gauge momenta. Section B explains why a single conjugation cannot turn the original ADM constraints into commuting momenta, and Sec. C shows how the second-order dressing is organized instead. Section D counts the matter content of the second-order terms. Section E proves the null-block identity (4.5), and Sec. F gives the Gaussian argument for the mutual and emission amplitudes, including the canonical endpoint contacts.

A. Local solution of the completed constraints

With $g_{\mu\nu} = \eta_{\mu\nu} + \kappa_g h_{\mu\nu}$, the gravitational curvature enters the unrescaled constraints divided by κ_g^2. Its linear part is therefore $O(\kappa_g^{-1})$ and its quadratic part $O(1)$, the same order as the matter stress. After rescaling the constraints so that their leading term is P_a, as in Eq. (2.35), both the quadratic gravitational terms and the matter sources enter at order κ_g. A truncation of the form $P_a - \kappa_g J_a - \kappa_g^2 \mathcal{T}_a^{\rm grav}$ is therefore not the general constraint expansion.

Since $\partial C_a/\partial P_b = \delta_a{}^b + O(\kappa_g)$, the equations $C_a = 0$ can be solved for $P_a = A_a(Q,z)$ as a formal series. Substituting $P = A$ into Eq. (2.35) and collecting powers of κ_g gives the recursion

$$A_a^{(1)} = -c_a^{(1)}(Q,0,z), \qquad A_a^{(2)} = -c_a^{(2)}(Q,0,z) - \left. \frac{\delta c_a^{(1)}}{\delta P_b} \right|_{P=0} A_b^{(1)}, \tag{B.1}$$

where repeated field indices include spatial integration. The coefficients are those of the full ADM constraints, Eq. (A.3), expanded to the required order; the recursion does not replace the gravitational self-stress by an unspecified

second-order source. The constraints $\bar{C}_a = P_a - A_a$ have the same zero set as the C_a, and $C_a = K_a{}^b \bar{C}_b$ with K invertible. Classically the $\bar{C}_a$ commute strongly, by the argument after Eq. (2.37). This classical statement does not fix the quantum operator ordering; that requires the common prescription of Sec. II E.

B. Why one conjugation cannot Abelianize the original constraints

For tangential generators normalized as $C[\alpha] = \kappa_g \mathcal{H}_i[\alpha^i]$, the ADM algebra gives, schematically and in a consistent quantum prescription,

$$[C[\alpha], C[\beta]] = i\kappa_g\, C[[\alpha, \beta]], \tag{B.2}$$

where $[\alpha, \beta]$ is the Lie bracket of the two vector fields. A common conjugation preserves commutators. If a single dressing E mapped every $C[\alpha]$ to $P[\alpha] + O(\kappa_g^3)$, the transformed commutator would be $O(\kappa_g^3)$, because the $P[\alpha]$ commute. The right side of Eq. (B.2), however, contains $i\kappa_g P[[\alpha, \beta]]$. A strong identity $E^\dagger C_a E = P_a + O(\kappa_g^3)$ therefore cannot hold for the original non-Abelian generators. What holds for a dressed observable $O_{\rm D}$ is a weak statement,

$$[C_a, O_{\rm D}] = \text{terms vanishing on the physical sector} + O(\kappa_g^3). \tag{B.3}$$

The equivalent Abelianized constraints $\bar{C}_a$ avoid the obstruction. The distinction between a strong operator identity and a physical-sector identity is essential. In particular, the order-G argument of Sec. F does not claim that the first-order dressed operator has an $O(\kappa_g^3)$ commutator with every constraint.

C. Flatness and the second-order dressing

Suppose an appropriately ordered Abelianized constraint has the form $\bar{C}_a = -i\partial_a - A_a(Q)$, with $\partial_a = \partial/\partial Q_a$ and $A_a = \kappa_g A_a^{(1)} + \kappa_g^2 A_a^{(2)} + \cdots$ an operator on the matter and TT space. A physical state $E\psi$ with Q-independent ψ requires

$$\bar{C}_a E\psi = 0 \quad \Longleftrightarrow \quad \partial_a E = iA_a E. \tag{B.4}$$

For one E to satisfy all of these equations, the connection must be flat:

$$\partial_a A_b - \partial_b A_a - i[A_a, A_b] = O(\kappa_g^3). \tag{B.5}$$

Knowing the right sides of functional differential equations does not guarantee this condition. It follows classically from the completed first-class system after solving for P, and quantum mechanically it requires the compatible ordering and measure of Sec. II E.

For the special case $A_a^{(1)} = J_a$ independent of Q, write $E = \exp(i\kappa_g D_1 + i\kappa_g^2 D_2)$ with $D_1 = D = \int Q_a J_a$. The Baker–Campbell–Hausdorff expansion gives

$$E^\dagger \bar{C}_a E = P_a + \kappa_g^2 \left(-A_a^{(2)} + \partial_a D_2 - \frac{i}{2}[J_a, D_1] \right) + O(\kappa_g^3), \tag{B.6}$$

so the second-order dressing must satisfy

$$\partial_a D_2 = A_a^{(2)} + \frac{i}{2}[J_a, D_1]. \tag{B.7}$$

Taking two derivatives exposes the necessary condition

$$\partial_a A_b^{(2)} - \partial_b A_a^{(2)} = i[J_a, J_b], \tag{B.8}$$

which is Eq. (B.5) at second order. This integrability condition is what is missing from the argument that a Green function automatically supplies D_2.

Once flatness holds on a star-shaped local gauge patch, the dressing can be constructed by integrating along sQ, $0 \le s \le 1$, with $E[0] = 1$. Define $B_n(s) = \int d^3x\, Q_a(\mathbf{x}) A_a^{(n)}(sQ, \mathbf{x})$. The path-ordered solution and its Magnus expansion give

$$\begin{aligned} E[Q] &= \mathcal{P} \exp\left(i \int_0^1 ds \int d^3x\, Q_a A_a(sQ) \right), \\ D_1 &= \int_0^1 ds\, B_1(s), \qquad D_2 = \int_0^1 ds\, B_2(s) + \frac{i}{2} \int_0^1 ds \int_0^s du\, [B_1(s), B_1(u)]. \end{aligned} \tag{B.9}$$

This is a constructive formula for the chosen Abelianized connection, not a fully evaluated D_2 of the original ADM generators. Its inputs include the completed gravitational sources and the quantum ordering; replacing them by the matter-only J_a does not do that work.

A two-dimensional toy model shows how the constraint completion and the dressing completion go together. Take gauge coordinates (x, y) with $J_1 = \sigma_x$ and $J_2 = \sigma_y$. Setting $A^{(2)} = 0$ violates Eq. (B.8). The completion $A_1^{(2)} = y\sigma_z$, $A_2^{(2)} = -x\sigma_z$ satisfies it, since $\partial_x A_2^{(2)} - \partial_y A_1^{(2)} = -2\sigma_z = i[\sigma_x, \sigma_y]$. For this completed system Eq. (B.7) is solved by $D_2 = 0$: the dressing $E = \exp[i\kappa_g(x\sigma_x + y\sigma_y)]$ is already exact through second order.

D. Matter content of the second-order terms

For the minimally coupled two-species model, each matter contribution to the full constraint is quadratic in the fields of one species, and there is no direct A–B matter interaction in the action. At the first two orders,

$$c_a^{(1)} = c_{a,g}^{(1)} - J_{a,A} - J_{a,B}, \qquad c_a^{(2)} = c_{a,g}^{(2)} + c_{a,A}^{(2)} + c_{a,B}^{(2)}. \tag{B.10}$$

Here $c_{a,g}^{(1)}$ is quadratic in the gravitational variables, $c_{a,g}^{(2)}$ is cubic, and each $c_{a,\alpha}^{(2)}$ is linear in a gravitational perturbation and quadratic in the fields of species α.

The recursion (B.1) then shows that $A_a^{(2)}$ contains a pure-gravity part and terms involving one matter species at a time. Differentiating $c_a^{(1)}$ with respect to a constrained momentum acts only on its gravitational part, since the flat-space J_a does not depend on P. Multiplying that derivative by $A^{(1)}$ therefore still introduces only one matter source. A product of two matter sources from substituting $P = O(\kappa_g J)$ into a quadratic gravitational constraint first appears at the next order. This counting concerns the constraint and dressing expansion. It does not remove the order-κ_g^2 Newtonian interaction, which is generated by two linear matter–gravity vertices and is retained in H_C.

The Magnus commutator in Eq. (B.9) introduces no A–B source either. The matter parts of $A^{(1)}$ contain commuting gauge coordinates and one species' operators, so their cross-species commutator vanishes, and a commutator with a pure-gravity part still contains only one species. With the reference normalization $E[0] = 1$, the independent second-order coefficient therefore has the form

$$D_2 = D_{2,g} + D_{2,A} + D_{2,B}, \tag{B.11}$$

where $D_{2,\alpha}$ may also involve gravitational variables. Products involving both species do occur in the expansion of the exponential, through the known term $-\frac{1}{2}\kappa_g^2 D_1^2$ of Eq. (2.31). They must be kept, and they are kept in the Gaussian argument of Sec. F. An additional gauge-invariant A–B integration constant would change the reference identification and is excluded by $E[0] = 1$.

The consequences for the leading block are as follows. At order κ_g^2, an independent D_2 insertion is paired with the zeroth-order product state of matter and TT vacuum. Tracing its gravitational factor gives a one-body matter amplitude correction, which is removed from the leading block by Eq. (4.5). Insertions of D_2 against the order-κ_g matter–TT correlation start at order κ_g^3. The same test disposes of the other omitted terms. A seagull vertex $h^2 T_\alpha$ contributes its vacuum expectation as a one-body Hamiltonian at this order, while its two-graviton transition probability starts at κ_g^4. Pure-gravity vacuum contributions cancel from normalized matter probabilities, and gravitational tadpoles are removed by the renormalization conditions that keep Minkowski space a solution: the matter stress is normal ordered and the renormalized cosmological constant vanishes. A pair-changing transition outside the retained number sector

has no selected gain, and at this order its selected loss and virtual return are same-species terms. These observations do not estimate the error of discarding pair sectors in the full channel, but they show why such terms do not enter the leading block. The counting assumes the minimal two-scalar action and the declared reference. Additional interspecies interactions in the effective action, a changed gravitational state, or a changed detector definition would require a separate analysis.

E. The null-block identity

Let $\rho_0 = |00\rangle\langle 00|$ be the product input in the local basis of Eq. (F.1). Then $\rho_0^{T_B} = \rho_0$, and $P_\perp = 1 - \rho_0$ projects onto the span of $|01\rangle, |10\rangle, |11\rangle$. Consider first $X = X_A \otimes 1$. Since the partial transpose acts only on B,

$$\big(X\rho_0\big)^{T_B} = X_A|0\rangle\langle 0| \otimes |0\rangle\langle 0|, \tag{B.12}$$

whose matrix elements between basis states $|ab\rangle$ and $|cd\rangle$ require $b = d = 0$. In the null subspace this forces $a \neq 0$ and $c \neq 0$, and then the element contains $\langle 0|c\rangle = 0$. For $X = 1 \otimes X_B$, the partial transpose gives $|0\rangle\langle 0| \otimes |0\rangle\langle 0|X_B^T$, whose elements require $a = c = 0$; in the null subspace this forces $b, d \neq 0$, and the element contains $\langle b|0\rangle = 0$. The same argument applies to $\rho_0 Y$. This proves Eq. (4.5). Two-sided terms $X\rho_0 Y^\dagger$ with both factors nontrivial, such as the gain $L\rho_0 L^\dagger$, are not removed; this is why the local gains ν_α survive in D.

F. The Gaussian argument with canonical endpoint contacts

Let Φ denote all linear gravitational histories, including the canonical momenta and the lapse and shift perturbations. For species α, write the completed vertex of Eq. (2.46) as

$$\ell_\alpha[\Phi] = S_{1,\alpha} - D_\alpha(t_f) + D_\alpha(t_i), \qquad \ell_\alpha[R\xi] = 0, \tag{B.13}$$

where $R\xi$ is a free gauge direction; the second equation is the content of Eq. (2.46). Choose coordinates v on the gauge quotient of these histories and a gauge coordinate η. This is an off-shell quotient of histories: it retains the nondynamical combinations that mediate the constraint response, not just on-shell TT waves. On any admissible linear section, $\Phi = \Phi_\chi(v) + R\eta$. The quadratic action with its matched free-state boundary kernels and each ℓ_α depend only on v, and the linear Faddeev–Popov determinant cancels the η integration, exactly as in Eq. (2.40). The normalized Gaussian generating functional is therefore

$$Z_\chi[s_A, s_B] = \frac{\displaystyle\int \mathcal{D}v\, e^{iS_{2,\mathrm{phys}}[v] + is_A\ell_A[v] + is_B\ell_B[v]}}{\displaystyle\int \mathcal{D}v\, e^{iS_{2,\mathrm{phys}}[v]}}, \tag{B.14}$$

independent of χ. The expression stands for either an amplitude with its boundary kernels or the doubled closed-time-path integral; the same physical free state is used in the numerator and the denominator. Gauge-volume factors cancel, and this Gaussian step has no interacting ordering anomaly. Regulated endpoint products are defined by the canonical integral before the sharp-time limit is taken.

The coefficient of $s_A s_B$ proves that the mutual contraction of the two completed vertices does not depend on the section. Equivalently, for gauge-fixed phase-space covariances $\mathscr{D}_\chi$ and $\mathscr{D}_0$ describing the same state,

$$\ell_A\big(\mathscr{D}_\chi - \mathscr{D}_0\big)\ell_B = 0. \tag{B.15}$$

This statement includes all canonical endpoint contacts, and it covers any linear history-space section, de Donder included. The contacts arise because D contains canonical momenta. Replacing a momentum by a time derivative inside a time-ordered product generates equal-time contact terms. Equivalently, integrating out the momenta in a path integral with endpoint sources generates quadratic endpoint terms. These terms can involve both species. They belong to the known product of two first-order vertices and must not be discarded as an unknown local D_2 correction. In a metric-only formulation they appear as the contacts generated when the momenta are eliminated; conservation of the sources does not justify omitting them.

Each source enters the mutual contraction once, and the two species commute, so no same-species stress-commutator contact arises there. On the radiative shell, $\ell_\alpha[R\xi] = 0$ also shows that the first-order emission amplitude decouples from gauge directions. In a physical TT mode $Q_a = 0$, so the dressing does not contribute, leaving the TT emission vertex of Eq. (3.12). Its squared amplitudes supply the gains and the cross loss. The off-shell response still contains the constrained interaction: source conservation does not remove the Newtonian term. Evaluating the same completed Gaussian functional in the canonical reference gauge gives the constraint-plus-TT propagation of Sec. III. This is what a covariant endpoint calculation must reproduce (Appendix C); it is not an independent evaluation of the specific de Donder endpoint kernels.

For same-species time-ordered products, differentiating the time ordering produces contact terms that involve stress commutators. The linear identity alone does not determine all of them. They are local and drop out of the leading block by Eq. (4.5), but their full form belongs to the common quantum prescription of Sec. II E.

Appendix C: The covariant (closed-time-path) calculation and endpoint matching

What this appendix does. Section III computes the reduced density matrix by first solving the constraints and treating only the TT modes as dynamical. A covariant calculation proceeds differently. It keeps all ten components of $h_{\mu\nu}$ in de Donder gauge, couples them to matter only during the observation window $[0, t_f]$, and integrates the field out. Both calculations describe the same physics, so they must agree. They agree, however, only if the covariant calculation also includes the gravitational field already present in the initial state and the dressing of the final detector (Secs. II D and III C). This appendix makes that statement precise for the one object that matters for entanglement: the order-G *mutual* (A–B) part of the reduced density matrix. It establishes two identities:

- **Response.** Covariant retarded exchange in the slab, plus endpoint terms, equals retarded TT exchange plus the equal-time constraint evolution $-i[\mathsf{K}_C^{AB}, \rho]$. This is Eq. (C.9).
- **Noise.** Covariant symmetric (fluctuation) terms in the slab, plus endpoint terms, equal the TT noise alone. The non-TT fluctuations cancel. This is Eq. (C.12).

Status of these identities. Equations (C.9) and (C.12) are identities. They show which endpoint terms the covariant slab needs in order to reproduce the reduced channel. That the dressed preparation and the dressed detector supply exactly these terms follows from the section-independence argument of Sec. II E, with matched physical boundary data, and from the Gaussian argument of Appendix B F. It is a consistency condition, not an independent evaluation of the endpoint kernels. A Ward identity alone would not fix them, because a gauge-invariant boundary functional can be added without violating it. For static sources, Eq. (3.19) checks the initial-edge term independently. An independent evaluation in general would contract the de Donder covariances of $Q_0 = \nabla^{-2}(-\dot{h}_{kk} + 2\partial_k h_{0k})$ and of Q_i with the bulk vertices at $t = 0$ and $t = t_f$, using the edge integrals $\mathcal{J}$ and $\mathcal{B}_\partial$ defined below.

The appendix proceeds as follows. Sec. A reviews second-order reduced dynamics. Sec. B turns the time integrals into frequency kernels. Sec. C splits the covariant numerator into TT and constraint parts. Secs. D and E prove the two identities, and Sec. F assembles the packet-projected density matrix and records checks.

A. Second-order reduced dynamics of a system coupled to a free field

Consider matter coupled linearly to a free field in a Gaussian state (here the graviton vacuum), with interaction-picture Hamiltonian $H_I(t) = -\int d^3x\, h_{\mu\nu}(t,\mathbf{x}) X^{\mu\nu}(t,\mathbf{x})$, where $X^{\mu\nu} = \frac{\kappa_g}{2}\chi(t) T^{\mu\nu}$ is the windowed matter stress. Expand $\rho_{\rm tot}$ to second order and trace over the field. Only the field two-point function survives. Split it into its commutator and anticommutator parts,

$$
\begin{aligned}
G_R(t,t') &= i\langle [h(t), h(t')]\rangle \quad (t > t'),\\
G_H(t,t') &= \tfrac{1}{2}\langle \{h(t), h(t')\}\rangle,\\
\langle h(t)h(t')\rangle &= G_H(t,t') - \tfrac{i}{2} G_R(t,t') \quad (t > t'),
\end{aligned}
$$

where spatial arguments and tensor indices are suppressed. The standard result [50, 55] is then

$$
\delta\rho = \frac{i}{2}\int_0^{t_f} dt \int_0^t dt'\, G_R(t,t')\,\big[X(t), \{X(t'), \rho\}\big] - \int_0^{t_f} dt \int_0^t dt'\, G_H(t,t')\,\big[X(t), [X(t'), \rho]\big]. \tag{C.1}
$$

(It follows from the second-order term $-\iint_{t>t'} \mathrm{Tr}_h[H_I(t),[H_I(t'),\rho\otimes\rho_h]]$ after inserting the split of $\langle hh\rangle$.)
The two terms.

- The **response** term contains the *state-independent* retarded Green function. The earlier source enters through an anticommutator, i.e. through the average of its action on ket and bra: it acts as a classical source. The later source enters through a commutator, i.e. as a generator of evolution. So this term is "the field produced at t' acts on the system at t".
- The **noise** term contains the *state-dependent* symmetric function, i.e. vacuum fluctuations of the field. The double commutator makes it a decoherence (fluctuation) term.

Mutual terms. For two particles, $X = X_A + X_B$. The A–B cross terms of Eq. (C.1) are:

- the *directed response* of A to B, $\frac{i}{2}\iint_{t>t'} G_R\,[X_A(t),\{X_B(t'),\rho\}]$, together with the reverse, $A \leftrightarrow B$;
- the *mutual noise*, $-\iint_{[0,t_f]^2} G_H(t,t')\,[X_A(t),[X_B(t'),\rho]]$. Here the two orderings combine into the full square because A and B operators commute.

Expanding the commutators shows where each operator sits relative to ρ:

$$\begin{aligned}[X_A,\{X_B,\rho\}] &= X_AX_B\rho + X_A\rho X_B - X_B\rho X_A - \rho X_BX_A,\\ [X_A,[X_B,\rho]] &= X_AX_B\rho - X_A\rho X_B - X_B\rho X_A + \rho X_BX_A.\end{aligned}$$

In the four placements, both operators act on the ket, A on the ket and B on the bra, the reverse, or both on the bra. The sign patterns $(+,+,-,-)$ and $(+,-,-,+)$ reappear as $U+V-W-Z$ and $U-V-W+Z$ in Eq. (C.15).
The field kernels. In de Donder gauge, $\langle h_{\mu\nu}h_{\rho\sigma}\rangle = \mathcal{P}^{\mathrm{dD}}_{\mu\nu,\rho\sigma}\times$ (the corresponding function of a canonically normalized massless scalar), with $\mathcal{P}^{\mathrm{dD}}$ from Appendix A E. For one Fourier mode of wavenumber k, the scalar's retarded function and symmetric function are those of a harmonic oscillator:

$$G_R \to g_k(t-t') = \frac{\sin k(t-t')}{k}\quad (t>t'), \qquad G_H \to \frac{\cos k(t-t')}{2k}.$$

Restricting $\mathcal{P}^{\mathrm{dD}}$ to TT components gives Π^{TT}. The TT part of Eq. (C.1) is then exactly the TT exchange, gain, and loss of the reduced channel (3.14). The question is what happens to the *non-TT* part of $\mathcal{P}^{\mathrm{dD}}$.

B. From time integrals to frequency kernels

In the interaction picture every matter transition carries a phase $e^{i\omega t}$ (Eq. (3.4)). In a mutual term, A receives $+\mathbf{k}$ with recoil energy ω_A and B receives $-\mathbf{k}$ with recoil energy ω_B (Eq. (E.2)). For each mode and each set of momenta, the time integrals in Eq. (C.1) therefore reduce to two kernels. The retarded kernel has the later vertex at t and the earlier at t'; the symmetric kernel has no ordering. With generic frequency arguments ω_1 (later vertex) and ω_2 (earlier vertex),

$$\begin{aligned}\mathcal{A}(z) &= \int_0^{t_f} e^{izt}dt, \qquad g_k(t) = \frac{\sin kt}{k},\\ K_R(\omega_1,\omega_2;k) &= \int_0^{t_f} dt \int_0^{t} dt'\, e^{i\omega_1 t + i\omega_2 t'}\, g_k(t-t'),\\ K_N(\omega_1,\omega_2;k) &= \int_0^{t_f} dt \int_0^{t_f} dt'\, e^{i\omega_1 t + i\omega_2 t'}\, \frac{\cos[k(t-t')]}{2k}.\end{aligned} \tag{C.2}$$

K_R occupies the retarded time triangle $0<t'<t<t_f$, whereas K_N occupies the full time square. Both are elementary. Closed forms useful for numerical evaluation are

$$\begin{aligned}\mathcal{B}(a,b) &= \int_0^{t_f} dt\, e^{iat} \int_0^{t} dt'\, e^{ibt'} = \frac{\mathcal{A}(a+b)-\mathcal{A}(a)}{ib},\\ K_R(a,b;k) &= \frac{\mathcal{B}(a+k,b-k)-\mathcal{B}(a-k,b+k)}{2ik},\\ K_N(a,b;k) &= \frac{\mathcal{A}(a+k)\mathcal{A}(b-k)+\mathcal{A}(a-k)\mathcal{A}(b+k)}{4k}.\end{aligned} \tag{C.3}$$

All apparent singularities at zero frequency or zero k are removable. The defining integrals or their Taylor series are used in those regions.

With these kernels, the *bare covariant slab* contributions of one mode to the mutual density-matrix terms are proportional to

$$\text{response of } A \text{ to } B: \quad \tfrac{i}{2}\big(\tfrac{\kappa_g}{2}\big)^2 (T_A \mathcal{P}^{\mathrm{dD}} T_B)\, K_R(\omega_A,\omega_B;k), \qquad \text{noise}: \quad -\big(\tfrac{\kappa_g}{2}\big)^2 (T_A \mathcal{P}^{\mathrm{dD}} T_B)\, K_N(\omega_A,\omega_B;k),$$

with T_α replaced by the vertices τ_α and multiplied by packet amplitudes (Sec. F). The remaining task is to split the numerator $T_A \mathcal{P}^{\mathrm{dD}} T_B$ into a TT part and a non-TT part and to find what the non-TT part becomes after the time integrals.

C. Splitting the covariant numerator

From here on, put $a = \omega_A$ and $b = \omega_B$. (In this appendix a and b are recoil frequencies, not the impact separation.) Stress-tensor conservation at each vertex, Eq. (3.3), reads $q_\mu \tau^{\mu\nu} = 0$ with four-momentum transfer $q_A = (a, \mathbf{k})$ for A and $q_B = (b, -\mathbf{k})$ for B. In components, with $\mathbf{n} = \mathbf{k}/k$:

$$k\mathbf{n}\cdot\mathbf{j}_A = a\varepsilon_A, \quad k\mathbf{n}\cdot\mathbf{j}_B = -b\varepsilon_B, \quad \mathsf{T}_A\mathbf{n} = \frac{a}{k}\mathbf{j}_A, \quad \mathsf{T}_B\mathbf{n} = -\frac{b}{k}\mathbf{j}_B. \tag{C.4}$$

These relations let us trade longitudinal components of the current and stress for the energy density and recoil frequencies. Define the auxiliary combination $\mathcal{Q}$ and the non-TT part of the numerator,

$$\mathcal{Q} = \frac{a\varepsilon_A \operatorname{tr}\mathsf{T}_B + b(\operatorname{tr}\mathsf{T}_A)\varepsilon_B}{2k^2}, \qquad \mathcal{D}_{\mathrm{long}} = T_A \mathcal{P}^{\mathrm{dD}} T_B - \mathsf{T}_A \Pi^{\mathrm{TT}} \mathsf{T}_B. \tag{C.5}$$

Inserting Eq. (C.4) into the two contractions $T_A \mathcal{P}^{\mathrm{dD}} T_B$ and $\mathsf{T}_A \Pi^{\mathrm{TT}} \mathsf{T}_B$ gives the independent-frequency identity

$$\mathcal{D}_{\mathrm{long}} = (k^2 + ab)\mathcal{C} - (a+b)\mathcal{Q}. \tag{C.6}$$

where $\mathcal{C}$ is exactly the constraint kernel (3.5). This is the key algebraic step: *the non-TT part of the covariant numerator is built from the constraint kernel.* Two limits clarify it. For an infinitely long time interval, energy conservation would impose $b = -a$. Then $\mathcal{D}_{\mathrm{long}} = (k^2 - a^2)\mathcal{C}$, and dividing by the propagator denominator $k^2 - a^2$ gives Eq. (A.16): the non-TT exchange is instantaneous. At finite time, $a + b \neq 0$ in general, and the $\mathcal{Q}$ term must be kept. Setting $b = -a$ before the time integrals would impose an infinite-time relation on a finite interval.

D. Response: retarded slab plus endpoint terms

We now need $\mathcal{D}_{\mathrm{long}} K_R = [(k^2+ab)\mathcal{C} - (a+b)\mathcal{Q}]K_R$. The combinations $(k^2+ab)K_R$ and $(a+b)K_R$ can be evaluated by integrating by parts on the retarded triangle. The factors a and b come from differentiating e^{iat} and $e^{ibt'}$, and k^2 from $\ddot{g}_k = -k^2 g_k$. The boundary contributions come from the three edges of the triangle: the diagonal $t = t'$, where $g_k = 0$ and $\dot{g}_k = 1$; the edge $t' = 0$ (the initial slice); and the edge $t = t_f$ (the detection time). Two auxiliary integrals appear:

$$\begin{aligned}\mathcal{J}(z;k) &= \int_0^{t_f} dt\, e^{izt} g_k(t) = \frac{\mathcal{A}(z+k) - \mathcal{A}(z-k)}{2ik},\\ \mathcal{B}_\partial(a,b;k) &= e^{i(a+b)t_f}\,\mathcal{J}(-b;k).\end{aligned} \tag{C.7}$$

The identities are

$$\begin{aligned}(k^2+ab)K_R &= \mathcal{A}(a+b) - e^{iat_f} g_k(t_f) - ib\mathcal{B}_\partial + ia\mathcal{J}(a;k),\\ (a+b)K_R &= -i[\mathcal{B}_\partial - \mathcal{J}(a;k)].\end{aligned} \tag{C.8}$$

The diagonal $t = t'$ supplies the *equal-time* term $\mathcal{A}(a+b)$, which is exactly the time integral of an instantaneous interaction. All other terms sit on the edges $t' = 0$ and $t = t_f$; they are endpoint contributions. Combining Eqs. (C.6) and (C.8) and collecting the endpoint terms into $\mathcal{R}_\partial$ gives

$$\begin{aligned}\mathcal{R}_\partial(a,b;k) &= \mathcal{C}e^{iat_f} g_k(t_f) + i(b\mathcal{C} - \mathcal{Q})\mathcal{B}_\partial(a,b;k)\\ &\quad + i(\mathcal{Q} - a\mathcal{C})\mathcal{J}(a;k),\\ (T_A\mathcal{P}^{\mathrm{dD}}T_B)K_R + \mathcal{R}_\partial &= (\mathsf{T}_A\Pi^{\mathrm{TT}}\mathsf{T}_B)K_R + \mathcal{C}\mathcal{A}(a+b).\end{aligned} \tag{C.9}$$

The last line is the finite-time Ward (gauge) matching identity. The covariant retarded slab plus its endpoint term $\mathcal{R}_\partial$ equals the TT retarded slab plus the equal-time constraint evolution, whose time integral is $\mathcal{C}\,\mathcal{A}(a+b)$, as in Eq. (E.5). By the reduction of Sec. II E with matched boundary data, this is the term that the dressed preparation and detector must supply; we have not evaluated those contractions independently. Reversing the directed response interchanges the species and the recoil frequencies.

E. Noise: symmetric slab and cancellation of non-TT fluctuations

The same steps apply to the symmetric kernel on the full square. The needed combinations are $(k^2+ab)K_N$ and $(a+b)K_N$. Define

$$\begin{aligned}\mathcal{Z}(z;k) &= \frac{\mathcal{A}(z+k) + \mathcal{A}(z-k)}{4k},\\ \mathcal{X}(z;a+b;k) &= e^{i(a+b)t_f}\,\mathcal{Z}(-z;k) - \mathcal{Z}(z;k),\\ \mathcal{E}_0(a,b;k) &= \frac{e^{i(a+b)t_f} + 1 - (e^{iat_f} + e^{ibt_f})\cos(kt_f)}{2k}.\end{aligned} \tag{C.10}$$

Integration by parts on the square now has no diagonal term, only edges. It gives

$$\begin{aligned}(k^2+ab)K_N &= \mathcal{E}_0 - ia\mathcal{X}(a) - ib\mathcal{X}(b),\\ (a+b)K_N &= -i[\mathcal{X}(a) + \mathcal{X}(b)].\end{aligned} \tag{C.11}$$

Combining these with Eq. (C.6) gives the endpoint noise coefficient

$$\mathcal{N}_\partial = \mathcal{C}\mathcal{E}_0 + i(\mathcal{Q} - a\mathcal{C})\mathcal{X}(a) + i(\mathcal{Q} - b\mathcal{C})\mathcal{X}(b) = \mathcal{D}_{\mathrm{long}}K_N. \tag{C.12}$$

Equation (C.12) states that the endpoint noise term is exactly the non-TT part of the covariant noise. In the density matrix it enters with the opposite sign to the bare slab noise $-(\kappa_g/2)^2(T_A\mathcal{P}^{\mathrm{dD}}T_B)K_N$ (Sec. B), so the sum is $-(\kappa_g/2)^2(\mathsf{T}_A\Pi^{\mathrm{TT}}\mathsf{T}_B)K_N$. Only the TT covariance remains as physical gravitational noise. That the constrained initial field is not an independent longitudinal decoherence bath follows from the canonical reduction of Appendix A; Eq. (C.12) is consistent with it.

F. Packet projection, assembly, and checks

Packet projection. To obtain density-matrix entries in the selected modes, each vertex is sandwiched between packet amplitudes as in Eq. (E.1). For selected mode indices, define the packet weight $\mathcal{W}$ and the three projected endpoint integrals (B^1 for the response of A to B, B^2 for the reverse, B^n for noise):

$$\begin{aligned}
\mathcal{W}_{r_A s_A, r_B s_B} &= w^*_{A r_A}(\mathbf{p}_A + \mathbf{k}) w_{A s_A}(\mathbf{p}_A) w^*_{B r_B}(\mathbf{p}_B - \mathbf{k}) w_{B s_B}(\mathbf{p}_B), \\
B^1 &= \int_{\mathbf{k},\mathbf{p}_A,\mathbf{p}_B} \mathcal{W}\, \mathcal{R}_\partial(\omega_A, \omega_B; k), \\
B^2 &= \int_{\mathbf{k},\mathbf{p}_A,\mathbf{p}_B} \mathcal{W}\, \mathcal{R}_\partial(\omega_B, \omega_A; k), \\
B^n &= \int_{\mathbf{k},\mathbf{p}_A,\mathbf{p}_B} \mathcal{W}\, \mathcal{N}_\partial(\omega_A, \omega_B; k).
\end{aligned} \tag{C.13}$$

Placement bookkeeping. Let a density-matrix entry have row (a, b) and column (c, d). (Here $a, b, c, d \in \{+, -\}$ are mode labels.) The four placements of the two vertices in Sec. A are encoded by Kronecker deltas:

$$U = \delta_{a r_A}\delta_{b r_B}, \quad V = \delta_{a r_A}\delta_{d s_B}, \quad W = \delta_{b r_B}\delta_{c s_A}, \quad Z = \delta_{c s_A}\delta_{d s_B}. \tag{C.14}$$

U means both vertices act on the ket, so their outgoing labels r_A, r_B fix the row. Z means both act on the bra from the right, so their incoming labels s_A, s_B fix the column. V and W are the mixed placements. For the real even cats, every entry of ρ_0 equals $\frac{1}{4}$. The sum over the undetermined initial labels then only produces that factor, which is absorbed into the coefficients: $\frac{i}{2}(\kappa_g/2)^2 \times \frac{1}{4} = i\kappa_g^2/32$, and $(\kappa_g/2)^2 \times \frac{1}{4} = \kappa_g^2/16$. The endpoint contribution to every selected mutual entry is

$$\begin{aligned}
(\delta\widetilde{\rho}^{AB}_\partial)_{ab,cd} = \sum_{r_A, s_A, r_B, s_B} \Big[&\frac{i\kappa_g^2}{32} B^1(U + V - W - Z) + \frac{i\kappa_g^2}{32} B^2(U + W - V - Z) \\
&+ \frac{\kappa_g^2}{16} B^n(U - V - W + Z) \Big].
\end{aligned} \tag{C.15}$$

Substituting the right side of Eq. (C.9) in both directed slots makes the first two terms exactly $-i[\mathsf{K}_C^{AB}, \rho_0]$. Equation (C.12) leaves the TT covariance. This gives the full sixteen-entry correspondence with the reduced channel, given the identification of the endpoint terms discussed at the beginning of this appendix.

For static sources, $a = b = 0$ and $\mathcal{Q} = 0$. The radial Fourier integral

$$\int_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}} \frac{\sin(kt_f)}{k^3} = \frac{\min(t_f, R)}{4\pi R} \tag{C.16}$$

gives the boundary phase $g\min(t_f, R)/R$. Adding the bare-slab term $g(t_f - R)_+/R$ gives gt_f/R, as required when the static field is already present at the initial slice. Static energy density has no TT stress, so Eq. (C.12) cancels its complete covariant-slab noise.

For the moving point-history benchmark, the constraint phase is given below, where $\gamma_\alpha = (1 - v_\alpha^2)^{-1/2}$ are the Lorentz factors and $\hat{\mathbf{r}}$ is the unit vector along the branch separation (Sec. III A derives this integrand from Eq. (3.5)):

$$\Phi_C^{s_A s_B} = g\gamma_A\gamma_B \int_0^{t_f} \frac{dt}{r(t)} \left[1 + v_A^2 + v_B^2 - \frac{7}{2}\mathbf{v}_A \cdot \mathbf{v}_B - \frac{1}{2}(\mathbf{v}_A \cdot \widehat{\mathbf{r}})(\mathbf{v}_B \cdot \widehat{\mathbf{r}}) \right]. \tag{C.17}$$

Together with the independent TT response in Appendix D, this gives $\Theta_C = 5.15685553309 \times 10^{-6}$, $\Theta_{\mathrm{TT}} = -7.47969827735 \times 10^{-11}$, and the matched point-history value $\Theta_{\mathrm{point}} = 5.15678073611 \times 10^{-6}$ for the ordinary benchmark. The difference from the bare-slab entangler is $4.82530243 \times 10^{-9}$ and is the endpoint contribution required by the matched preparation and detector.

The branchwise comparison underlying these signed combinations is

Subtracting the bare-slab phase from the matched constraint-plus-TT phase gives

Branch	Φ_C	Φ_{TT}	Bare-slab Φ_{bulk}
$++$	0.00525090386259	$4.393368825 \times 10^{-10}$	0.00515158285442
$+-$	0.00486719534263	$5.758104856 \times 10^{-10}$	0.00476768632552
$-+$	0.00561398496042	$6.020512104 \times 10^{-10}$	0.00551487156160
$--$	0.00525090386259	$4.393368825 \times 10^{-10}$	0.00515158285442

$$\Delta\Phi_\partial = (9.93214475024,\ 9.95095929195,\ 9.91140008756,\ 9.93214475024) \times 10^{-5}. \tag{C.18}$$

The common part of this vector drops out of the density matrix. Its signed two-particle combination gives

$$\Delta\Theta_\partial = 4.82530243 \times 10^{-9}. \tag{C.19}$$

Appendix D: Point histories and geometric limits

This appendix collects the small-arm analysis of the Newtonian phase and the point-history calculations used to interpret the packet result. They compare approximations to the declared experiment; none replaces the selected Gaussian channel.

A. Newtonian branch phases and cancellation during the incoming flight

Closed-form branch phases. For the planar geometry of Section II, write the branch distances of Eq. (3.20) with the remaining fraction of the flight u and the relative arm displacement $\delta_{s_A s_B}$ introduced in Eq. (2.15):

$$u = 1 - \frac{t}{T}, \qquad \delta_{s_A s_B} = \frac{s_A d_A - s_B d_B}{2}, \qquad r^2_{s_A s_B} = (2L)^2 u^2 + (b + \delta_{s_A s_B} u)^2. \tag{D.1}$$

Every branch phase is therefore $\Phi_N^{s_A s_B} = gTF(\delta_{s_A s_B})$ (using $dt = -T\,du$), with the same geometric integral

$$\begin{aligned} F(\delta) &= \int_0^1 \frac{du}{\sqrt{(2L)^2 u^2 + (b + \delta u)^2}} \\ &= \frac{1}{\sqrt{(2L)^2 + \delta^2}} \left[\operatorname{arsinh} \frac{(2L)^2 + \delta^2 + b\delta}{2Lb} - \operatorname{arsinh} \frac{\delta}{2L} \right]. \end{aligned} \tag{D.2}$$

Completing the square in the denominator gives the second line; no expansion in arm length has been used. The entangling phase is the signed sum in Eq. (3.21). Substituting Eq. (D.2) gives

$$\Theta_N = \frac{gT}{4} \left[F\left(\frac{d_A - d_B}{2}\right) + F\left(-\frac{d_A - d_B}{2}\right) - F\left(\frac{d_A + d_B}{2}\right) - F\left(-\frac{d_A + d_B}{2}\right) \right]. \tag{D.3}$$

Small arms. The entangling phase is a signed combination of four positive interaction integrals, so it can be small even when their common part is appreciable. Its geometric origin is clearest for small branch-separation vectors $\boldsymbol{\xi}_A, \boldsymbol{\xi}_B$ about a mean separation $\mathbf{R}$. Expanding $g/|\mathbf{R} + s_A\boldsymbol{\xi}_A/2 - s_B\boldsymbol{\xi}_B/2|$ in the displacements, the first term containing the product $s_A s_B$ gives the rate of change of Θ_N:

$$\frac{d\Theta_N}{dt} = -\frac{g}{4} \xi_A^i \xi_B^j \partial_i \partial_j \frac{1}{R} = -\frac{g}{4R^3} \left[3(\boldsymbol{\xi}_A \cdot \widehat{\mathbf{R}})(\boldsymbol{\xi}_B \cdot \widehat{\mathbf{R}}) - \boldsymbol{\xi}_A \cdot \boldsymbol{\xi}_B \right]. \tag{D.4}$$

The minus sign arises because B's displacement enters the relative coordinate with the opposite sign. The angular dependence is the mixed second derivative (tidal tensor) of the ordinary Newtonian potential. It can vanish even though the potential itself remains nonzero.

The free branches carry exactly the dipole differences needed for this term. Write $\Delta T_\alpha = T_{\alpha+} - T_{\alpha-}$ for the difference between the stress tensors of the two branches. Since both branches have equal speeds, the mass difference $\Delta M_\alpha = \int \Delta T_\alpha^{00} d^3x$ vanishes. The mass-dipole difference $\Delta\mathbf{D}_\alpha$ and momentum difference $\Delta\mathbf{P}_\alpha$ do not, and at nonrelativistic order they are

$$\begin{aligned} \Delta D_\alpha^i &= \int d^3x\, x^i \Delta T_\alpha^{00}, \quad \Delta\mathbf{D}_\alpha(t) = m_\alpha d_\alpha (1 - t/T)\mathbf{e}_y, \\ \Delta P_\alpha^i &= \int d^3x\, \Delta T_\alpha^{0i}, \qquad \Delta\mathbf{P}_\alpha = -m_\alpha d_\alpha \mathbf{e}_y/T, \qquad \frac{d\Delta\mathbf{D}_\alpha}{dt} = \Delta\mathbf{P}_\alpha. \end{aligned} \tag{D.5}$$

Free conservation relates these differences but does not set them to zero. The declared model contains no apparatus carrying a compensating branch dipole or momentum. The common monopole phase drops out of Eq. (3.21), leaving the R^{-3} entangling term of Eq. (D.4). A source whose complete preparation cancels both monopole and dipole differences has a different multipole expansion; a quadrupole–quadrupole R^{-5} result for such a source cannot be substituted here.

In the present planar geometry, $\boldsymbol{\xi}_\alpha = d_\alpha u \mathbf{e}_y$. The curvature relevant to Eq. (D.3) is

$$\partial_\delta^2 \frac{1}{\sqrt{(2L)^2u^2 + (b+\delta u)^2}}\bigg|_{\delta=0} = \frac{u^2[2b^2 - (2L)^2u^2]}{[(2L)^2u^2 + b^2]^{5/2}}. \tag{D.6}$$

It changes sign when the longitudinal separation $2Lu$ passes through $\sqrt{2}\,b$. Thus early and late portions of the incoming motion contribute with opposite signs. There is no outgoing flyby segment in the interval $[0, T]$. Integrating this curvature over the flight, with the variable $y = 2L/b$, gives

$$\begin{aligned} \Theta_N &= -\frac{gTd_Ad_B}{4(2L)^3} H(2L/b) + O(d^4), \\ H(y) &= \int_0^y \frac{x^2(2-x^2)}{(1+x^2)^{5/2}}\, dx \\ &= \frac{y}{\sqrt{1+y^2}} + \frac{y^3}{(1+y^2)^{3/2}} - \operatorname{arsinh} y, \\ H(y_*) &= 0, \qquad y_* = 2.919847688300\ldots. \end{aligned} \tag{D.7}$$

Here $O(d^4)$ denotes total arm degree at least four at fixed L, b. The condition $(d_A + d_B)/(4L) \ll 1$ is sufficient for the expansion uniformly over the flight. When L is small, one may instead require $d_A, d_B \ll b$ or use Eq. (D.3) directly. The limiting forms are

$$\begin{aligned} 2L \gg b: &\quad \Theta_N \simeq \frac{gTd_Ad_B}{4(2L)^3}\left[\log\frac{4L}{b} - 2\right], \\ 2L \ll b, \quad d_A, d_B \ll b: &\quad \Theta_N \simeq -\frac{gTd_Ad_B}{6b^3}. \end{aligned} \tag{D.8}$$

The root in Eq. (D.7) cancels the leading small-arm contribution. It does not cancel the common phase, higher arm orders, finite-width transitions, or the full constraint matrix element. Equal arms alone impose none of those cancellations. Section IV C returns to the planar family and tests the full selected constraint contribution for a changed input.

Rotating both branch separations to z, perpendicular to both the approach and the impact directions, removes this competition at leading small-arm order. The corresponding result is

$$\Theta_{N,\perp} = \frac{gTd_Ad_B}{4(2L)^3}\left[\operatorname{arsinh}(2L/b) - \frac{2L}{\sqrt{(2L)^2 + b^2}}\right] + O(d^4). \tag{D.9}$$

The bracket is positive for $L/b > 0$, since its derivative with respect to $2L/b$ is $y^2/(1+y^2)^{3/2}$. Section B below gives the exact integral and the parallel-flight limit; the main text retains the original planar geometry.

Finite retardation provides a further check on a small signed phase. For prescribed inertial point histories, the exact response generated only by the sources inside the time slab ("bare slab") retains the light-cone delay, its Jacobian, and the first causal arrival. Its reciprocal phase can even have the opposite sign to Eq. (D.3) when the duration is only a few light-crossing times; the complete comparison is in Sec. C below. That onset belongs to the bulk source admitted after the initial slice. The constrained preparation of Section II already carries its initial field and has its own matched constraint-plus-TT phase. Neither point-history construction includes Gaussian recoil or transitions; Section IV evaluates those effects in the selected quantum state.

B. Alternative branch geometries

If both species split along z, perpendicular to the approach and impact directions, the branch distance and exact geometric integral are

$$r^2_{s_A s_B} = b^2 + [(2L)^2 + \delta^2_{s_A s_B}]u^2, \qquad F_\perp(\delta) = \frac{\operatorname{arsinh}(\sqrt{(2L)^2+\delta^2}/b)}{\sqrt{(2L)^2+\delta^2}}. \tag{D.10}$$

Substitution in the signed branch sum gives Eq. (D.9). Its bracket is positive for every $L/b > 0$, so the cancellation of the planar geometry is absent. Parallel mean flight follows by taking $L \to 0$:

$$\Theta_{N,\perp} \longrightarrow \frac{gTd_Ad_B}{12b^3}, \qquad d_A, d_B \ll b. \tag{D.11}$$

The increased interaction time near the impact separation must still satisfy the weak-deflection and packet-separation conditions in Section II.

C. Exact retarded response of inertial histories

For one joint branch, treat both particles as classical point masses on the inertial center histories of Eq. (2.2), and put

$$\mathbf{r}_0 = \mathbf{a}_A - \mathbf{a}_B, \qquad \mathbf{w} = \mathbf{v}_A - \mathbf{v}_B, \qquad \mathbf{r}(t) = \mathbf{r}_0 + \mathbf{w}t. \tag{D.12}$$

The signal from B received at A at time t was emitted a time λ_B earlier, where λ_B solves the light-cone condition $\lambda_B = |\mathbf{r} + \mathbf{v}_B\lambda_B|$. The positive solution and the retarded-distance (Liénard–Wiechert) Jacobian are

$$\begin{aligned} D_B(t) &= \sqrt{(1-v_B^2)r(t)^2 + [\mathbf{r}(t)\cdot\mathbf{v}_B]^2}, \\ \lambda_B(t) &= \frac{\mathbf{r}(t)\cdot\mathbf{v}_B + D_B(t)}{1-v_B^2}, \\ R_{\rm ret}(1-\mathbf{n}_{\rm ret}\cdot\mathbf{v}_B) &= D_B(t). \end{aligned} \tag{D.13}$$

This varying delay fixes the integrand. A separate onset time ensures that the emission lies after the initial slice:

$$\begin{aligned} t^{AB}_{\rm on} &= \frac{\mathbf{r}_0\cdot\mathbf{v}_A + \sqrt{(\mathbf{r}_0\cdot\mathbf{v}_A)^2 + (1-v_A^2)r_0^2}}{1-v_A^2}, \\ J^{s_A s_B}_{AB} &= gC_{s_A s_B}\,\mathbf{1}_{\{t_f > t^{AB}_{\rm on}\}}\int_{t^{AB}_{\rm on}}^{t_f}\frac{dt}{D_B(t)}. \end{aligned} \tag{D.14}$$

Here $C_{s_A s_B}$ is the complete point-particle contraction of energy, current, and spatial stress, i.e. the analogue of the numerator $T_A\mathcal{P}^{\rm dD}T_B$ for two point particles, normalized so that $C = 1$ for static particles. The indicator function $\mathbf{1}$ and the lower limit $t^{AB}_{\rm on}$ ensure that only signals emitted after $t = 0$ contribute. The reverse response exchanges A and B. Since D_B^2 is quadratic in time, the integral is elementary. With

$$M_B = (1 - v_B^2)I + \mathbf{v}_B\mathbf{v}_B^T, \quad \mathcal{A}_B = \mathbf{w}^T M_B \mathbf{w}, \quad \mathcal{B}_B = \mathbf{r}_0^T M_B \mathbf{w}, \quad \mathcal{C}_B = \mathbf{r}_0^T M_B \mathbf{r}_0, \tag{D.15}$$

one has

$$\int \frac{dt}{D_B(t)} = \frac{1}{\sqrt{\mathcal{A}_B}} \operatorname{arsinh} \frac{\mathcal{A}_B t + \mathcal{B}_B}{\sqrt{\mathcal{A}_B \mathcal{C}_B - \mathcal{B}_B^2}}. \tag{D.16}$$

For static sources Eq. (D.14) reduces to $g(t_f - r_0)_+/r_0$. This is the response generated inside the slab. The initial constraint field supplies the complementary term of Appendix C and Eq. (3.19).

The signed directed responses are

$$\begin{aligned} \mathcal{J}_{AB} &= \frac{1}{4}\sum_{s_A,s_B} s_A s_B J_{AB}^{s_A s_B}, \quad \mathcal{J}_{BA} = \frac{1}{4}\sum_{s_A,s_B} s_A s_B J_{BA}^{s_A s_B}, \\ \Theta_{\mathrm{bulk}} &= \frac{\mathcal{J}_{AB} + \mathcal{J}_{BA}}{2}, \qquad \zeta_{\mathrm{bulk}} = \frac{\mathcal{J}_{AB} - \mathcal{J}_{BA}}{2}. \end{aligned} \tag{D.17}$$

The corresponding density-matrix phase is

$$\Theta_{\mathrm{bulk}}(s_A s_B - s'_A s'_B) + \zeta_{\mathrm{bulk}}(s_A s'_B - s'_A s_B). \tag{D.18}$$

The first term has the form of a reciprocal joint Hamiltonian phase. The second couples a ket label of one species to a bra label of the other. Equal arms make it vanish by inversion and species exchange.

For $b = 1$, $L = 2$, and $d_A = d_B = 0.2$, exact evaluation gives

T	$\Theta_N/(gT)$	$\Theta_{\mathrm{bulk}}/(gT)$
20	$3.31165463 \times 10^{-5}$	$-1.24199362 \times 10^{-5}$
50	$3.31165463 \times 10^{-5}$	$2.55232811 \times 10^{-5}$
100	$3.31165463 \times 10^{-5}$	$3.11865153 \times 10^{-5}$
200	$3.31165463 \times 10^{-5}$	$3.26297702 \times 10^{-5}$
1000	$3.31165463 \times 10^{-5}$	$3.30969326 \times 10^{-5}$

The sign reversal at $T = 20$ belongs to the reciprocal phase of the bare slab. It is not a negative entanglement measure or a waiting-time claim for the constrained initial state.

D. Finite-slice TT virtual response

For two constant velocities define the TT angular contraction

$$\begin{aligned} F_{\mathrm{TT}}(\mathbf{n}) &= v_A^i v_A^j \Pi_{ij,kl}^{\mathrm{TT}}(\mathbf{n}) v_B^k v_B^l \\ &= [\mathbf{v}_A \cdot \mathbf{v}_B - (\mathbf{n}\cdot\mathbf{v}_A)(\mathbf{n}\cdot\mathbf{v}_B)]^2 \\ &\quad - \frac{1}{2}[v_A^2 - (\mathbf{n}\cdot\mathbf{v}_A)^2][v_B^2 - (\mathbf{n}\cdot\mathbf{v}_B)^2]. \end{aligned} \tag{D.19}$$

For a nonzero vector $\mathbf{r}$, choose $\mathbf{e}_1, \mathbf{e}_2$ perpendicular to $\widehat{\mathbf{r}}$ and put

$$\mathbf{n}_{\mathbf{r},z}(\varphi) = z\widehat{\mathbf{r}} + \sqrt{1 - z^2}(\mathbf{e}_1 \cos\varphi + \mathbf{e}_2 \sin\varphi). \tag{D.20}$$

Radial integration of the retarded TT propagator gives

$$J_{AB}^{\mathrm{TT}} = \frac{g\gamma_A\gamma_B}{\pi}\int_0^{t_f} dt\left[\frac{1}{r(t)}\int_0^{2\pi} d\varphi\,\frac{F_{\mathrm{TT}}(\mathbf{n}_{\mathbf{r}(t),0})}{1-[\mathbf{n}_{\mathbf{r}(t),0}\cdot\mathbf{v}_B]^2} - \frac{\mathbf{1}_{\{t<r_A(t)\}}}{r_A(t)}\int_0^{2\pi} d\varphi\,\frac{F_{\mathrm{TT}}(\mathbf{n}_{\mathbf{r}_A(t),t/r_A(t)})}{1-\mathbf{n}_{\mathbf{r}_A(t),t/r_A(t)}\cdot\mathbf{v}_B}\right]. \tag{D.21}$$

The second line enforces zero TT mean and first time derivative on the initial slice. The reciprocal phase and entangling sum are

$$\Phi_{\mathrm{TT}}^{s_As_B} = \frac{J_{AB}^{\mathrm{TT}}+J_{BA}^{\mathrm{TT}}}{2}, \qquad \Theta_{\mathrm{TT}} = \frac{1}{4}\sum_{s_A,s_B} s_As_B\Phi_{\mathrm{TT}}^{s_As_B}. \tag{D.22}$$

Because Π^{TT} is an orthogonal projector, $|F_{\mathrm{TT}}| \le v_A^2v_B^2/2$, which gives

$$|J_{AB}^{\mathrm{TT}}| \le g\gamma_A\gamma_Bv_A^2v_B^2\left[\frac{1}{1-v_B^2}\int_0^{t_f}\frac{dt}{r(t)} + \frac{1}{1-|\mathbf{v}_B|}\int_0^{\min(t_f,t_{\mathrm{on}}^{AB})}\frac{dt}{r_A(t)}\right]. \tag{D.23}$$

For the ordinary benchmark, the largest reciprocal branch bound is 9.88×10^{-10}. Adaptive time quadrature with 64, 128, and 256 angular nodes changes each evaluated phase by less than 2×10^{-24}.

Appendix E: Gaussian matrix elements and numerical evaluation

This appendix writes out the selected matrix elements and records the packet reduction, numerical quadrature, complete selected matrix, and checks supporting Section IV.

A. Matrix elements between selected modes

One-particle matrix elements. By Eq. (2.8), the selected mode states are $|s\rangle_\alpha = c_{\alpha s}^\dagger|0_\alpha\rangle = \int_{\mathbf{p}} w_{\alpha s}(\mathbf{p})\,|\mathbf{p}\rangle_\alpha$. With the normalization $\langle\mathbf{q}|\mathbf{p}\rangle = (2\pi)^3\delta^3(\mathbf{q}-\mathbf{p})$, this gives $\langle\mathbf{p}|s\rangle_\alpha = w_{\alpha s}(\mathbf{p})$. Every Fourier component (3.4) has the form $O = \int_{\mathbf{p}} o(\mathbf{p}+\mathbf{k},\mathbf{p})\,|\mathbf{p}+\mathbf{k}\rangle\langle\mathbf{p}|$, and for any such operator

$$\langle r|O|s\rangle_\alpha = \int_{\mathbf{p}} w_{\alpha r}^*(\mathbf{p}+\mathbf{k})\; o(\mathbf{p}+\mathbf{k},\mathbf{p})\; w_{\alpha s}(\mathbf{p}). \tag{E.1}$$

Read from right to left, this is the amplitude that the particle in mode s has momentum $\mathbf{p}$, times the vertex for absorbing $\mathbf{k}$, times the amplitude that $\mathbf{p}+\mathbf{k}$ belongs to mode r. The diagonal element of $\varepsilon_\alpha(t,\mathbf{k})$ is $\langle s|\varepsilon_\alpha(t,\mathbf{k})|s\rangle = \int d^3x\, e^{i\mathbf{k}\cdot\mathbf{x}}\langle s|\varepsilon_\alpha(t,\mathbf{x})|s\rangle$, the Fourier transform of the packet's energy density *at time t*. The recoil phase $e^{i\omega t}$ in Eq. (3.4) is what makes this profile move with the packet. The off-diagonal elements ($r\neq s$) are transitions between the two selected modes.

Recoil energies. In a mutual transition, A receives momentum $+\mathbf{k}$ and B receives $-\mathbf{k}$ (Sec. III A). For incoming momenta $\mathbf{p}_A, \mathbf{p}_B$, the two separate recoil energies are

$$\omega_A = E_{A,\mathbf{p}_A+\mathbf{k}} - E_{A,\mathbf{p}_A}, \qquad \omega_B = E_{B,\mathbf{p}_B-\mathbf{k}} - E_{B,\mathbf{p}_B}. \tag{E.2}$$

Derivation of the constraint matrix element. Start from $\mathsf{K}_C^{AB} = \int_0^{t_f} H_C^{AB}(t)\,dt = -\frac{\kappa_g^2}{4}\int_0^{t_f} dt\int_{\mathbf{k}}\mathcal{C}_{AB}(t;\mathbf{k},-\mathbf{k})$ (Eq. (3.6)).

- *Operator form.* $\mathcal{C}_{AB}$ is a fixed linear combination of products of one A-component (ε_A, $\mathbf{j}_A$, or T_A at $+\mathbf{k}$) and one B-component (at $-\mathbf{k}$). The coefficients depend on $\mathbf{k}$ through $1/k^2$ and through the direction $\mathbf{n}$ used in the longitudinal/transverse split. Inserting Eq. (3.4) for each factor gives

$$\mathcal{C}_{AB}(t;\mathbf{k},-\mathbf{k}) = \int_{\mathbf{p}_A,\mathbf{p}_B} \mathcal{C}_{AB}[\tau_A(p_A+\mathbf{k},p_A),\tau_B(p_B-\mathbf{k},p_B);\mathbf{k}]\, e^{i(\omega_A+\omega_B)t}\, |\mathbf{p}_A+\mathbf{k}\rangle\langle\mathbf{p}_A| \otimes |\mathbf{p}_B-\mathbf{k}\rangle\langle\mathbf{p}_B|. \tag{E.3}$$

- *The kernel with vertices.* $\mathcal{C}_{AB}[\tau_A,\tau_B;\mathbf{k}]$ is Eq. (3.5) with each operator replaced by its one-particle kernel. Writing $\boldsymbol{\tau}^0_\alpha = (\tau^{01}_\alpha,\tau^{02}_\alpha,\tau^{03}_\alpha)$ for the momentum-density vector and splitting it into parts along and transverse to $\mathbf{n}$,

$$\mathcal{C}_{AB}[\tau_A,\tau_B;\mathbf{k}] = \frac{1}{k^2}\Big[\tfrac{1}{2}\tau^{00}_A\tau^{00}_B - 2\,\boldsymbol{\tau}^0_{A,T}\cdot\boldsymbol{\tau}^0_{B,T} - \tfrac{3}{2}\,\boldsymbol{\tau}^0_{A,L}\cdot\boldsymbol{\tau}^0_{B,L} + \tfrac{1}{2}\big(\tau^{00}_A\,\tau^{ii}_B + \tau^{ii}_A\,\tau^{00}_B\big)\Big]. \tag{E.4}$$

- *Time integral.* The two vertices act at the same time, so the only time dependence is $e^{i(\omega_A+\omega_B)t}$, and $\int_0^{t_f} e^{i(\omega_A+\omega_B)t}dt = \mathcal{A}_{t_f}(\omega_A+\omega_B)$ with $\mathcal{A}_{t_f}$ from Eq. (3.11).
- *Matrix element.* Taking $\langle r_A,r_B|\cdots|s_A,s_B\rangle$ and applying Eq. (E.1) to each species gives the exact mutual constraint generator:

$$\begin{aligned}(\mathsf{K}^{AB}_C)_{r_Ar_B,s_As_B} = &-\frac{\kappa_g^2}{4}\int_{\mathbf{k},\mathbf{p}_A,\mathbf{p}_B} w^*_{Ar_A}(\mathbf{p}_A+\mathbf{k})w_{As_A}(\mathbf{p}_A)w^*_{Br_B}(\mathbf{p}_B-\mathbf{k})w_{Bs_B}(\mathbf{p}_B)\\ &\times \mathcal{C}_{AB}[\tau_A(p_A+\mathbf{k},p_A),\tau_B(p_B-\mathbf{k},p_B);\mathbf{k}]\,\mathcal{A}_{t_f}(\omega_A+\omega_B).\end{aligned} \tag{E.5}$$

Each transition is weighted by its incoming and outgoing packet amplitudes. The same integral therefore retains energy density, currents, spatial stress, exact square-root energies, and off-diagonal mode transitions. The last of these are absent from a four-phase description, even when the external packets have negligible overlap. Note that the finite observation time supplies no relation setting the two recoil energies equal, or identifying either with a graviton frequency.

Check against Sec. III D. Keep only the energy density, $\tau^{00}\simeq m$, so that $\mathcal{C}_{AB}\simeq m_Am_B/(2k^2)$. For narrow, well-separated packets keep only the diagonal elements. Eq. (E.5) then becomes $-\frac{\kappa_g^2}{8}m_Am_B\int_0^{t_f}dt\int_{\mathbf{k}}F_A(\mathbf{k},t)F_B(-\mathbf{k},t)/k^2$, where $F_\alpha(\mathbf{k},t)$ is the Fourier transform of the packet's normalized density at time t. For point-like packets this is $-g\int_0^{t_f}dt/r(t) = \mathsf{K}_N$, the branch phase of Eq. (3.20). Section C below gives the exact Gaussian version of this Newtonian limit, including off-diagonal elements.

TT matrix elements. The TT operators are evaluated in the same way, with the emission vertex in place of the constraint kernel. From Eq. (3.12), $L_{\alpha,\lambda\mathbf{k}} = \frac{i\kappa_g}{2\sqrt{2k}}\int_{\mathbf{p}} e^{\lambda *}_{ij}\tau^{ij}_\alpha(p-\mathbf{k},p)\,\mathcal{A}_{t_f}(\Delta_\alpha)\,|\mathbf{p}-\mathbf{k}\rangle\langle\mathbf{p}|_\alpha$, so

$$\langle r|L_{\alpha,\lambda\mathbf{k}}|s\rangle_\alpha = \frac{i\kappa_g}{2\sqrt{2k}}\int_{\mathbf{p}} w^*_{\alpha r}(\mathbf{p}-\mathbf{k})\,e^{\lambda *}_{ij}\tau^{ij}_\alpha(p-\mathbf{k},p)\,\mathcal{A}_{t_f}\big(\Delta_\alpha(\mathbf{p},\mathbf{k})\big)\,w_{\alpha s}(\mathbf{p}). \tag{E.6}$$

For the ordered exchange operator of Eq. (3.13), the mutual part has matrix elements

$$\langle r_Ar_B|\mathsf{V}^{AB}_{\rm TT}|s_As_B\rangle = \sum_\lambda\int_{\mathbf{k}}\int_0^{t_f}dt\int_0^t dt'\Big[\langle r_A|\mathsf{v}^\dagger_{A,\lambda\mathbf{k}}(t)|s_A\rangle\langle r_B|\mathsf{v}_{B,\lambda\mathbf{k}}(t')|s_B\rangle + (A\leftrightarrow B)\Big], \tag{E.7}$$

in which one particle emits at t' and the other absorbs at $t>t'$. As in Eq. (3.13), its anti-Hermitian part gives $i\mathsf{K}^{AB}_{\rm TT}$ and its Hermitian part gives $\frac{1}{2}\Gamma_{\rm cross}$. The time integrals are ordered double integrals of the type treated in Appendix C. Equations (E.5)–(E.7) are the quantities evaluated numerically below: Sec. B reduces them to normalized Gaussian averages, and Sec. C gives an analytic convolution for the nonrelativistic Newtonian specialization, retaining the same mode transitions. The independent-frequency CTP representation and its endpoint matching are developed in Appendix C.

B. Exact Gaussian averages

Every mutual matrix element, such as Eq. (E.5), is an integral over $(\mathbf{k}, \mathbf{p}_A, \mathbf{p}_B)$ of a product of four Gaussian packet amplitudes times a smooth kernel. The product of the four Gaussians is itself a Gaussian in these nine variables. We therefore write each matrix element as its total Gaussian weight C times the average of the kernel over a normalized nine-dimensional Gaussian distribution, and evaluate that average by quadrature or quasi-Monte Carlo sampling. For raw packet indices, define

$$\begin{aligned}
\Delta\mathbf{p}_A &= \mathbf{p}_{Ar_A} - \mathbf{p}_{As_A}, \quad \Delta\mathbf{a}_A = \mathbf{a}_{Ar_A} - \mathbf{a}_{As_A},\\
\Delta\mathbf{p}_B &= \mathbf{p}_{Br_B} - \mathbf{p}_{Bs_B}, \quad \Delta\mathbf{a}_B = \mathbf{a}_{Br_B} - \mathbf{a}_{Bs_B},\\
q_A &= \ell_A^2, \quad q_B = \ell_B^2, \quad \mathbf{k}_0 = \frac{q_A\Delta\mathbf{p}_A - q_B\Delta\mathbf{p}_B}{q_A + q_B},\\
C_{r_As_A,r_Bs_B} &= \frac{\exp[-q_Aq_B|\Delta\mathbf{p}_A + \Delta\mathbf{p}_B|^2/(2(q_A+q_B))]}{(2\pi)^{3/2}(q_A+q_B)^{3/2}}.
\end{aligned} \tag{E.8}$$

Here $\mathbf{k}_0$ is the mean momentum transfer and C is the total Gaussian weight. Samples of the nine-dimensional Gaussian are generated from independent standard normal three-vectors $\mathbf{z}_0, \mathbf{z}_A, \mathbf{z}_B$:

$$\begin{aligned}
\mathbf{k} &= \mathbf{k}_0 + \frac{\mathbf{z}_0}{\sqrt{q_A+q_B}},\\
\mathbf{p}_A &= \frac{\mathbf{p}_{Ar_A} + \mathbf{p}_{As_A} - \mathbf{k}}{2} + \frac{\mathbf{z}_A}{2\ell_A},\\
\mathbf{p}_B &= \frac{\mathbf{p}_{Br_B} + \mathbf{p}_{Bs_B} + \mathbf{k}}{2} + \frac{\mathbf{z}_B}{2\ell_B},\\
\varphi &= \mathbf{p}_A\cdot\Delta\mathbf{a}_A + \mathbf{p}_B\cdot\Delta\mathbf{a}_B + \mathbf{k}\cdot(\mathbf{a}_{Ar_A} - \mathbf{a}_{Br_B}).
\end{aligned} \tag{E.9}$$

where φ collects the position-dependent phases of the packets. A mutual coefficient is then a normalized nine-dimensional expectation value,

$$\widetilde{I}^{Q}_{r_As_A,r_Bs_B} = C_{r_As_A,r_Bs_B}\,\mathbb{E}\left[e^{i\varphi}\,\tau_A^{\mu\nu}\mathcal{P}^{\mathrm{dD}}_{\mu\nu,\rho\sigma}\tau_B^{\rho\sigma}K_Q\right]. \tag{E.10}$$

Here $\mathcal{P}^{\mathrm{dD}}$ is the de Donder numerator of Appendix A E, and K_Q is the finite-time kernel of the contribution being evaluated, labeled by Q (for example the retarded kernel K_R or the symmetric kernel K_N of Eq. (C.2), with the recoil energies (E.2) as frequency arguments). Applying R_A and R_B on all external indices gives the orthonormal selected coefficients. Exact recoil, tensor vertices, packet phases, and finite-time functions all remain inside the average.

C. Analytic Newtonian specialization

For one species, let the following be the mean position, mean momentum, and differences associated with a pair of raw branches (r, s). (The vector $\mathbf{h}_{rs}$ here is unrelated to the graviton field.)

$$\begin{aligned}
\overline{\mathbf{p}}_{rs} &= \frac{\mathbf{p}_r + \mathbf{p}_s}{2}, \qquad \Delta\mathbf{p}_{rs} = \mathbf{p}_r - \mathbf{p}_s, \qquad \Delta\mathbf{a}_{rs} = \mathbf{a}_r - \mathbf{a}_s,\\
\mathbf{R}_{rs}(t) &= \frac{\mathbf{a}_r + \mathbf{a}_s}{2} + \frac{\overline{\mathbf{p}}_{rs}t}{m}, \quad \mathbf{h}_{rs}(t) = \ell^2\Delta\mathbf{p}_{rs} - \frac{t\Delta\mathbf{a}_{rs}}{4m\ell^2}.
\end{aligned} \tag{E.11}$$

Suppressing the species label, let $\mathcal{S}_{rs}$ denote the overlap in Eq. (2.6). The transition-density transform is

$$F_{rs}(\mathbf{k}, t) = \mathcal{S}_{rs}\exp[-\sigma_x(t)^2k^2/2 + \mathbf{k}\cdot\mathbf{h}_{rs}(t) + i\mathbf{k}\cdot\mathbf{R}_{rs}(t)]. \tag{E.12}$$

The exact Gaussian convolution of the Newtonian potential is

$$\begin{aligned}\widetilde{\mathcal{V}}_{r_A s_A, r_B s_B}(t) &= g\mathcal{S}_{A,r_A s_A}\mathcal{S}_{B,r_B s_B}\frac{\operatorname{erf}(\sqrt{\mathbf{Z}\cdot\mathbf{Z}}/(\sqrt{2}s))}{\sqrt{\mathbf{Z}\cdot\mathbf{Z}}},\\ s^2 &= \sigma_{x,A}^2+\sigma_{x,B}^2,\\ \mathbf{Z} &= \mathbf{R}_{A,r_A s_A}-\mathbf{R}_{B,r_B s_B}-i(\mathbf{h}_{A,r_A s_A}-\mathbf{h}_{B,r_B s_B}).\end{aligned} \tag{E.13}$$

The dot product contains no complex conjugation. For stable off-diagonal evaluation, combine the overlap exponent Λ with the convolution:

$$\widetilde{\mathcal{V}}(t)=\frac{2g}{\sqrt{\pi}}\int_0^{1/(\sqrt{2}s(t))}dz\,\exp[\Lambda-(\mathbf{Z}\cdot\mathbf{Z})z^2]. \tag{E.14}$$

After the Gram transform, $\mathsf{K}_N=-\int_0^{t_f}\mathcal{V}(t)dt$ and $\Phi_N=-\mathsf{K}_N$.

D. Exact constraint and TT quadrature

For midpoint momenta $\mathbf{u}_A=\mathbf{p}_A+\mathbf{k}/2$ and $\mathbf{u}_B=\mathbf{p}_B-\mathbf{k}/2$, arrange the constraint kernel as follows. (In this subsection X and Y are scalar coefficients, not Pauli operators.)

$$\mathcal{C}_{AB}=\frac{X}{k^2}+\frac{Y}{k^4},\quad X=\frac{1}{2}\varepsilon_A\varepsilon_B-2\mathbf{j}_A\cdot\mathbf{j}_B+\frac{1}{2}(\varepsilon_A\operatorname{tr}\mathsf{T}_B+\operatorname{tr}\mathsf{T}_A\varepsilon_B),\quad Y=\frac{1}{2}(\mathbf{j}_A\cdot\mathbf{k})(\mathbf{j}_B\cdot\mathbf{k}). \tag{E.15}$$

Laplace representations of k^{-2} and k^{-4}, followed by completion of the Gaussian squares, use

$$q(t)=\ell^2+\frac{t^2}{4m^2\ell^2},\qquad z=\frac{1}{2\sqrt{q+\beta}},\qquad 0<z<\frac{1}{2\sqrt{q}}. \tag{E.16}$$

Tensor Gauss–Hermite quadrature treats midpoint and transfer variables; Gauss–Legendre quadrature treats t and z. Every node uses the exact scalar vertex and energy. The stable recoil residual is

$$\Delta E-\frac{\mathbf{u}\cdot\mathbf{k}}{m}=-\frac{\mathbf{u}\cdot\mathbf{k}}{m(E_++E_-)}\left[\frac{\mathbf{p}_+^2}{E_++m}+\frac{\mathbf{p}_-^2}{E_-+m}\right],\qquad \mathbf{p}_\pm=\mathbf{u}\pm\mathbf{k}/2. \tag{E.17}$$

The Gaussian contour deformation remains far from the energy branch points. Bounds at transfer radius 400 and midpoint residual radius 600 give $|\operatorname{Im}\mathbf{p}_\pm|<330\ll m=5000$, $\operatorname{Re}E_\pm>4989$, and a tail contribution below 10^{-30} after a generous polynomial allowance.

Mutual TT exchange and selected emission use scrambled Sobol averages. Same-species loss retains the full intermediate momentum. For $p=|\mathbf{p}|$, $E=\sqrt{m^2+p^2}$, $\mu=\widehat{\mathbf{p}}\cdot\widehat{\mathbf{k}}$, and $u=E_{\mathbf{p}-\mathbf{k}}+k-E$,

$$k=\frac{u(2E+u)}{2(E-p\mu+u)},\qquad \frac{k\,dk}{E_{\mathbf{p}-\mathbf{k}}}=\frac{u(2E+u)}{2(E-p\mu+u)^2}\,du. \tag{E.18}$$

The range is $0<u<\infty$. Analytic sine and cosine integrals evaluate the radial oscillations without a cutoff. The one-body matrices are

$$\Gamma_{\alpha,\mathrm{self}}\simeq\begin{pmatrix}7.94095402485\times10^{-11} & -1.30680241160\times10^{-18}\\ -1.30680241160\times10^{-18} & 7.94095402485\times10^{-11}\end{pmatrix},\quad \mathsf{K}_{\alpha,\mathrm{tr}}\simeq\begin{pmatrix}4.72499825141\times10^{-12} & -7.77568927575\times10^{-20}\\ -7.77568927575\times10^{-20} & 4.72499825141\times10^{-12}\end{pmatrix}. \tag{E.19}$$

The symmetric local cat is an eigenvector of both matrices, so its transient Hermitian commutator vanishes. Full self loss remains in p_{sel}.

E. Dissipative bound and complete matrix

For the benchmark packets, a one-particle emission bound $q_* < 3.4\times10^{-9}$, including a Gaussian tail below 5.9×10^{-22}, gives

$$\|\delta\rho_{\mathrm{TT,diss,cond}}\|_1 < 1.23\times10^{-7}, \qquad \|\delta\rho^{T_B}_{\mathrm{TT,diss,cond}}\|_1 < 2.46\times10^{-7}. \tag{E.20}$$

This bound is much looser than the directly evaluated shift and does not constrain Hermitian exchange.
For the ordinary benchmark, in the basis $(++,+-,-+,--)$,

$$\begin{aligned}\delta\rho \simeq 10^{-10}&\begin{pmatrix} 0.050234 & 0.927408 & -0.992453 & 0.018806\\ 0.927408 & 1.869509 & -0.148691 & 0.927495\\ -0.992453 & -0.148691 & -1.970183 & -0.992336\\ 0.018806 & 0.927495 & -0.992336 & 0.050439\end{pmatrix}\\ &+ i\,10^{-4}\begin{pmatrix} 0 & 0.959494232 & -0.907912143 & 0\\ -0.959494232 & 0 & -1.867406375 & -0.959494232\\ 0.907912143 & 1.867406375 & 0 & 0.907912143\\ 0 & 0.959494232 & -0.907912143 & 0\end{pmatrix}.\end{aligned} \tag{E.21}$$

The event data are

$$p_{\mathrm{sel}} = 1 - 7.62766\times10^{-11} + O(G^2), \quad \mathrm{Tr}(\text{selected TT gain}) = 1.34927\times10^{-10}, \quad \langle\Gamma_{\mathrm{TT}}\rangle_0 = 2.11203\times10^{-10}. \tag{E.22}$$

F. Convergence and reproducibility

Contribution and quadrature	Result	Empirical standard error
Constraint, Hermite 3 and time/auxiliary 96/96	$\lambda_C = -5.158282922176\times10^{-6}$	deterministic
Constraint, Hermite 5 and time/auxiliary 64/64	$\lambda_C = -5.158282922120\times10^{-6}$	deterministic
TT Hermitian, 2^{19} points and 8 scramblings	$\Delta\lambda_H = 7.44089096\times10^{-11}$	2.669×10^{-13}
TT Hermitian, 2^{22} points and 8 scramblings	$\Delta\lambda_H = 7.40100782\times10^{-11}$	3.277×10^{-14}
TT dissipative, 2^{18} points and 8 scramblings	$\Delta\lambda_{\mathrm{diss}} = 6.49366378\times10^{-12}$	1.365×10^{-15}
TT dissipative, 2^{20} points and 8 scramblings	$\Delta\lambda_{\mathrm{diss}} = 6.49415077\times10^{-12}$	5.153×10^{-15}

The quoted absolute uncertainty 10^{-12} is deliberately larger than the observed refinements and five times the final scramble standard error. It does not estimate omitted $O(G^2)$ physics. The assembled channel was checked for Hermiticity, conditional trace, selected loss, and agreement with the block formula, Eq. (4.2).

G. Full-precision benchmark data

The table separates comparisons made in the Newtonian model from the contributions to the exact-recoil channel. In the additive rows, the constraint result is followed by the Hermitian TT shift and then by the conditional TT dissipative shift. All eigenvalues refer to the first-order projected spectrum (the smallest eigenvalue of D) at the single observation time of Eq. (4.7).

Numerically, the entries of the block (4.2) are $\mathcal{E} \simeq -5.15820891203\times10^{-6} + i\,1.00670\times10^{-11}$, $\nu_A \simeq 6.49577\times10^{-12}$, $\nu_B \simeq 6.49277\times10^{-12}$, and $\gamma_{\mathrm{cross}} \simeq -6.68303\times10^{-12} - i\,2.24\times10^{-16}$. The last imaginary part lies within the stochastic

Calculation or contribution	Negative eigenvalue, shift, or probability
Newtonian finite-width conservative comparison	$-5.45972935 \times 10^{-6}$
Full exact-recoil constraint interaction	$-5.15828292 \times 10^{-6}$
Hermitian TT shift	$+7.40101 \times 10^{-11}$
Full Gaussian Hermitian eigenvalue λ_H	$-5.15820891 \times 10^{-6}$
Conditional TT dissipative shift	$+6.49415 \times 10^{-12}$
Full conditional eigenvalue λ_-	$-5.15820242 \times 10^{-6}$
Selection probability $p_{\rm sel}$	$1 - 7.62766 \times 10^{-11} + O(G^2)$

numerical uncertainty. The full selected matrix and its component integrals are given in Sec. E above. Some of its imaginary entries are of order 10^{-4}, because local phases need not be as small as the entangling amplitude.

The matched point-history constraint-plus-TT result (Appendix C) provides a separate Hermitian reference, $\lambda_{\rm point} = -5.15678073611 \times 10^{-6}$. Define

$$\begin{aligned} \epsilon_{\rm Gauss} &= \lambda_H - \lambda_{\rm point} \simeq -1.42818 \times 10^{-9}, \\ \lambda_- &= \lambda_{\rm point} + \epsilon_{\rm Gauss} + \delta\lambda_{\rm TT,diss}, \\ \delta\lambda_{\rm TT,diss} &\simeq 6.49415 \times 10^{-12}. \end{aligned} \tag{E.23}$$

The first difference includes the declared-input velocity shift, finite width, and recoil. It is not a width correction at artificially matched exact velocities. The second is obtained by evaluating the physical selected gain, unrestricted loss, and conditioning. An independent TT jump bound gives a partial-transpose trace-norm correction below 2.46×10^{-7} for this preparation, including its Gaussian momentum tails. That bound is much looser than the directly evaluated shift and does not bound Hermitian exchange.

The full-precision eigenvalues of D are $-5.15820241789 \times 10^{-6}$, 0, and $+5.15821540643 \times 10^{-6}$, so that $\mathcal{N}(\rho_{\rm det}) = 5.1582024 \times 10^{-6} + O(G^2)$ for the declared preparation.

The estimated absolute integration uncertainty in the negative eigenvalue and $\epsilon_{\rm Gauss}$ is 10^{-12}; the selection-loss uncertainty is approximately 10^{-13}. Section F above distinguishes these empirical numerical allowances from omitted $O(G^2)$ terms and additional scalar Fock sectors. The result is for the number-preserving scalar channel and finite-slice TT vacuum specified in Sections II and III.

Appendix F: Perturbative partial transpose

This appendix derives the leading partial-transpose test used in Sections IV A and IV C, and gives a diagonal-noise specialization that separates conservative phase, correlated fluctuations, and local noise.

A. Projection onto the initially empty subspace

Let the input be a pure product $|\psi_A, \psi_B\rangle$, and choose local orthogonal complements $|\psi_A^\perp\rangle$, $|\psi_B^\perp\rangle$. In the local basis

$$|00\rangle = |\psi_A, \psi_B\rangle, \quad |01\rangle = |\psi_A, \psi_B^\perp\rangle, \quad |10\rangle = |\psi_A^\perp, \psi_B\rangle, \quad |11\rangle = |\psi_A^\perp, \psi_B^\perp\rangle. \tag{F.1}$$

Conditioning. The selected block must first be conditioned on selection. Write $\widetilde{\rho}_{\rm sel} = \rho_0 + \delta\widetilde{\rho}_{\rm sel}$ and $p_{\rm sel} = 1 + \mathrm{Tr}\,\delta\widetilde{\rho}_{\rm sel}$. The first-order correction to $\rho_{\rm det} = \widetilde{\rho}_{\rm sel}/p_{\rm sel}$ is then $\delta\rho_{\rm det} = \delta\widetilde{\rho}_{\rm sel} - \rho_0\,\mathrm{Tr}\,\delta\widetilde{\rho}_{\rm sel}$.

Degenerate perturbation theory. The initial state is a pure product, so $\rho_0^{T_B}$ is itself a rank-one projector, with one eigenvalue 1 and a threefold-degenerate eigenvalue 0. Entanglement at first order means that one of the three zero eigenvalues is pushed *negative*. By degenerate perturbation theory, the first-order values of those eigenvalues are the eigenvalues of the perturbation restricted to the degenerate (null) subspace. Mixing with the eigenvalue 1, separated by a gap of order one, affects them only at second order. With $V_{\rm null}$ the 4×3 matrix whose columns form an orthonormal basis of the null subspace, the relevant 3×3 matrix is

$$\mathsf{D} = V_{\rm null}^\dagger \left(\delta\widetilde{\rho}_{\rm sel} - \rho_0\,\mathrm{Tr}\,\delta\widetilde{\rho}_{\rm sel}\right)^{T_B} V_{\rm null}. \tag{F.2}$$

For the real even cats, $|\psi_\alpha\rangle = |+_x\rangle$, so the local basis (F.1) is the X basis: $|0\rangle = |+_x\rangle$ and $|1\rangle = |-_x\rangle$. (These $|0\rangle, |1\rangle$ are *not* the branch states $|\pm\rangle$.) Then $\rho_0 = |00\rangle\langle 00|$, and the columns of $V_{\rm null}$ are $(|01\rangle, |10\rangle, |11\rangle)$, in that order. Working with this projection avoids assigning second-order meaning to eigenvalues obtained by directly diagonalizing a density matrix that is known only through first order.

Which entries of $\delta\rho$ are needed. By Eq. (3.23), the partial transpose rearranges indices as $\langle ij|\sigma^{T_B}|kl\rangle = \langle il|\sigma|kj\rangle$. The six independent entries of the Hermitian block D are therefore six specific entries of $\delta\rho \equiv \delta\rho_{\rm det}$:

Entry of D (row, column)	Entry of $\delta\rho$	Meaning
$(01, 01)$	$\langle 01\|\delta\rho\|01\rangle$	population with only B flipped
$(10, 10)$	$\langle 10\|\delta\rho\|10\rangle$	population with only A flipped
$(11, 11)$	$\langle 11\|\delta\rho\|11\rangle$	population with both flipped
$(01, 10)$	$\langle 00\|\delta\rho\|11\rangle$	double-flip coherence
$(01, 11)$	$\langle 01\|\delta\rho\|11\rangle$	single-flip/double-flip coherence
$(10, 11)$	$\langle 11\|\delta\rho\|10\rangle$	double-flip/single-flip coherence

Evaluating them: the rank-one argument. Insert $\rho_0 = |00\rangle\langle 00|$ into Eq. (3.14). For selected states x, y,

$$\langle x|\delta\widetilde{\rho}_{\rm sel}|y\rangle = -i\langle x|\mathsf{K}|00\rangle\delta_{y,00} + i\,\delta_{x,00}\langle 00|\mathsf{K}|y\rangle - \tfrac{1}{2}\langle x|\Gamma_{\rm TT}|00\rangle\delta_{y,00} - \tfrac{1}{2}\delta_{x,00}\langle 00|\Gamma_{\rm TT}|y\rangle + \sum_\lambda \int_{\mathbf{k}} \langle x|L_{\lambda\mathbf{k}}|00\rangle\langle 00|L^\dagger_{\lambda\mathbf{k}}|y\rangle, \quad \text{(F.3)}$$

and the conditioning term $-\rho_0 \,\mathrm{Tr}\,\delta\widetilde{\rho}_{\rm sel}$ contributes only to $x = y = 00$. Three observations settle every entry of the table.

1. *Hermitian and loss terms need $|00\rangle$ on one side.* Among the six entries, only the double-flip coherence $\langle 00|\delta\rho|11\rangle$ qualifies. It receives $i\langle 00|\mathsf{K}|11\rangle - \frac{1}{2}\langle 00|\Gamma_{\rm TT}|11\rangle$.

2. *Only cross-species operators can flip both particles.* A one-body operator $O_A \otimes I$ has $\langle 00|O_A \otimes I|11\rangle = \langle 0|O_A|1\rangle\langle 0|1\rangle = 0$, and likewise for $I \otimes O_B$. So local Hamiltonians, self-energy transients, and local losses drop out of the double-flip coherence. Only $\mathsf{K}^{AB} = \mathsf{K}^{AB}_C + \mathsf{K}^{AB}_{\rm TT}$ and $\Gamma_{\rm cross} = \sum_\lambda \int_{\mathbf{k}} (L^\dagger_A L_B + L^\dagger_B L_A)$ survive.

3. *The gain from one-body jumps reaches only singly flipped states.* Because $L = L_A \otimes I + I \otimes L_B$, one has $\langle 11|L|00\rangle = \langle 1|L_A|0\rangle\langle 1|0\rangle + \langle 1|0\rangle\langle 1|L_B|0\rangle = 0$. The gain term therefore contributes nothing to the $|11\rangle$ population, to the two single-flip/double-flip coherences, or to the double-flip coherence, since each of these contains the factor $\langle 11|L|00\rangle$ or its complex conjugate. It does contribute the two single-flip populations, $\sum\int |\langle 01|L|00\rangle|^2 = \sum\int |\langle 1|L_B|0\rangle|^2$ and the same with A.

So D is fixed by the five numbers of Eq. (4.1). In terms of the branch-basis matrix elements of Appendix E A, and using $\langle 1|r\rangle = r/\sqrt{2}$ and $\langle s|0\rangle = 1/\sqrt{2}$ for the X basis, the double-flip amplitude is a signed sum over the branch-basis matrix,

$$\mathcal{E}_C = \frac{1}{4}\sum_{r_A, r_B, s_A, s_B = \pm} r_A r_B\, (\mathsf{K}^{AB}_C)_{r_A r_B, s_A s_B}, \qquad \gamma_{\rm cross} = \sum_\lambda \int_{\mathbf{k}} \Big[\langle 0|L_{A,\lambda\mathbf{k}}|1\rangle^* \langle 1|L_{B,\lambda\mathbf{k}}|0\rangle + (A \leftrightarrow B)\Big], \quad \text{(F.4)}$$

and similarly for $\mathcal{E}_{\rm TT}$. For a diagonal, branch-preserving generator $(\mathsf{K}^{AB}_C)_{s_A s_B, s_A s_B} = -\Phi_{s_A s_B}$, the first formula gives $\mathcal{E}_C = -\Theta$, the entangling phase of Eq. (3.21).

B. Diagonal phase-and-noise model

A useful analytic specialization assumes that the interaction preserves the branches (it is diagonal in the branch basis $|s_A, s_B\rangle$), that the response is reciprocal, and that each branch phase fluctuates with Gaussian statistics. The noise is characterized by an equal local phase-noise parameter ν_0 for each particle and a cross covariance ν_c between the phase noises of A and B. The resulting decoherence factors are (here A, B, D are numbers, not particle labels)

$$A = e^{-2\nu_0}, \qquad B = e^{-4(\nu_0 + \nu_c)}, \qquad D = e^{-4(\nu_0 - \nu_c)}. \quad \text{(F.5)}$$

After local phases are removed, the density matrix in the branch basis is

$$\rho = \frac{1}{4}\begin{pmatrix} 1 & Ae^{2i\Theta} & Ae^{2i\Theta} & B \\ Ae^{-2i\Theta} & 1 & D & Ae^{-2i\Theta} \\ Ae^{-2i\Theta} & D & 1 & Ae^{-2i\Theta} \\ B & Ae^{2i\Theta} & Ae^{2i\Theta} & 1 \end{pmatrix}. \tag{F.6}$$

In the Bell basis, four times the partial transpose separates into

$$\begin{pmatrix} 1+D & 2A\cos 2\Theta \\ 2A\cos 2\Theta & 1+B \end{pmatrix}, \qquad \begin{pmatrix} 1-D & -2iA\sin 2\Theta \\ 2iA\sin 2\Theta & 1-B \end{pmatrix}. \tag{F.7}$$

The eigenvalues are

$$\begin{aligned} \lambda_{\pm}^{(+)} &= \frac{2+B+D \pm \sqrt{(B-D)^2+16A^2\cos^2(2\Theta)}}{8}, \\ \lambda_{\pm}^{(-)} &= \frac{2-B-D \pm \sqrt{(B-D)^2+16A^2\sin^2(2\Theta)}}{8}. \end{aligned} \tag{F.8}$$

The first pair is nonnegative. Hence

$$\mathcal{N} = \frac{1}{8}\left[\sqrt{(B-D)^2+16A^2\sin^2(2\Theta)} - (2-B-D)\right]_+ = \left[\sqrt{\Theta^2+\nu_c^2}-\nu_0\right]_+ + O(G^2). \tag{F.9}$$

For unequal local noise parameters ν_{AA} and ν_{BB} (which reduce to ν_0 in the equal case), the projected first-order block gives

$$\mathcal{N} = \left[\sqrt{\Theta^2+\nu_c^2+\frac{1}{4}(\nu_{AA}-\nu_{BB})^2} - \frac{\nu_{AA}+\nu_{BB}}{2}\right]_+ + O(G^2). \tag{F.10}$$

The leading entanglement condition is $\Theta^2+\nu_c^2 > \nu_{AA}\nu_{BB}$. A directed bra–ket response ζ (the antisymmetric part of the two directed retarded responses, Eq. (D.17)) does not shift these initially zero partial-transpose eigenvalues at first order. Positivity of the untransposed state, however, requires $\nu_{AA}\nu_{BB} \geq \nu_c^2+\zeta^2$. This is why a retarded term without its matching covariance and endpoint state is not a physical channel.

Appendix G: Constraint zeros: root search and sensitivity

This appendix records the root definitions, symmetry checks, and numerical sensitivity supporting Section IV C.

A. Root function and symmetry

For the phase choice $\theta_A^{\rm in} = \theta_B^{\rm in} = \pi/2$, define the real root function

$$f_C(L,T) = \langle \psi_A^{\perp}, \psi_B^{\perp} | P_{\rm sel} \mathsf{K}_C(L,T) P_{\rm sel} | \psi_A, \psi_B \rangle. \tag{G.1}$$

Inversion followed by species exchange obeys $U_{\rm IX}|\psi\rangle = -|\chi\rangle$ and commutes with the full selected constraint operator. Therefore f_C is real before numerical integration. The implementation nevertheless evaluates the complex matrix element and keeps its imaginary residual as a symmetry diagnostic.

The simple and stationary roots solve

$$f_C(L_*,100) = 0, \qquad f_C(L_\dagger,T_\dagger) = 0, \qquad \partial_L f_C(L_\dagger,T_\dagger) = 0, \tag{G.2}$$

where the derivative changes positions and central momenta together through Eq. (2.2). The stored representatives are

$$L_* = 1.42994819774, \qquad (L_\dagger, T_\dagger) = (2.3975949, 22.5036458). \tag{G.3}$$

At the simple point, an independent Hermite-order-4 evaluation gives a residual 7.23×10^{-17}; the quoted root tolerance is 10^{-15}. At the stationary representative, the residual is below 10^{-13}.

B. Channel data at the roots

The full finite-width outputs are

Quantity	Simple root	Stationary root
$\mathcal{E}_C$	7.23×10^{-17} in the stored check	below 10^{-13}
$\mathcal{E}_{\mathrm{TT}}$	4.4610×10^{-11}	7.91754×10^{-9}
ν_A	2.78149×10^{-12}	3.65307×10^{-9}
ν_B	2.78141×10^{-12}	3.65317×10^{-9}
γ_{cross}	-3.52228×10^{-12}	-2.56962×10^{-9}
λ_-	-4.18635×10^{-11}	-4.36799×10^{-9}
$1 - p_{\mathrm{sel}}$	2.23687×10^{-11}	5.53669×10^{-8}

The dissipative channel alone has no negative eigenvalue. At the stationary root, setting $\mathcal{E}_{\mathrm{TT}} = 0$ gives the nonzero values $(2.3683, 4.9379) \times 10^{-9}$. The negative eigenvalues in the table therefore come from Hermitian TT exchange.

The root is not a zero of the Newtonian contribution or of the constraint operator. At the simple point, the finite-width Newtonian entangling phase is 3.76866×10^{-7}, the common constraint phase is 0.00620829, and the norm of the remaining nonlocal selected constraint operator after subtracting identity and one-body terms is about 8.0×10^{-11}. At the stationary point the corresponding Newtonian and common phases are 1.16738×10^{-6} and 0.00114049. Near the simple root, the leading small-arm and finite-arm point-Newtonian roots, $L = 1.45992384415$ and $L = 1.41506033267$, differ from the full selected root.

C. Local sensitivity

Along the prescribed geometry, the simple root has

$$\partial_L f_C(L_*, 100) \simeq -2.47227 \times 10^{-5}. \tag{G.4}$$

Keeping renewed constraint contamination below $0.1|\mathcal{E}_{\mathrm{TT}}|$ gives $|\delta L| \lesssim 1.8 \times 10^{-7}$.

Near the stationary point, the evaluated local expansion is

$$f_C \simeq -1.0289 \times 10^{-7}\,\delta T + 3.8661 \times 10^{-7}\,(\delta L)^2 + 2.3436 \times 10^{-6}\,\delta b - 3.3424 \times 10^{-8}\,\delta d + \cdots . \tag{G.5}$$

The resulting isolated tolerances at the same ten-percent criterion are

Variation	Approximate allowance
L, with momenta changed according to Eq. (2.2)	$\lvert\delta L\rvert \lesssim 0.045$
T, with momenta changed according to Eq. (2.2)	$\lvert\delta T\rvert \lesssim 0.0077$
impact parameter b	$\lvert\delta b\rvert \lesssim 3.4 \times 10^{-4}$
initial L with momenta fixed	$\lvert\delta L\rvert \lesssim 1.6 \times 10^{-4}$

The last row uses $\partial_L f_C|_{\mathbf{p}} \simeq -4.84 \times 10^{-6}$. The stationary derivative in Eq. (G.2) therefore does not protect against an independent position error. Simultaneous variations must be combined in Eq. (G.5).

D. Reflection comparison and perturbative scope

A geometry with mean flight along x, the A arms along y, and the B arms along z has two separate reflection symmetries for even cats. Each selected Hamiltonian and the loss operator commute with the corresponding local branch flip. Their double-flip matrix elements vanish in both the constraint and TT sectors, leaving only nonnegative local gain in the leading block. This construction shows that a symmetry of the whole setup can cancel the TT entangler together with the constraint entangler; it does not forbid other tuned preparations.

The refinement records separate root-location error from quadrature error:

Constraint calculation	Residual $\mathcal{E}_C$
Simple root, Hermite 3 and time/auxiliary 48/48	-7.40×10^{-19}
Same point, Hermite 4 and 64/64	$+7.23 \times 10^{-17}$
Stationary representative, Hermite 3 and 48/48	$+2.11 \times 10^{-16}$
Same representative, Hermite 4 and 64/64	-5.42805×10^{-15}
Same representative, Hermite 5 and 64/64	-5.42696×10^{-15}

The TT-channel refinement is

Evaluation	Negative eigenvalue	Scramble standard error
Simple point, 2^{18} samples and 4 scramblings	-4.18249×10^{-11}	6.66×10^{-14}
Simple point, 2^{21} samples and 8 scramblings	-4.18635×10^{-11}	1.52×10^{-14}
Stationary point, 2^{18} samples and 4 scramblings	-4.33221×10^{-9}	2.19×10^{-11}
Stationary point, 2^{20} samples and 4 scramblings	-4.36799×10^{-9}	7.35×10^{-12}

These records verify Hermiticity, trace, the inversion-exchange symmetry, and agreement with the two-dimensional formula in Eq. (4.15). Their numerical uncertainty does not bound the omitted $O(G^2)$ constraint contribution. At the quoted finite coupling, no such bound below the surviving TT signal has been established.

Appendix H: Position statistics at exact overlap

This appendix gives the position-statistics argument summarized in Sec. V: at the designed crossing time, ideal position detection cannot distinguish the gravitational phase state from a separable mixture. It uses the nonrelativistic detection envelope of the field modes. The interaction calculation itself never makes this approximation.

A. Spatial envelopes and the information they carry

Connection to ordinary wavefunctions. Position space enters only after the field-theoretic state and detector have been fixed. For a one-particle field state, the function $\langle 0|\phi(t,\mathbf{x})|f\rangle$ is its positive-frequency wavefunction. In the nonrelativistic limit, after the rest-energy phase e^{-imt} and the normalization $1/\sqrt{2m}$ are removed, it becomes the usual Schrödinger wavefunction. For a raw branch:

$$\begin{aligned}
\mathcal{U}_{\alpha s}(t,\mathbf{x}) &= \langle 0_\alpha|\phi_\alpha(t,\mathbf{x})|f_{\alpha s}\rangle = \int_{\mathbf{p}} \frac{f_{\alpha s}(\mathbf{p})}{\sqrt{2E_{\alpha,\mathbf{p}}}} e^{-iE_{\alpha,\mathbf{p}}t + i\mathbf{p}\cdot\mathbf{x}}, \\
\mathcal{U}_{\alpha s}(t,\mathbf{x}) &= \frac{e^{-im_\alpha t}}{\sqrt{2m_\alpha}} u_{\alpha s}(\mathbf{x},t)\left[1 + O\left(\frac{p^2_{\text{support}}}{m_\alpha^2}\right)\right], \\
\langle f_{\alpha s}| : T^{00}_\alpha(t,\mathbf{x}) : |f_{\alpha s}\rangle &= m_\alpha |u_{\alpha s}(\mathbf{x},t)|^2 \left[1 + O\left(\frac{p^2_{\text{support}}}{m_\alpha^2}\right)\right].
\end{aligned} \tag{H.1}$$

Here $p_{\rm support}$ denotes a momentum scale bounding the region of appreciable packet weight, and $:T^{00}_\alpha:$ is the normal-ordered energy density. The last line confirms that, to leading order, the packet's energy density is $m_\alpha|u_{\alpha s}|^2$, the mass density of the familiar one-particle wavefunction. Superpositions include the corresponding cross terms. Thus $u_{\alpha s}$ is an effective one-particle detection envelope derived from the field state. It is not being promoted to a fundamental relativistic position wavefunction.

Using $E_{\alpha,\mathbf{p}} - m_\alpha \simeq \mathbf{p}^2/(2m_\alpha)$, the Gaussian integral gives the familiar freely spreading Gaussian,

$$\begin{aligned} u_{\alpha s}(\mathbf{x},t) &= \frac{(2\pi\ell_\alpha^2)^{-3/4}}{(1+i\eta_\alpha)^{3/2}} \exp\left[-\frac{|\mathbf{x}-\mathbf{a}_{\alpha s}-\mathbf{p}_{\alpha s}t/m_\alpha|^2}{4\ell_\alpha^2(1+i\eta_\alpha)} + i\mathbf{p}_{\alpha s}\cdot(\mathbf{x}-\mathbf{a}_{\alpha s}) - i\frac{\mathbf{p}_{\alpha s}^2 t}{2m_\alpha}\right], \\ \eta_\alpha &= \frac{t}{2m_\alpha\ell_\alpha^2}, \qquad \sigma_{x,\alpha}(t)^2 = \ell_\alpha^2 + \frac{t^2}{4m_\alpha^2\ell_\alpha^2}, \\ v_{\alpha s}(\mathbf{x},t) &= \sum_{r=\pm} u_{\alpha r}(\mathbf{x},t)(R_\alpha)_{rs}. \end{aligned} \tag{H.2}$$

where η_α is the dimensionless spreading time and $\sigma_{x,\alpha}(t)$ the rms position width at time t. The rest-energy phase has already been separated in Eq. (H.1). The final line applies the same Gram transformation as Eq. (2.8) and defines the orthonormal nonrelativistic detection amplitudes $v_{\alpha s}$ used below. Equation (H.2) is used only for this spatial detector model. The relativistic interaction calculation continues to use the exact Fock-space modes and square-root energies of Eqs. (2.3), (2.5), and (2.12).

What happens at the crossing time. At $t = T$, the two branches of each particle sit at the same place (in this approximation their density envelopes coincide exactly) but still move with different transverse momenta. Their superposition therefore shows interference fringes. Restoring $\hbar$ in the spatial scales, the momentum difference, the fringe wavelength, and the packet width are

$$\Delta\mathbf{p}_\alpha = -\frac{m_\alpha d_\alpha}{T}\mathbf{e}_y, \qquad \lambda_{\rm fringe,\alpha} = \frac{2\pi\hbar T}{m_\alpha d_\alpha}, \qquad \sigma_{x,\alpha}(t)^2 = \ell_\alpha^2 + \frac{\hbar^2 t^2}{4m_\alpha^2\ell_\alpha^2}. \tag{H.3}$$

The overlap inferred from the common envelope is $\exp[-\sigma_{x,\alpha}(T)^2|\Delta\mathbf{p}_\alpha|^2/(2\hbar^2)] = S_\alpha$, equal to the Fock-space commutator in Eq. (2.6). Spatially coincident densities therefore do not make the two one-particle field modes identical: the modes remain distinguishable through their momenta. In the dispersive limit, $\sigma_{x,\alpha}(T)/\lambda_{\rm fringe,\alpha} \to d_\alpha/(4\pi\ell_\alpha)$. Minimizing the final width over ℓ_α gives $\ell_\alpha^2 = \hbar T/(2m_\alpha)$ and $\sigma_{x,\alpha,\min}(T) = \sqrt{\hbar T/m_\alpha}$.

Let $v_{\alpha\pm}(\mathbf{x}, t_f)$ be the orthonormal nonrelativistic detection amplitudes defined in Eq. (H.2). They are the spatial readout approximation to the outgoing one-particle field modes, including the exact Gram transformation of the raw Gaussians. If particle α is in the selected state $\sum_s c_s|s\rangle_\alpha$, its position wavefunction is $\sum_s c_s v_{\alpha s}(\mathbf{x})$, and the detection probability density is $\sum_{r,s} c_r^* c_s\, v_{\alpha r}^*(\mathbf{x}) v_{\alpha s}(\mathbf{x})$. Written as an operator on the two-dimensional space, this defines the position POVM (positive operator-valued measure) $E_\alpha(\mathbf{x})$ of ideal number-resolving position detection. For two particles the joint position distribution is

$$\begin{aligned} E_\alpha(\mathbf{x}) &= \sum_{r,s=\pm} v_{\alpha r}^*(\mathbf{x},t_f) v_{\alpha s}(\mathbf{x},t_f)|r\rangle\langle s|, \qquad \int d^3x\, E_\alpha(\mathbf{x}) = I, \\ P(\mathbf{x}_A,\mathbf{x}_B) &= \mathrm{Tr}[\rho_{\rm det} E_A(\mathbf{x}_A)\otimes E_B(\mathbf{x}_B)]\,. \end{aligned} \tag{H.4}$$

Any operator on a qubit can be expanded in Pauli matrices. Suppressing the species and time labels temporarily,

$$\begin{aligned} E(\mathbf{x}) &= a_0 I + a_x X + a_y Y + a_z Z, \\ a_0 &= \tfrac{1}{2}(|v_+|^2 + |v_-|^2), & a_z &= \tfrac{1}{2}(|v_+|^2 - |v_-|^2), \\ a_x &= \mathrm{Re}(v_+^* v_-), & a_y &= -\,\mathrm{Im}(v_+^* v_-). \end{aligned} \tag{H.5}$$

We use $Y = \left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$. The coefficient a_0 describes the average envelope, a_z its branch-density difference, and a_x, a_y the two interference quadratures. A coefficient of the density matrix can be recovered from position data only when its associated spatial pattern is distinguishable from the others. In particular, joint interference does not by itself establish access to all two-qubit correlations.

B. A separable mimic at exact overlap

At $t_f = T$, the two raw local envelopes have the same magnitude, $|u_{\alpha+}| = |u_{\alpha-}|$. Symmetric orthonormalization preserves this equality: the real matrix $R_\alpha = r_0 I + r_x X$ of Eq. (2.9) cannot produce a Z component when it acts on a raw POVM containing only I, X, and Y. Thus $a_{\alpha z} = 0$ exactly, even with finite raw overlap. Position data retain interference but lose all branch-population information. In other words, at exact overlap position detection is only sensitive to operators of the form $\sigma_\mu \otimes \sigma_\nu$ with $\mu, \nu \in \{0, x, y\}$.

Choose local equatorial axes that absorb the acquired one-body phases. The ideal conservative phase state of Eq. (3.22) and a separable mixture are

$$\begin{aligned} |\psi_\Theta\rangle &= e^{i\Theta Z_A Z_B} |+_x, +_x\rangle \\ &= \cos\Theta\, |+_x, +_x\rangle + i \sin\Theta\, |-_x, -_x\rangle, \\ \rho_{\text{mimic}} &= \cos^2\Theta\, |+_x, +_x\rangle\langle +_x, +_x| \\ &\quad + \sin^2\Theta\, |-_x, -_x\rangle\langle -_x, -_x|. \end{aligned} \tag{H.6}$$

Both states give the same values for every accessible correlation. Using $e^{-i\Theta Z_A Z_B} X_A e^{i\Theta Z_A Z_B} = X_A e^{2i\Theta Z_A Z_B}$ (because X_A anticommutes with Z_A), one finds for $|\psi_\Theta\rangle$ that $\langle X_A\rangle = \langle X_B\rangle = \cos 2\Theta$, while $\langle X_A X_B\rangle = 1$ because $X_A X_B$ commutes with $Z_A Z_B$. Every other nontrivial correlation built from X and Y vanishes. The mixture ρ_{mimic} gives $\langle X_A\rangle = \cos^2\Theta - \sin^2\Theta = \cos 2\Theta$, $\langle X_A X_B\rangle = 1$, and zero for the rest. The two states therefore yield the same entire joint position distribution, yet one is entangled and the other is separable. Correlated random local Z rotations preserve this ambiguity, because they preserve the accessible local operator span and the separability of the mimic. Prepared relative phases merely rotate the equatorial axes for this phase-state family.

The obstruction applies to the ideal diagonal phase state and the indicated dephasing, rather than to every entangled state. It nevertheless shows why the intended gravitational phase cannot be certified from exact-overlap positions, even with perfect resolution and trusted two-mode support.

Away from overlap, the local branch centers separate again. The density difference then restores the population information while interference remains visible. Reconstructing Pauli correlations from positions, however, requires the four local spatial patterns to be linearly independent and calibrated, and contamination from motional modes outside the selected spaces to be bounded. These conditions are not part of the calculation in this paper.

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